Landowner Expectations of Wildfire Risk Have Accelerated Harvest, Reduced Volume, and Altered Fuel Loads in Western U.S. Forests

Cade White[*1]
Christopher Mihiar[2,†]
David J. Lewis[1]

Abstract

Wildfire risk has intensified across the western United States, generating hundreds of billions of dollars in economic damages each year. Economic theory shows that changing wildfire risk should induce landowners to reduce risk exposure through forest management, though there is little empirical evidence of such landowner adaptation. This study provides the first large-scale empirical evidence on how increasing wildfire risk affects plot-level forest management decisions on non-federal land. Using confidential plot-level data from the U.S.D.A. Forest Service Forest Inventory and Analysis program linked with wildfire perimeters and fire-related weather conditions, we examine how nearby wildfires and hotter, drier fire seasons influence the choice among harvest types and decisions regarding planting density on private and state timberland in Oregon, California, and Washington. Controlling for plot and climate factors, we find that increased wildfire exposure and a warmer, drier climate lead landowners to harvest more and to lower their planting densities, revealing adaptive responses to changing fire risk consistent with predictions from natural resource economic theory. A bio-economic simulation shows that landowner adaptation to the changing fire regime since 2001 has reduced standing volume in non-federal western forests by roughly 7 to 10 percent. We quantify two of the externalities that landowner adaptation created on observed wildfires from 2020 and 2024. Holding observed fires fixed, adaptation's impact on tree volume, surface fuels and crown fuels raised $PM_{2.5}$ emissions which resulted in damages from the 2020 fire season of $63 to $72 million, which outweighed the lower CO_2 emissions of $28 to $58 million.

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[*]Corresponding author: whiteja6@oregonstate.edu.
[†]Formerly Forest Economics and Policy, USDA Forest Service, Southern Research Station.

[1]Department of Applied Economics, Oregon State University.
[2]CTrees.

1 Introduction

Large wildfires have become more frequent and severe across the western United States in recent decades, largely driven by anthropogenic climate change and increased fuel loads (Abatzoglou and Williams, 2016). The social costs of this trend are large and growing, with nearly 50 million homes now in the wildland-urban interface (Burke et al., 2021). Recent estimates place the total annual economic cost of wildfire in the United States in the hundreds of billions of dollars (U.S. Congress Joint Economic Committee, 2023), and the mortality costs of wildfire smoke alone are projected to exceed the climate-driven damages from all other causes combined (Qiu et al., 2025). Smoke output from wildfires depends on many factors, including the management choices of forest landowners that determines the density and structure of forests, as denser and more homogeneous stands have more fuel and burn more severely (Agee and Skinner, 2005; Zald and Dunn, 2018).

Long-standing predictions in natural resource economic theory hold that fire risk should shape forest management. Reed (1984) shows that an increase in wildfire risk acts like an increase in the discount rate, pushing managers to harvest sooner and lowering land values. The intuition is that unharvested volume is exposed to future fire, and higher risk lowers the expected value of letting forests grow because landowners' probability of being able to harvest unburned future forests is lower. When other choice variables are introduced to the Reed framework, Amacher et al. (2005) predict that landowners facing elevated fire risk should also plant at lower densities and invest in fuel reduction, while Patto and Rosa (2022) show that more frequent fire can move landowners toward commercial thinning earlier in the rotation. Because denser, higher-volume stands burn more severely (Agee and Skinner, 2005), removing volume can both lower the damage a fire would do to the stand and allow the landowner to generate revenue. While recent work shows that fire risk is capitalized into timberland prices (Wang and Lewis, 2024), we are aware of no direct empirical evidence on how landowners adapt harvest and replanting decisions in response to changing wildfire. How private forest landowners adapt to a changing fire regime influences future fire risk

alongside their effects on smoke output, forest composition, timber supply, and the many ecosystem services that forests provide.

This paper estimates how changing wildfire risk induces adaptation from private forest management in the U.S. Pacific states along two primary margins. The first is the choice among leaving a stand to grow, partial cutting, and clear-cutting. The second is the density at which landowners establish new stands, which Amacher et al. (2005) predict will fall as fire risk rises. Landowners act on their perceptions of fire risk, which we do not observe. We therefore measure risk with two signals that are potentially salient to landowners. The first is recent fire activity near the stand. The second is short-run weather shocks to fire-season vapor pressure deficit (VPD), a measure of hot and dry weather conditions that drives fire activity (Abatzoglou and Williams, 2016; Zhuang et al., 2021) and reduces forest growth (Restaino et al., 2016). Because fire is stochastic, realized fires are a noisy measure of the underlying hazard, and VPD lets us capture weather shocks that elevate risk in places where fire has not yet arrived. Non-federal owners manage primarily for timber revenue, and the U.S. Pacific coast region's spatial and temporal variation in fire activity and fire-related weather conditions provides the setting needed to identify management responses.

We combine management data from the U.S. Forest Service Forest Inventory and Analysis (FIA) program with wildfire perimeters from the Monitoring Trends in Burn Severity (MTBS) database and climate data from the Parameter-elevation Regressions on Independent Slopes Model (PRISM). We measure fire activity around FIA plots using confidential data with true plot locations. Publicly available plot coordinates are shifted to protect landowner privacy, which would introduce measurement error into any measure of nearby fire activity. Using true plot locations, we construct measures of fire activity around each plot, and we relate these measures to the annual harvest choice with a multinomial logit model that uses a correlated random effects specification to control for time-invariant forest stand heterogeneity (Train, 2009; Mundlak, 1978). We relate the same fire and weather measures to planting density in a model that controls for unobserved heterogeneity through

county and planting-period fixed effects.

We find novel evidence that western U.S. landowners adapt through harvest, as both increased fire activity and hotter, drier fire seasons raise the likelihood of tree volume removal through both partial cutting and clear-cutting. This pattern is consistent with Reed (1984), who predicts landowners respond to a higher perceived wildfire arrival rate by removing volume that would otherwise remain exposed to fire. We also find adaptation through planting density, which falls as recent nearby fire activity rises, consistent with predictions from Amacher et al. (2005). This response is driven by changes in fire activity and fire-related weather outcomes over time near a stand.

Taken together, the forest management adaptations we estimate result in the removal of forest volume from the landscape. We use our estimates in a bio-economic simulation that defines a counterfactual by holding fire and weather at their levels at the start of the century, and show that the changing fire and weather regime since 2001 has reduced standing volume across non-federal western forests by roughly 7 to 10 percent. This landowner adaptation carries both social benefits and social costs. Less standing volume means fewer crown fuels, but harvests also leave slash that adds surface fuels. In order to translate our main results to changes in wildfire-related outcomes, we use the FIA to estimate how surface fuels change in response to harvest. Holding the extent and severity of the 2020 fires fixed, we find that adaptation raised $PM_{2.5}$ emissions from those fires which led to additional smoke damages of \$63 to \$72 million. In contrast, we find that adaptation's effect on biomass led to lower CO_2 emissions and corresponding climate benefits of \$28 to \$58 million. Less volume also means less carbon stored on the landscape, an increase in short-run timber supply, and changes in wildlife habitat that can benefit species that prefer early-seral forest and damage species that prefer older forests. Forest management adaptations to wildfire risk therefore generate a range of positive and negative externalities.

This paper makes several contributions. First, it provides the first large-scale empirical evidence that private forest landowners adapt harvest and replanting management choices

to rising fire risk, broadly in the direction that natural resource economic theory predicts. Reed (1984) shows that a higher wildfire risk acts like an increase in the discount rate, pushing landowners to harvest sooner. Patto and Rosa (2022) show that increasing risk moves landowners toward commercial thinning earlier in the rotation, and Amacher et al. (2005) show that managers may respond to higher risk by lowering planting densities. We test these theoretical and numerical predictions directly with observed management decisions. Because forest management alters the fuels that drive future fire, documenting how landowners adapt is a first step toward understanding how private responses will shape the future fire regimes.

Second, our work complements recent evidence that fire risk is capitalized into timberland prices (Wang and Lewis, 2024) by identifying behavioral channels that accompany that capitalization, showing that the same fire activity that lowers land values also reshapes management decisions that alter landscapes and yield positive and negative externalities. This paper also extends a growing body of empirical work on forest adaptation to climate change. Prior empirical work on forest adaptation has examined how landowners adapt to climate and weather variability by altering the tree species chosen for replanting (Hashida and Lewis, 2019; Johnson and Lewis, 2024). Additionally, Hashida et al. (2025) finds that forest landowners adapt to wildfire risk by increasing prescribed burning acreage in the southeastern U.S. This paper extends that work to estimate how harvest and replanting decisions are used to adapt private forests to wildfire risk in the western United States where prescribed burning is uncommon on private land.

Third, a large literature projects the future costs of wildfire and smoke under climate change, with recent estimates suggesting smoke may become among the costliest consequences of global warming (Qiu et al., 2025). These projections examine how climate drives fire and fire drives smoke, but hold forest management fixed. The landowner adaptation we document reshapes the fuels that determine fire behavior and smoke. Characterizing that behavioral response is a necessary input to understanding how damages from fire will change over time. We connect our estimated forest management adaptations to wildfire

emissions in two steps. First, we build a bio-economic simulation that tracks how standing volume, aboveground biomass, and surface fuels change across the broader landscape, using difference-in-differences estimates of how clearcuts and partial cuts change six classes of surface fuels. Second, we link these simulated fuel changes to the First Order Fire Effects Model (FOFEM), a model developed by the U.S. Forest Service and widely used by fire managers to predict fuel consumption and smoke emissions from wildfires (Reinhardt et al., 1997; Lutes, 2020), allowing us estimate how adaptation changed $PM_{2.5}$ and CO_2 emissions from recent fire seasons. We find that adaptation leads to higher $PM_{2.5}$ emissions but lower CO_2 emissions, leading to net damages from the single 2020 fire season of \$14 to \$34 million.

Finally, our results speak to the design of fuel-treatment policy. Fuel-treatment programs often subsidize the removal of volume from private forests. We find that landowners already remove volume in response to rising fire risk, without any policy incentive. Subsidizing activities landowners would undertake on their own has additionality problems (Mason and Plantinga, 2013) and may point toward targeting cost-sharing at activities that generate no revenue for the landowner (Amacher et al., 2005). Our results point to one such activity, known as site preparation, which removes logging slash that increase surface fuels after a clearcut but is costly to landowners. Site preparation is relatively uncommon, occurring on roughly 30 percent of recently clearcut plots in our sample. In our simulation, universal site preparation following every clearcut would have reduced $PM_{2.5}$ emissions from the 2020 fires by 687 metric tons, avoiding roughly \$37 million in smoke damages, and would have turned the net emissions damages of adaptation into a net benefit, before accounting for the private cost of site preparation. Whether such programs are the most cost-effective way to reduce fire risk depends on how much each activity lowers risk, which our results do not measure. Evidence from public lands suggests fuel treatments can reduce fire damages (Strabo et al., 2026) and lower suppression costs (Strabo and Reimer, 2026), though whether private fuel reduction yields comparable returns is an open question. Because fire crosses ownership boundaries, landowner adaptation can lower risk for neighboring landowners (Lauer et al.,

2017), lowering future fire risk and smoke for the broader public.

The remainder of the paper is organized as follows. Section 2 describes forestry and wildfire in the western United States. Section 3 presents the theoretical framework, and Section 4 outlines our empirical strategy. Section 5 describes the data, and Section 6 presents our estimates. Section 7 simulates the effect of the changing fire regime on forest volumes, biomass, and fuels, and quantifies the resulting changes in wildfire emissions. Section 8 discusses the implications and concludes.

2 Study Context

This paper studies private forest management decisions across Washington, Oregon, and California over the period 2001 to 2024. This setting is chosen to provide substantial variation in wildfire activity and in the climate conditions associated with fire risk, while holding fixed a setting in which landowners manage primarily for timber revenue. The region is one of the largest national producers of timber products in the country, owing to a mild and wet climate west of the Cascade Mountains that is well suited to growing commercially valuable species such as Douglas-fir. Figure 1 shows the study region and the division between the west side and the interior that we refer to throughout.

Forest land in the region is an ownership mosaic of public and private land. Public ownership is predominantly federal land managed by the USDA Forest Service as National Forests along with the Bureau of Land Management and is intermixed with private and state timberland. Roughly 45 percent of the forested landscape in our study region is held in non-federal ownership (Harris et al., 2025),[3] and non-federal timberland is concentrated on the west side and along the productive areas of the interior, as shown in figure 5. Although non-federal owners hold less than half of the forested landscape, they account for the majority

[3]Computed from Harris et al. (2025) as the share of forest land in California, Oregon, and Washington in non-federal ownership: 27.4 million acres of family, corporate, tribal, state, and local forest out of 60.6 million acres of forest land whose ownership is determined. A further 1.8 million acres of forest are of undetermined ownership and are not counted. Ownership is measured as of roughly 2022.

of the timber harvests in the region. Following the passage of the federal Northwest Forest Plan in the early 1990s, harvest operations on federal lands have been greatly reduced, while harvests on private lands have remained relatively stable. Private and state timberland owners are primarily managing their land for timber production and most closely represent the profit maximizing behavior standard economic theory describes. Federal ownership has much different management objectives, such as providing recreation and habitat preservation. For this reason, we restrict our analysis to the actively managed private and state timberland.

The west side and interior areas of our study region differ substantially in their climate, their forest types, and their forest management strategies. The climate on the west side is much cooler and wetter, supporting more rapid growth and more productive land. Outside of the Western Cascades and Coast Range, in what we refer to throughout as the east side or interior, the climate is far drier and hotter. Douglas-fir is prevalent on the west side, along with western hemlock, Sitka spruce, and coastal redwood, these being among the most productive and commercially valuable forest types in the country. Ponderosa pine, lodgepole pine, and less productive mixed conifer stands are more common in the interior.

These climate and species differences shape which management strategies are economically viable in each region, though the set of harvest choices available to a landowner is the same throughout. A manager decides both when to harvest and what type of harvest to conduct. A clear cut harvest removes essentially all merchantable timber off the land, while a partial cut, also known as a commercial thin, removes some merchantable volume but leaves a residual stand behind to continue to grow. Because a clear cut leaves no residual trees to provide seed, and because reforestation following harvest is required by state forest practices rules throughout the region, it is standard to replant following a clear cut, where the landowner will determine how densely to replant. This replanting can incur substantial costs for the landowner. In part for this reason, clear cutting and replanting is a less common strategy on the east side, where productivity is significantly lower.[4] Douglas-fir is also

[4]Using data from Wang and Lewis (2024), average timberland values in Oregon and Washington west of the Cascades are \$5,320 per acre (n = 3,965), compared with \$3,207 per acre elsewhere in the region

shade intolerant, so completely clear cutting the stand and allowing the young trees ample sunlight is a beneficial strategy. Stands more common on the east side, such as ponderosa pine, are much more shade tolerant, and their less dense canopies allow more sunlight to reach the forest floor, so a landowner can remove some of the trees and let those trees seed additional trees in the understory, a process known as natural regeneration. Figure 2 shows the resulting pattern. Clearcut harvests are concentrated on the west side, while partial cuts are more common throughout the interior. Qualitatively, however, managers in both regions are doing very similar things: both a partial cut and a clear cut are ways for a landowner to remove and sell vegetation off of their land.

Landowners also have practices available that generate no immediate revenue at all. A pre-commercial thin cuts young, unmerchantable trees to promote growth on the remaining stand. Another practice is prescribed burning, which removes surface fuels and can lower the severity of a future burn (Agee and Skinner, 2005). It is common on private forest land in the Southeast (Melvin, 2022), where studies of private fire adaptation have accordingly focused on that margin (Hashida et al., 2025), but it remains rare on private timberland in the western United States, owing to liability concerns, drier conditions, and landowner risk perceptions (Miller et al., 2020). Because prescribed burning is rare on western private timberland, we do not examine it in this analysis, and we focus instead on the harvest and planting density margins.

These management activities all change the volume of standing timber on a stand. Standard reforestation practice in the region has emphasized dense, regularly spaced planting, but higher planting density can elevate the risk of future mortality from fire and drought (North et al., 2019). More generally, treatments that reduce stand density and fuels are associated with lower subsequent fire severity (Jain et al., 2020; Prichard et al., 2021). Through harvest decisions and planting density, a landowner controls how much volume, and therefore how much fuel, is available during a future fire. Privately optimal management should therefore

(n = 5,180). The substantially lower value of interior timberland limits the returns available to justify the up-front cost of establishing a planted stand.

respond to changes in fire risk.

Fire risk in the region has not been constant. Partly driven by the hot and dry climate on the east side, the eastern interior is a far more fire active region than the Western Cascades and Coast Range. Figure 3(a) plots the share of regional non-federal timberland burned each year over the 1984 to 2024 period for which perimeter data are available, separately for the west side and the interior. Fire activity is higher in the interior throughout this time period, and in both regions the level rises sharply after roughly 2014. Fire on the west side remains comparatively rare and is concentrated in a small number of spike years, most notably 2020.

The region has also seen increases in drought and heat stress. A widely used measure of these conditions is vapor pressure deficit (VPD), which combines temperature and humidity into a single measure of atmospheric aridity, where higher values indicate hotter and drier air.[5] VPD is closely linked to forest productivity, with higher VPD associated with reduced growth in Douglas-fir and in vegetation more broadly (Restaino et al., 2016; Yuan et al., 2019). It is also strongly linked to wildfire, and increases in VPD have been identified as the dominant driver of the recent growth in summer forest fire area across the western United States (Abatzoglou and Williams, 2016; Williams et al., 2019). VPD has risen across the region and is projected to continue rising under climate change (Ficklin and Novick, 2017). Figure 3(b) plots mean fire-season VPD on timberland by region, where fire season is defined as June through September. VPD is substantially higher in the interior than on the west side in every year, mirroring the regional difference in fire activity in Figure 3(a), and it has risen in the interior over the sample period. On the west side the upward trend is smaller, but there is considerable year to year variation in VPD. Together, these figures show substantial variation in fire activity and in fire-related climate conditions, both across the region and over time. The following section develops the theoretical framework linking fire risk to management decisions.

[5] VPD is the difference between the amount of moisture the air can hold at saturation and the amount of moisture it currently holds.

3 Theoretical Framework of Forest Management under Wildfire Risk

This section develops a simple model of forest management under wildfire risk. We build on Reed (1984), who extends the Faustmann rotation problem to catastrophic fire risk, and on Hashida and Lewis (2019), who write forest management as a discrete choice among harvest types. The model generates testable hypotheses about how forest management responds to rising wildfire risk.

3.1 The landowner's management choice

Consider a risk-neutral landowner who manages an existing stand to maximize its expected present value. The stand is described by its age a, and it has merchantable volume $v(a, C)$, where C denotes local climate conditions. Volume grows with age, $v_a > 0$, and climate shifts the growth function, so hotter and drier conditions reduce growth at a given age. The landowner faces a constant per-unit stumpage price P and discounts the future at rate γ, with discount factor $\delta = 1/(1+\gamma)$.

Each period, the landowner chooses one of three actions, indexed by j, using the harvest types described in Section 2. The landowner can clearcut ($j = C$), removing all merchantable volume and replanting the stand at density D at a regeneration cost $R(D)$. The landowner can partial cut ($j = P$), removing a portion of the volume and leaving a residual stand. Finally, the landowner can choose not to harvest ($j = 0$), leaving the stand to grow another period. Net revenue from a harvest of type j is

$$\pi^j(a, C) \;=\; P\, v^j(a, C) \;-\; c^j, \qquad j \in \{P, C\}, \tag{1}$$

where v^j is the volume removed under harvest type j, with $v^C(a, C) > v^P(a, C)$, and c^j denotes harvesting costs (Hashida and Lewis, 2019).

Following Reed (1984), fire arrives as a Poisson process with arrival rate λ. Fire is catastrophic, destroying the standing stock with no salvage value, so a stand that burns leaves the landowner holding bare land worth V^{bare}. The landowner chooses the action that maximizes the value of the land,

$$V_t(a, C, \lambda) \;=\; \max \begin{cases} \pi^C(a, C) - R(D) \;+\; \delta\, \mathbb{E}\big[V_{t+1}^{bare}(1, C, \lambda)\big] & \text{clearcut} \\ \pi^P(a, C) \;+\; \delta\, \mathbb{E}\big[V_{t+1}(a' + 1, C, \lambda)\big] & \text{partial cut} \\ \delta\, \mathbb{E}\big[V_{t+1}(a + 1, C, \lambda)\big] & \text{no harvest,} \end{cases} \tag{2}$$

Equation (2) says that a harvest gives the landowner a one-time net revenue today, received with certainty, along with a land value next period that depends on the trees the landowner leaves standing. If the landowner clearcuts, she removes all merchantable volume, receives $\pi^C(a, C)$, and pays the regeneration cost $R(D)$ to replant the stand, so all trees are of age 1 in period $t + 1$. If the landowner partial cuts, she removes some of the volume, receives the smaller net revenue $\pi^P(a, C)$, and the trees left standing continue to grow, where a' is the age of those trees at the time of harvest. If the landowner chooses not to harvest, she receives no revenue and the stand grows another period.

The future value of the stand in period $t+1$ always depends on the wildfire arrival rate λ, no matter what harvest choice they make in period t. However, the harvest choice in period t has an important effect on the landowner's short-run exposure to financial loss from wildfire. If a wildfire arrives in period $t+1$ the landowner faces the largest financial losses if they chose not to harvest in period t - $V_{t+1}^{bare}(1, C, \lambda) - V_{t+1}(a, C, \lambda)$ - because all of their commercial stock will be destroyed. If a wildfire arrives in period $t+1$ the landowner faces essentially zero financial losses if they chose to clearcut harvest in period t - $V_{t+1}^{bare}(1, C, \lambda) - V_{t+1}^{bare}(1, C, \lambda)$- since they have already removed all commercial volume in period t. If a wildfire arrives in period $t + 1$ the landowner faces a moderate financial loss if they chose to partial cut harvest in period t - $V_{t+1}^{bare}(1, C, \lambda) - V_{t+1}(a' + 1, C, \lambda)$- because they left some commercial

stock following the harvest in period t that will be destroyed by the fire. In this simplified model, we assume the wildfire arrival rate is exogenous to the landowner's harvest choice. The larger the wildfire arrival rate, the more exposed the stand is to short-run financial loss from choosing not to harvest.

3.2 Perceived wildfire risk

The arrival rate λ is the probability that fire arrives on the stand in a given period and destroys the standing stock of trees. Following Reed (1984), we assume this loss is catastrophic, with no salvage value recovered from the burned timber.

We treat λ as the landowner's perception of fire risk rather than an objective hazard rate. Landowners form this perception from what they observe. We write

$$\lambda_{it} \;=\; \lambda(\mathit{Fire}_{it},\, C_{it})\,, \qquad \lambda_{\mathit{Fire}} > 0, \quad \lambda_C > 0, \tag{3}$$

where Fire_{it} is recent fire activity near the stand and C_{it} measures hot and dry weather conditions correlated with fire risk, proxied by VPD. Nearby fire is a salient signal of an active fire regime, and because fire spreads across ownership boundaries, fire on adjacent land raises the risk to the stand itself (Lauer et al., 2017). As described in Section 2, hot and dry conditions drive fire activity, so a landowner can perceive elevated risk even where fire has not yet arrived. Equation (3) follows Wang and Lewis (2024), who model wildfire expectations in timberland markets as a function of observed fire and climate.

Climate therefore enters the landowner's problem in two ways. First, climate shifts perceived risk through equation (3). Second, climate shifts growth through $v(a, C)$. Both margins have the same result, since lower growth incentivizes the landowner to harvest sooner. We return to this in Section 4.

We treat expectations as reactive rather than anticipatory. Landowners update λ from fire and climate they have observed, as opposed to forecasts of future climate and wildfire

conditions. This assumption follows Massetti and Mendelsohn (2018), who argue that climate adaptation is likely to be reactive, and is consistent with evidence that forest landowners in the Pacific Northwest do not account for anticipated future climate in their management.[6]

3.3 Management regimes

Equation (2) is written for a landowner choosing among all three actions in every period. In practice, a stand's management regime often determines which harvest type is available, reducing the landowner's decision to whether to harvest as opposed to what harvest choice to make. We describe three common management regimes in pacific northwest forestry below, along with what theory predicts for each as perceived wildfire risk rises.

3.3.1 Clearcut regimes

Consider a landowner who manages a uniformly aged stand and removes it in one harvest. The stand is planted, left to grow, and clearcut at the age that maximizes net revenue, at which point this even-aged rotation cycle begins again. The landowner never makes a partial cut, so the choice set reduces to $j \in \{0, C\}$ and the decision left for the landowner is when to clearcut. Without fire risk, this is the classic forest rotation problem outlined by Faustmann (1849), and equation (2) reduces to the setting in Reed (1984). It is the most common management regime on productive Douglas-fir forests in western Oregon and Washington, where high growth rates and land values justify the cost of replanting after a clearcut. Reed (1984) shows that a higher wildfire arrival rate acts like an increase in the discount rate applied to future timber rents, which shortens the optimal rotation and raises the probability of a clearcut in any given period.

[6] Forest managers in the region are trained to value forestland using a static expectations Faustmann formula (Hashida and Lewis, 2019).

3.3.2 Partial cut regimes

Consider instead a landowner who never harvests the full stand. Volume is taken through repeated harvests, each of which leaves trees standing to grow and to regenerate the stand. The landowner never makes a clearcut, so the choice set reduces to $j \in \{0, P\}$ and the decision left for the landowner is when to make the next partial cut. We argue that the effect of fire risk on harvesting decisions in this regime is qualitatively the same as outline in Reed (1984). The only difference is what the landowner holds after a harvest, a stand of remaining trees rather than bare land, so a higher wildfire arrival rate again raises the probability of a harvest in any given period. This regime is common in the ponderosa pine forests of eastern Washington and Oregon, where, as described in Section 2, natural regeneration is feasible and lower land values limit the returns to replanting a clearcut stand.

3.3.3 Mixed regimes

Finally, consider a landowner who chooses among both harvest types. The stand is partial cut one or more times as it grows and then clearcut at the end of a rotation that runs longer than in the clearcut only regime, so the full choice set $j \in \{0, P, C\}$ applies and the landowner decides both when to harvest and which harvest to make. Landowners choose longer rotations to grow larger, higher-quality timber for niche markets, and the intermediate partial cut harvests generate revenue for the landowner prior to the clearcut. This regime appears on some west side forests in California, Oregon, and Washington, and is common in the mixed conifer forests of the Sierra Nevada mountain range. Figure 2 depicts where certain management regimes are most common. Patto and Rosa (2022) extend the Reed model to this setting, where the timing of partial cuts and the final clearcut harvest are chosen jointly. Patto and Rosa (2022) find that a higher wildfire arrival rate leads landowners to partial cut earlier in the rotation while the effect on the rotation age is ambiguous, and can go in either direction depending on how the partial cut alters the growth and fire risk of the remaining stand.

3.3.4 Regime choice

We do not observe which regime landowners choose. Instead, we observe the management actions consistent with these management regimes. Landowners can also switch management regimes, which is also unobserved in our setting. Our empirical models control for the regional differences that shape which management regimes are common, as described in Section 4.

3.4 Other margins of adaptation

Harvest timing is not the only margin on which landowners can respond to fire risk. Amacher et al. (2005) extend the Reed framework to examine the effect of fire risk on planting density and pre-commercial thinning, two additional management actions that are common in our setting, particularly under clearcut regimes.

Planting density is a decision the landowner makes following a clearcut, paying the regeneration cost $R(D)$ today in exchange for volume at the end of the next rotation. That volume is realized only if fire does not arrive first, so a higher arrival rate lowers the expected return to the marginal seedling while leaving its cost unchanged. Amacher et al. (2005) predict that landowners facing higher risk establish stands at lower planting densities.

Pre-commercial thinning removes young trees before they are large enough to sell on the market. Unlike the harvest decisions in equation (2), pre-commercial thinning generates no revenue and is a pure cost to the landowner. Pre-commercial thinning is typically done to increase growth on the remaining stand and lower future fire risk. Amacher et al. (2005) model thinning as protective effort, which raises the salvage value of the stand if fire arrives. They predict that landowners will undertake more pre-commercial thinning as wildfire risk rises. Their application is to the southeastern United States, where forest types, ownership, and fire regimes differ from our setting. We report results for pre-commercial thinning in Appendix C.

3.5 Testable hypotheses

The regimes above give predictions that differ by the choice set a landowner faces. Under a clearcut regime, a higher wildfire arrival rate raises the likelihood of clearcutting. Under a partial cut regime, a higher arrival rate raises the likelihood of partial cutting. Under a regime that uses both harvest types, a higher arrival rate raises the likelihood of partial cutting, with an ambiguous effect on clearcutting. Across all regimes, a higher arrival rate lowers the density at which landowners establish new stands, and raises the likelihood of pre-commercial thinning.

4 Empirical Strategy

In this section, we link the predictions from Section 3 to the data by estimating two models, one for each adaptation margin in our theory. First, we estimate a harvest choice model using a discrete-choice econometric framework. This tests how landowners' harvest decisions respond to changes in fire activity and fire-related weather conditions. Second, we estimate a planting density model using a linear econometric framework. This tests whether fire and weather affect the density at which landowners plant new stands. Section 5 describes how we measure each term in these models.

Our theory is written in terms of λ, the perceived fire arrival rate. Landowner perceptions of λ are unobserved, and it is unknown how landowners form these expectations. We proxy for λ with two observed signals: fire activity near the stand and fire-season vapor pressure deficit (VPD). Fire is a stochastic event, so any single fire is a noisy signal of the underlying hazard. Repeated fire activity, however, can reasonably signal to a landowner that risk is higher. It can be argued that it is irrational for landowners to read nearby fire as a signal of higher risk, since fire consumes fuel and may lower risk. However, fire repeatedly occurs around the plots in our sample. On average, stands in our harvest sample experienced 3.97 nearby fires over the preceding 15 years (Table 1). In the interior, the average is 6.24

(Figure 10). In places where fires are this common, increasing nearby fire activity may plausibly signal an increase in the probability that a wildfire will arrive on a stand.

The VPD measure of hot and dry weather enters as a second signal of risk. We do not assume landowners observe VPD directly. Instead, we argue that hot, dry fire seasons are salient, especially in relation to a landowner's own average climate. However, VPD can also affect management through changes in forest growth (Restaino et al., 2016), and decreasing yields would also push landowners to harvest sooner. Our preferred specifications measure VPD over a short, three-year window of recent fire seasons, which we argue is a signal of fire risk more so than long-run productivity.

Theory also implies that adaptation can only be observed when risk is changing. Where the fire regime is stable, risk should already be capitalized into management decisions. Differences in fires from year-to-year in nearby areas could reflect randomness, not changes in the underlying hazard. In stable risk environments, we would expect to find little management response to fire, even in fire-prone areas. It is the change in fire risk that creates the variation needed to study adaptation. As Section 2 shows, the western United States has seen substantial change in fire risk over our study period.

4.1 Harvest Choice

We model harvest as an annual discrete choice. Each year, the landowner managing forest stand i chooses among three actions: no harvest ($j = 0$), partial cut harvest ($j = P$), or clearcut harvest ($j = C$). Following Hashida and Lewis (2019), we pair the value functions in Section 3 with a random utility model. The landowner chooses the action with the highest value,

$$U_{ijt} \;=\; V_{ijt} \;+\; \varepsilon_{ijt}, \tag{4}$$

where V_{ijt} is the part of the value of action j that we observe, and ε_{ijt} is unobserved. We specify V_{ijt} as a reduced-form linear index of the drivers of value according to our economic

theory. Landowner-specific factors, such as expectations, management skill, and objectives, enter the unobserved error. If ε_{ijt} is i.i.d. Type I extreme value, the model is a multinomial logit (Train, 2009).

Normalizing the value of choosing no harvest, $V_{i0t} = 0$, the observed value of each harvest action $j \in \{P, C\}$ is

$$V_{ijt} = \alpha_j + \beta_j \, Fire_{it} + \delta_j \, VPD_{it} + \gamma_j' \, Age_{it} + \theta_j \, West_i + \tau_{jt} + u_{ij} \tag{5}$$

where $Fire_{it}$ is a measure of wildfire activity near stand i in the years before t, and is our proxy for perceived risk, λ_{it}. VPD_{it} is recent fire-season VPD at the stand. It proxies for perceived wildfire risk, as well as for climate conditions that shape growth, $v(a, C)$. Age_{it} controls flexibly for stand age, which captures where the stand sits on its growth curve. $West_i$ indicates stands west of the Cascades (Figure 1), where forest types, harvest systems, and fire regimes differ from the interior. τ_{jt} are action-specific period fixed effects that absorb shocks common to all stands in a time period, such as timber prices, demand, interest rates, and trade policy. Finally, u_{ij} is an action-specific, time-invariant stand effect. It captures stable features of the stand and its owner that we do not observe, including baseline fire risk and site productivity. Section 5 describes how we measure each variable.

The fire coefficient, β_j, measures how nearby fire shifts the value of action j relative to not harvesting. In our theory, fire affects this value through perceived risk. More fire activity raises a landowners perceived arrival rate, and a higher arrival rate raises the value of cutting relative to waiting, in line with our predictions in section 3

The stand effect, u_{ij}, may be correlated with fire exposure and climate, since stands in fire-prone areas may differ in ways that also affect harvest. However, we cannot estimate u_{ij} as a separate fixed effect for each stand due to the incidental parameters problem. In a nonlinear model, estimating a parameter for every unit leads to inconsistent estimates of the other coefficients (Neyman and Scott, 1948; Lancaster, 2000). We instead follow a correlated

random effects (CRE) approach (Mundlak, 1978; Wooldridge, 2019) and model the stand level effect as

$$u_{ij} \;=\; \xi_j \,\overline{Fire}_i \;+\; \psi_j \,\overline{VPD}_i \;+\; \eta_{ij}, \tag{6}$$

where overbars denote group means, the average of each variable over the years we observe stand i. These means absorb the part of u_{ij} that is correlated with a stand's typical fire exposure and weather outcomes. The stand means capture each stand's baseline risk, which should already be capitalized in management decisions. Conditional on these group means, β_j and δ_j are identified from changes in fire and VPD within a stand over time, the variation our theory says should drive adaptation. We assume the remaining term, η_{ij}, is independent of the regressors, and we treat it as part of the logit error. Substituting equation (6) into equation (5), the probability that stand i conducts action j in year t is

$$\Pr(y_{it} = j) \;=\; \frac{\exp\left(\tilde{V}_{ijt}\right)}{\sum_{k\in\{0,P,C\}} \exp\left(\tilde{V}_{ikt}\right)}, \qquad \tilde{V}_{i0t} = 0, \tag{7}$$

where $\tilde{V}_{ijt}$ is equation (5) with u_{ij} replaced by $\xi_j\overline{Fire}_i + \psi_j\overline{VPD}_i$.

The CRE approach is a modeling assumption, and it requires choosing which variables receive group means. We include group means for fire and VPD, but not for stand age. Stand age rises by one each year, so its group mean reflects when we first observe a stand and how long we observe it, not a stable trait of the stand.[7] We therefore include stand age without its group mean. We discuss our treatment of stand-level effects and sensitivity of our results in Section 6.3.

We estimate equation (7) via maximum likelihood. We weight each stand-year by stand area, so larger stands count more.[8] We cluster standard errors by FIA plot, since stands on the same plot share fire and climate measures. We use multi-year period fixed effects rather

[7] Consider two stands that are both 40 years old today. If we first observe one seven years ago and the other one year ago, their age means are 36.5 and 39.5. With these means in the model, the two stands would receive different age contributions to the value of cutting based only on when they entered our sample.

[8] Section 5.1 describes how we measure stand area. Most stands cover a full FIA plot, and our results are essentially robust without weights.

than annual fixed effects because partial cuts and clearcuts are rare and are not observed in every year of our sample.

Our identifying assumption is that, conditional on the stand-level group means, period effects, and age controls, within-stand changes in nearby fire and fire-season VPD are unrelated to other time-varying drivers of management activity. Two features of our setting support this assumption. First, all fire and weather measures are taken strictly before the management year, so we observe management responses to prior fire and weather. Second, nearby fire is largely outside a landowner's control. Ignitions come from lightning and human sources, and fire spread is driven by weather, terrain, mitigation effort, and fuels across a mosaic of ownerships (Figure 5).[9]

Our theory predicts that higher perceived risk raises the value of both harvest types relative to not harvesting. In terms of equation (5), this implies $\beta_P > 0$ and $\beta_C > 0$. If fire-season VPD also signals fire risk to landowners, the VPD coefficients δ_P and δ_C should be positive as well. Because slower growth also favors earlier harvest, positive VPD coefficients are consistent with both channels, and we do not separate them.

4.2 Planting Density

Theory also predicts that landowners facing higher risk should plant fewer trees (Amacher et al., 2005). We test this prediction on recently planted stands. Unlike harvest, planting is not a repeated decision within our sample window. Rotations in the western United States typically exceed 40 years, so we never observe the same stand planted twice. Because of this, we estimate our planting density model on a pooled cross-section of recently planted stands.

For stand i planted in year t, we estimate

$$\begin{aligned} \ln TPA_{it} \;=\; & \beta_0 \;+\; \beta_1\, Fire_{it} \;+\; \beta_2\, VPD_{it} \;+\; \beta_3\, Burn_i \;+\; \beta_4\, TSP_i \;+\; \beta_5\, DF_i \\ & +\; \phi'\, Site_i \;+\; \omega'\, X_i \;+\; \mu_{c(i)} \;+\; \tau_t \;+\; \varepsilon_{it}, \end{aligned} \tag{8}$$

[9] Wang and Lewis (2024) find that 94% of large fires within 15 km of private timberland parcels in the Pacific states burned on non-private timberland, with most burning on other land use types.

where ln TPA_{it} is the log of trees per acre of the dominant species, our measure of planting density. $Fire_{it}$ is a measure of wildfire activity near stand i in the years before planting and VPD_{it} is recent fire-season VPD at the stand, both of which proxy for λ_{it} $Burn_i$ equals one if the stand itself burned before it was planted, which captures replanting after a fire destroyed the previous stand. DF_i indicates the Douglas-fir forest type, the dominant planted type in our sample. Douglas-fir and other species have different planting practices, and DF_i controls for this difference. $Site_i$ is a set of site class bins that capture site productivity, and X_i contains topographic controls including elevation, slope, and aspect. $\mu_{c(i)}$ are county fixed effects, τ_t are planting-period fixed effects, and ε_{it} is an error term. Section 5 describes how we measure each variable.

TSP_i is time since planting, the number of years between planting and FIA measurement. Using the FIA data, we observe trees per acre at the FIA visit, not at the time of planting by the landowner. During this gap between actual planting and measured TPA, planted trees may undergo mortality and natural regeneration. Observed density at FIA measurement is therefore a proxy for the planting density the landowner chose, and time since planting absorbs systematic changes in density over this time period. The gap in our estimating sample is 3.9 years on average. In Appendix B.4 (Table 13, column 7), we show that our results hold when we restrict the sample to stands measured soon after planting.

County fixed effects absorb baseline differences in fire risk across space, along with local geography and markets that vary little within counties (Wang and Lewis, 2024). Planting-period fixed effects absorb shocks common to all stands planted in the same period, such as seedling costs and timber markets. We use period bins because some years in our sample we observe few recently planted stands, so annual effects are imprecisely estimated. The fire coefficient is therefore identified from differences in pre-planting fire activity among stands planted in the same period and county. Our identifying assumption is that, conditional on these fixed effects and controls, the timing and location of nearby fire is unrelated to other drivers of planting density. We estimate equation (8) by weighted least squares, weighting

each stand by its size, and we cluster standard errors by county.

Theory from Amacher et al. (2005) and in Section 3 predicts $\beta_1 < 0$. If fire-season VPD also signals risk, β_2 could be negative as well. The expected sign of β_2 is less clear than in the harvest model, however. Hot, dry conditions also raise seedling mortality, and landowners may plant more trees to offset expected losses. The next section describes our data and how we measure fire, climate, and management.

5 Data

5.1 Forest Management Data

We are interested in the effect of wildfire risk on subsequent forest management decisions. Our management data come from the USDA Forest Service Forest Inventory and Analysis (FIA) program, a national inventory of permanent forest plots across all ownerships in the United States. FIA field crews physically visit each plot and record a suite of information, including management activity, forest type, stand age, and volume. Remote sensing data, by contrast, can capture canopy change and disturbance, but cannot reliably distinguish a partial cut from a clearcut at scale, detect management activity under the canopy, or observe decisions such as planting density. For this reason, we argue that this study can only be completed using this detailed inventory data from the FIA. In the West, about 10% of plots are measured each year, so each plot is revisited about once every 10 years. Our study period begins in 2001, when the current inventory design began (Bechtold and Patterson, 2005).

Each FIA plot consists of four subplots spread over about one acre. Field crews divide each plot into forest "conditions," which are areas within the plot with similar forest types, stand sizes, or ownership. Most plots contain a single condition. The management actions, stand ages, and forest types we use are all recorded at the condition level, so the condition is our unit of analysis. A condition is the empirical measurement of the forest stand in our theory and empirical strategy, and we refer to conditions as stands throughout the paper.

We restrict our sample to private and state forest land, whose owners are more likely to have profit-maximizing objectives, which fits our economic theory of forest management under changing fire risk. Federal forests are managed for a wide range of objectives beyond revenue (U.S. Congress, 1960; Loomis, 2002), so we exclude federal land from our analysis.

For each stand, crews record whether a clearcut, partial cut, or pre-commercial thinning occurred since the last visit, along with the estimated year of each activity.[10] Section 2 describes these practices. Our main analysis focuses on the choice among clearcutting, partial cutting, and not harvesting, and we report results for pre-commercial thinning in Appendix C. FIA crews also record disturbances, including crown fire, ground fire, insects, and wind, with estimated disturbance years. We measure whether a stand burned from these crew records rather than from fire perimeters.[11] We also drop any stand with a harvest coded as salvage. When a fire damages a stand, the landowner may salvage the remaining timber. Because fire activity near a stand is higher when the stand itself burns, keeping these harvests would produce a positive relationship between fire and harvest that is largely mechanical. Dropping them lets us focus on management decisions that respond to changing risk expectations rather than to realized damage.

To estimate our harvest choice model, we build an annual panel of stands measured between 2001 and 2024. Although plots are measured about once a decade, the variables in our harvest model are observed annually. Crews record the estimated year of each management action. Additionally, stand age is measured at each visit, and because we observe any action that would reset it, we can infer age in the years between visits.[12] Fire and climate come from sources outside the FIA and are measured every year. Together, these give us an annual

[10] To define clearcuts vs. partial cuts, we use FIA definitions. A clearcut leaves residual trees of all sizes with less than 25 percent crown cover; these residual trees are typically low-value hardwoods. The FIA separates large and small partial cuts, which we combine into a single partial cut measure. For more information, see the FIA field manual for California, Oregon, and Washington: https://research.fs.usda.gov/pnw/understory/2026-pnw-fia-california-oregon-and-washington-field-manual.

[11] Fire severity and spread vary greatly within a fire perimeter, and some areas inside a perimeter do not burn. A crew visit is a better indicator of whether a stand itself burned. A stand is recorded as burned if crews find that fire occurred between the previous visit and the current one, about 10 years on average.

[12] For example, a stand aged 40 at its 2004 visit, with no clearcut before its 2014 visit, was 41 in 2005, 42 in 2006, and so on.

panel of stand-years, which makes up our harvest choice estimating sample.

We first observe a stand at its first FIA measurement. We require at least two measurements, since management actions are recorded relative to the previous visit. We follow stands forward across visits. If an FIA plot has two stands at one visit and three at the next, we follow all three forward from that point up until the plots third visit. A stand leaves the panel at its final observed FIA measurement, or when it is clearcut, whichever comes first. A recently clearcut stand has essentially no chance of being clearcut again within our window, since it would need decades to regrow. We also restrict the panel to stand-years with a stand age of at least 30, since partial cuts and clearcuts rarely occur in younger stands. We show robustness to this threshold in Appendix B.3 (Table 9, column 6). Partial cuts, in contrast, can recur over time, so stands remain in the panel after a partial cut.[13] The final harvest sample is an unbalanced panel containing 51,451 stand-years from 5,277 stands on 4,367 plots (Figure 6).

To estimate our planting density model, we use a separate sample of recently planted stands. We identify recently planted stands using FIA treatment codes, which identify planting the area to establish a manageable stand. [14] Crews record the estimated planting year. For each planted stand, we observe trees per acre (TPA) at the FIA visit, not at planting, as discussed in Section 4.2.

We define planting density as the TPA of the stand's dominant species, based on the forest type assigned by field crews. Our measure of TPA sums seedlings, saplings, and trees.[15] As expected for young plantations, seedlings and saplings account for 98% of TPA in our sample. The final planting sample contains 556 planted stands on 530 plots on private

[13] In practice, repeat partial cuts are rare. Only three stands with a partial cut record a second partial cut in a later year. Nineteen stands record both a partial cut and a clearcut in the same year; we treat these as clearcuts, which leaves 447 stand-years with a partial cut.

[14] Planting typically follows a disturbance or a clearcut harvest.

[15] Seedlings are less than 1.0 inch in diameter, saplings are 1.0–4.9 inches, and trees are 5.0 inches or larger. FIA crews tally stems on fixed-area subplots and microplots, and each stem carries an expansion factor that converts it to trees per acre. Each expansion factor treats a stem as if it were sampled across the full plot. When a stand covers only part of the plot, we divide its summed expansion factors by the stand's share of the plot.

and state land, planted between 2004 and 2023 (Figure 7).

To protect landowner privacy, public FIA plot coordinates are "fuzzed" by up to 1 mile (1.6 km) from their true location. Additionally, the coordinates of some private plots are also swapped with those of similar plots within the same county (Bechtold and Patterson, 2005; McRoberts et al., 2005). Because we link each plot to nearby fire, this fuzzing and swapping process could add measurement error to our fire variables. To avoid this, we construct our fire variables using confidential true plot coordinates obtained through a joint venture agreement with the U.S. Forest Service.

5.2 Wildfire Data

Given we are interested in the effect of wildfire on subsequent forest management decisions, we need to link our FIA plot data to fire information. We measure fire activity using the Monitoring Trends in Burn Severity (MTBS) database (Eidenshink et al., 2007). In the Western U.S., MTBS maps all large fires greater than 1,000 acres dating back to 1984. For each fire, we observe its final perimeter, ignition date, and whether it was a wildfire or prescribed fire. Using the MTBS data, we link each fire perimeter to our true FIA plot locations and construct a series of wildfire metrics.

Economic theory does not tell us how to directly measure λ, which is the likelihood that a stand catastrophically burns in a given year (Reed, 1984). More importantly, our theory does not tell us how landowners form expectations over λ. We therefore construct several measures of fire activity around each plot, each of which could plausibly shape those risk expectations. Each measure describes fire activity within some distance of the stand, over some time window ending before the management decision. We report the sensitivity of our main results to choices of the time and distance windows in Appendix B.1 (Tables 6 and 7).

Building off (Wang and Lewis, 2024) our first measure is the number of fires around a plot. We argue this is an appropriate measure of the theoretical arrival rate, as it counts how often fire arrives near the stand. Our second measure is the acres burned around a

plot, which captures how much of the surrounding landscape burns rather than how often fire occurs. Third, we measure the change in the number of fires over time, which captures whether local fire activity is rising.[16] Each of these measures capture a different signal of the fire regime. Figure 8 and Appendix A describe the construction of these fire measures in more detail.

5.3 Climate Data

We measure climate using the Parameter-elevation Regressions on Independent Slopes Model (PRISM) at a 4 km resolution (Daly et al., 2008). Our preferred climate measure is vapor pressure deficit (VPD). As discussed in Section 2, higher VPD is associated with reduced growth in Douglas-fir and in vegetation more broadly (Restaino et al., 2016; Yuan et al., 2019), and is a key driver of fire activity (Abatzoglou and Williams, 2016; Zhuang et al., 2021). VPD also varies more within counties than temperature or precipitation (Wang and Lewis, 2024), which helps identify our planting density model using county fixed effects.

VPD gives us a second signal of λ, one that does not depend on realized fires. Fire occurrence is driven by stochastic events such as lightning strikes and fire weather, so observed fires may be a noisy measure of the underlying hazard a landowner faces. For example, a stand may be in a hot, dry environment where fires have not yet arrived but where the underlying wildfire risk has increased. Because VPD drives fire activity, hot and dry fire seasons can signal a higher λ even when no fire has occurred. VPD may also affect management through decreased forest growth (Restaino et al., 2016). Our preferred specifications use average fire-season maximum VPD in the three years preceding a management decision, where we define the fire season as June through September.[17] A three-year window captures recent fire seasons that are likely salient to landowners. Our results are not sensitive to the

[16] For example, suppose a stand had one fire within 30 km between 2002 and 2011, and two fires between 2012 and 2021. The change measure for a decision in 2022 equals one, describing one more fire than in the previous decade. Positive values indicate that recent fire activity exceeds that of the prior period.

[17] We build our climate measures from public FIA plot coordinates rather than true coordinates. Both our fire and climate measures are linked to plot locations, not stand locations, so they vary over time but are the same for all stands on a plot in a given year.

measurement of VPD, and we report sensitivity checks in Appendix B.2 (Table 8).

5.4 Summary Statistics

Table 1 summarizes the harvest choice sample. Harvests are rare in any given year: 97.89% of stand-years see no harvest action made, 0.87% record a partial cut, and 1.24% record a clearcut. The two harvest types also occur in different places (Figure 2). Clearcuts are concentrated on the west side, while partial cuts are more common in the interior. Fire exposure differs just as sharply across regions (Figure 10). On the west side, 80.8% of stand-years had no fire within 30 km over the prior 15 years, and the mean is 0.38 fires. In the interior, the mean is 6.24 fires, and only 6.0% of stand-years had no fire activity, highlighting that multiple fires can occur around a stand. Fire-season VPD is also higher in the interior, averaging 25.95 hPa compared with 16.04 hPa on the west side, but it varies widely in both regions (Figure 11). These regional differences motivate the west side control in our harvest model. Conditional on dropping all stands less than 30 years of age, the average stand in the harvest sample is 74.7 years old.

Table 2 summarizes the planting density sample. Most planted stands are in Oregon and Washington, largely on the west side, with a smaller group in interior northern California (Figure 7). Douglas-fir is the dominant planted type, accounting for 401 of 556 stands, with a mean density of about 402 TPA. On average, stands saw more fires in the decade before planting than in the decade before that, with a mean change of 0.50 fires. Mean fire-season VPD is 18.77 hPa, lower than in the harvest sample (22.12 hPa). This fits the location of planted stands in cooler, more productive forests. Finally, 7.2% of stands were planted following a direct burn. We now present our estimates of how landowners respond to changes in wildfire risk.

6 Results

In this section, we present estimates of the effect of wildfire and fire related climate conditions on subsequent forest management decisions.

6.1 Effects of wildfire exposure and weather shocks on harvest choice

Table 3 examines the effect of fire and climate conditions on the harvest choice, specifically the decision of whether to clear cut, partial cut, or not harvest, where not harvesting is the base action. In column (1) we measure the effect of fire through a cumulative count of nearby fires, in column (2) as the sum of nearby acres burned, and in column (3) as the change in nearby fire counts over time.

In columns (1) and (2), we estimate a positive and statistically significant effect of fire on both the decision to partial cut and the decision to clear cut. In column (3), we estimate no effect of changing fire activity on the decision to partial cut, but a positive and significant effect on the decision to clear cut. We are unable to test directly why the level and trend specifications differ. One possibility is that the irreversibility of clear cutting makes it responsive both to the level of nearby fire activity and to whether that activity is rising, while we find no evidence that partial cutting responds to the trend.

To interpret the magnitudes of these effects, we first estimate the baseline probability of clear cutting and partial cutting according to our multinomial logit by predicting the probability of each action for every observation in the sample and taking the average. This gives a reference point against which to compare the marginal effects. Using the results from column (1), the average annual probability of a partial cut is 0.96 percent and the average annual probability of a clear cut is 1.33 percent. In column (1), an additional nearby fire increases the probability of a partial cut by 0.08 percentage points and the probability of a clear cut by 0.34 percentage points, translating to relative increases of 8.3 and 25.5 percent.

In column (2), an increase of 10,000 nearby acres burned raises the probability of a partial cut by 0.06 percentage points and the probability of a clear cut by 0.12 percentage points, with relative increases of 6.6 and 8.8 percent. In column (3), the marginal effect of the fire trend on the decision to clear cut is smaller in magnitude than the effect in column (1) but remains statistically significant.

Across all three columns, we estimate positive and statistically significant effects of fire-season VPD on subsequent partial cutting and clear cutting decisions. These effects are similar in magnitude between the partial cut and clear cut equations and stable across all three specifications. For example, in column (2), a one unit increase in fire-season VPD raises the probability of a partial cut by 0.14 percentage points and the probability of a clear cut by 0.18 percentage points, corresponding to relative increases of 14.3 and 13.6 percent.

The remaining controls behave as expected. In line with the discussion in Section 2, clear cutting is more common on the west side than the interior as shown by the large positively signed estimated parameter on the west side dummy in the clearcut equation. We also estimate a positive effect on the probabality of clear cutting for stands aged 50 to 69 relative to the omitted category of 30 to 49, and a negative and statistically significant effect for stands aged 90 and older. This is consistent with a selection story in which these older stands are old because they have already been chosen not to be clear cut. We find a less clear relationship between stand age and the likelihood of partial cutting.

Taken together, we estimate positive effects on the probability of both partial cutting and clear cutting in response to elevated fire activity and fire related climate conditions. This is consistent with a Reed (1984) style framework, in which managers respond to elevated disturbance risk by cutting sooner and removing volume from the landscape. Removing volume also removes fuel, so this response lowers a landowner's exposure to future loss (Agee and Skinner, 2005).

6.2 Effects of wildfire exposure and weather shocks on planting density

Table 4 reports weighted least squares estimates of log planting density on wildfire exposure for recently planted stands. In column (1), fire exposure is measured as the change in nearby fire counts over time.[18] We estimate a negative and statistically significant effect of fire on subsequent planting density, providing evidence that managers facing rising fire activity respond by planting fewer trees. Using the estimates from column (1), we find that a one-fire increase in nearby fire activity is associated with an 8.7 percent reduction in planting density,[19] or roughly 50 fewer trees per acre at the sample mean of 525. We find no such effect using the count of nearby fires in column (2) or nearby acreage burned in column (3). This contrast is consistent with how our planting density and harvest choice models are identified. Because the planting density model includes county rather than stand fixed effects, the level measures in columns (2) and (3) compare plantings across locations with different long-run fire regimes. Conversely, the change measure in column (1) nets out each plot's own recent fire history, much like the group means in the harvest choice model. It therefore captures increases in fire activity relative to the plot's baseline.

We do not estimate any significant effect of fire-season VPD on subsequent planting densities in any specification. Appendix B.4 (Table 13, column 6) reports similarly null effects using alternative measures of fire related climate. One explanation is that planting density commits the landowner for an entire rotation, making it less responsive to the three-year window over which we measure VPD than to the longer-run change in fire activity in column (1).

The post-burn planting dummy, which measures if a planting year was preceded by a direct burn in the previous / 10 years, is negative but not statistically significant. Time since planting, defined as the number of years between the planting year and the subsequent FIA

[18] We use a ten-year difference because a fifteen-year difference would require thirty years of pre-planting fire history and would exclude a substantial portion of the estimating sample.

[19] The percent change is computed as $100 \times (\exp(\hat{\beta}) - 1) = 100 \times (\exp(-0.0993) - 1) = -9.5$.

measurement, is positive and statistically significant across all specifications. This controls for growth and mortality that occurs between planting and measurement. To examine the sensitivity of our results to the gap between planting and FIA measurement, Appendix B.4 (Table 13, column 7) restricts the sample to conditions measured within three years of their planting year and finds similar effects. We estimate negative and statistically significant effects of lower land productivity relative to the omitted base category of the most productive site classes. The topographic controls are not statistically significant conditional on county fixed effects, which may be due to the small spatial setting of the planting density sample, leaving little within-county variation in elevation, slope, and aspect.

Taken together, we interpret these results as suggestive evidence that planting decisions respond to changes in fire activity over time. Lower planting density leaves less volume growing on the stand and less fuel available to a future fire, so this margin operates in the same direction as the harvest responses estimated above (North et al., 2019). It also implies lower future timber supply from these stands. We treat the evidence as suggestive because the planting density sample is considerably smaller than the harvest choice sample and many planted conditions have had limited fire exposure.

6.3 Robustness checks

We conduct a range of robustness checks reported in the Appendix, and summarize the main points here.

We re-estimate the harvest choice model with wildfire exposure measured over five- and ten-year windows (Appendix B.1, Table 6). Our results are largely insensitive to the choice of time window. The fifteen-year window specification yields the highest log pseudolikelihood for both measures, which supports our choice of using the longer horizon as our preferred measure. We also test whether the harvest response depends on distance to fire (Table 7). Restricting exposure to fires within 15 km yields a larger clearcut response per fire (0.59 versus 0.34 percentage points) but fits the data worse than the 0–30 km measure. When

we include both the 0–15 km and 16–30 km bands, we cannot reject equal coefficients in either equation (p-values from 0.17 to 0.65), and fit improves only marginally over the pooled measure, despite adding extra parameters. Our sample has limited power to separate fire effects by distance, particularly for clearcuts. However, we find no evidence that the response differs between near and more distant fires, which supports our preferred specification pooling fire exposure within 30 km.

We also re-estimate the harvest choice model with VPD measured over alternative time windows (Appendix B.2, Table 8). We replace the June–September fire-season window with a growing-season (April–July) window, a shortened fire-season (July–September) window, or an annual measure. We also average VPD over one and five years rather than three. The VPD effect stays positive and statistically significant in every specification. The effect is smallest for April–July VPD, which excludes most of the fire season, and grows with the length of the time window, consistent with landowners responding to persistent weather shocks. All six specifications produce similar log pseudolikelihoods, and how we measure weather shocks does not drive our results.

Table 9 in Appendix B.3 reports additional robustness checks for the harvest choice model. In column (1), we replace period fixed effects with annual fixed effects. The wildfire effects are robust to this change, but the VPD effects are no longer significant at the 5% level. Much of VPD's year-to-year variation is common across stands in the region. Year fixed effects absorb these common annual shocks, leaving only each stand's deviation from the regional shock to identify the VPD effect. As a result, the standard errors on the VPD effects rise, and the point estimates fall. However, each estimate lies within the 95% confidence interval of the estimates in Table 3. Our results are also robust to replacing the period fixed effects with state-by-period fixed effects (column 2), adding a wildfire–VPD interaction (column 4), adding an indicator for whether the stand burned before the management year (column 5), and estimating the model on stands of all ages (column 6).

We do not observe stumpage prices for most of our sample, and short-run price shocks

can influence harvesting decisions. The annual fixed effects in column (1) of Table 9 absorb price shocks common to all species in a given year, and the wildfire effects are robust to their inclusion.[20] Our period fixed effects absorb longer-run price movements common to all species, and column (3) relaxes this by interacting the period fixed effects with forest type groups (Douglas-fir, ponderosa pine, and all other types), allowing price movements to differ across forest types. The wildfire and VPD effects are robust to this change.

The multinomial logit assumes independence of irrelevant alternatives, which may fail if partial cutting and clearcutting share unobserved determinants, such as a landowner's general willingness to harvest. We test this assumption by estimating a nested logit that groups the two harvest actions in a single nest above the no-cut alternative (Appendix B.3). In our main specification (Table 10), the dissimilarity parameter is poorly identified and lies outside the range consistent with random utility maximization (McFadden, 1977; Börsch-Supan, 1990). The west-side dummy enters the two harvest equations with large coefficients of opposite sign, and one explanation is that it leaves little residual variation to identify correlation within the nest. Without the west-side dummy (Table 11), the dissimilarity parameter is 1.35 (standard error 0.38), we cannot reject $\lambda = 1$, which corresponds to the multinomial logit ($p = 0.36$) (Train, 2009), and the wildfire and VPD coefficients are similar to the multinomial logit estimates, providing evidence that the independence assumption is justified.

Our main specification follows Wooldridge (2019) in including group means of time-varying regressors, averaged over the years each stand is observed in our estimating sample. Wooldridge (2019) also recommends that CRE models on unbalanced panels include group means of time dummies. In a balanced panel, every stand would be observed in the same years, so these group means are identical across stands and drop out. In an unbalanced

[20] Stumpage prices in Oregon move together over time. Proprietary quarterly stumpage prices for several major species in Oregon, available to the authors, differ in levels but share common trends over the last 30 years. We do not include these prices as controls because they are available only for Oregon and for a subset of species, which would discard most of our sample. Because prices appear to vary mainly over time rather than across space, time fixed effects absorb most of their variation.

panel, stands are observed in different years, so the group means vary across stands and can be correlated with the stand effect. Our main specification does not include group means of the period fixed effects. For stands that are clearcut, the years we observe them, and thus their group means, depend in part on when the clearcut occurred, since clearcut stands drop out of our estimating sample. The group means of the period dummies largely record this timing, so including them can partly absorb the outcome. Consistent with this, adding them significantly raises the log pseudolikelihood (column 2 of Table 12). Column (3) instead computes the wildfire and VPD group means over a fixed 2001–2024 window for every stand, regardless of when it is observed. The partial cut responses to wildfire and VPD are robust to both alternatives. The clearcut response to wildfire is not. Its marginal effect falls and is not statistically significant in either specification. The clearcut response to VPD holds with period group means but not with the fixed-window group means. We therefore view the clearcut results as more sensitive to how the stand effect is modeled.

Table 13 in Appendix B.4 reports robustness checks for the planting density model. The negative effect of recent fire activity on planting density is robust to replacing the planting-period bins with planting-year fixed effects (column 3), replacing the Douglas-fir indicator with forest type fixed effects (column 4), and measuring the change in fire activity over 15 rather than 10 years (column 5), which restricts the sample to stands planted from 2014 onward. Because we infer planting density from the trees present when FIA crews measure the stand, seedling mortality between planting and measurement could bias our outcome, as discussed in Section 4.2. Restricting the sample to stands measured within three years of planting (column 7) yields similar estimates. Replacing county with state fixed effects (column 2) reduces the estimate to -0.048, significant at the 10 percent level, suggesting that county fixed effects absorb local differences in planting practices that are correlated with fire exposure. To examine whether planting density responds longer-run climate conditions as opposed to short-run weather shocls, we replace fire-season VPD with the annual maximum VPD averaged over the 15 years before planting (column 6). As with fire-season VPD, we

find no significant effect of VPD on planting density, and the fire effect is unchanged.

We also estimate how pre-commercial thinning responds to wildfire exposure and fire-season VPD (Appendix C, Table 14). Pre-commercial thinning removes small, unmerchantable trees and generates no revenue. However, pre-commercial thinning can reduce surface fuels and may lower a landowner's risk to fire (Agee and Skinner, 2005). Amacher et al. (2005) model pre-commercial thinning as a protective effort activity that lowers a landowner's expected losses from fire, and predict that landowners should increase pre-commercial thinning as fire risk rises. We estimate a correlated random effects logit of annual thinning with the same fire and VPD measures, group means, and controls as the harvest choice model in Table 3. Because most thinning occurs in young stands, we drop the harvest model's stand age restriction and control for stand age bins, giving 205 pre-commercial thins across 71,088 stand-years. Nearby wildfire has no statistically significant effect on thinning under either the fire count or acreage burned specification. This may be because most pre-commercial thins occur on west side plots, in particular northwestern Washington, an area with little fire exposure during our sample. However, fire-season VPD increases the likelihood of doing a pre-commercial thin. A one-hPa increase in VPD raises the annual probability of a pre-commercial thin by 0.054 percentage points, about 18 percent of the 0.29 percent baseline. Landowners thus appear to respond to drier fire seasons with protective thinning, consistent with predictions from Amacher et al. (2005).

We also examine whether fire exposure changes the age at which stands are harvested, a more direct test of the predictions made by Reed (1984) and Patto and Rosa (2022) that fire risk shortens clearcut rotation ages and leads to partial cutting earlier in the rotation (Appendix E). We regress stand age at the time of harvest on wildfire exposure and fire-season VPD with the same controls as the harvest choice model. Each additional fire within 15 km is associated with partial cutting about 2.1 years earlier (Table 17), but the estimate is smaller and not statistically significant at 30 km, and neither fire exposure nor VPD is associated with younger clearcut ages (Table 18). We interpret these results as suggestive.

Our 24-year window never observes the same stand clearcut twice, and we rarely observe repeat partial cuts, so these regressions compare the ages of different stands at harvest rather than changes in a given stand's rotation. Because fire exposure also changes which stands are harvested (Table 3), the age at harvest reflects both earlier cutting and changes in which stands are cut. The harvest choice model is therefore our main evidence on how fire risk shifts harvest timing.

7 Impacts of Historical Changes in Wildfire Risk

Section 6 shows that nearby fire activity and hot and dry weather conditions raise the probability of partial cutting and clearcutting, and that rising fire activity lowers planting densities. Each of these forest management adaptations lowers the volume of standing timber on the landscape. In this section, we quantify how the observed change in fire activity and fire-related climate since the start of the century has altered forest management decisions, forest volumes, biomass, and surface fuels, and quantify how these changes impact smoke and carbon emissions from recent wildfire seasons.

7.1 Impacts of adaptation on forest volumes

To answer these questions, we conduct a bio-economic Monte Carlo simulation that accounts for our estimated forest management probabilities along with dynamics of tree growth. Figure 12 summarizes each annual time step of the simulation. Taking the first measurement of each stand in our harvest choice sample, we use our empirical estimates from column (1) of Table 3 to generate estimates of the annual probability of not cutting, partial cutting, and clear cutting, and then draw management actions for each stand in each year. If a stand is clearcut, we assume that the harvest removes all growing stock volume, and the stand is replanted at the density predicted by our planting density model estimated in column (1) of Table 4. If a stand is partial cut, a portion of growing stock volume is removed, and the

magnitude of this removal is predicted by an intensive margin model estimated on a sample of partial cuts from the harvest choice estimating sample. This intensive margin model regresses the share of wood volume removed on the fire and VPD measures used in Table 3, stand-level controls, and county and period fixed effects (Table 19). Fire activity has no statistically significant effect on the share of volume removed, and higher levels of fire-season VPD marginally increases the share of standing volume removed (p¡0.1). We interpret this as suggestive evidence that fire activity and dry weather primarily affect whether a landowner cuts, with minimal impacts on how much volume is removed.

When a stand is left uncut, we use estimated yield functions from Mihiar and Lewis (2021)[21] to predict forest volumes as a function of stand age across the landscape.

Additionally, we develop empirical predictions of aboveground biomass as a function of stand age.[22] Biomass differs from merchantable volume in that it includes bark and small trees that contain no merchantable volume. Because biomass maps directly to carbon, it measures changes in the standing stock of carbon on the landscape. We estimate von Bertalanffy biomass yield functions using FIA data, controlling for forest type, region, and site class (Table 21).[23] We assume a clearcut removes all aboveground biomass, and a partial cut removes a share predicted by an estimated intensive margin model of biomass removal (Table 20).

Following each clearcut and partial cut, the simulation also updates surface fuels on the stand using difference-in-differences estimates of how cutting changes surface fuels, which we describe in Section 7.2. At the end of each year, we update each stand's age, standing volume, aboveground biomass, and surface fuels, advance fire and VPD exposure to the next year, and repeat until 2024. We run this bio-economic simulation for 5,000 iterations, with

[21] Mihiar and Lewis (2021) estimate tree-level von Bertalanffy yield functions by county and forest type. Because the functions are estimated at the tree level, predicted changes in planting density carry through directly to post-clearcut yields.

[22] Aboveground biomass is the oven-dry weight of wood and bark in the aboveground portion of live and standing dead trees, excluding foliage.

[23] Because the biomass yield functions are estimated at the stand level, predicted changes in planting density do not affect post-clearcut biomass growth.

each iteration drawing a parameter vector from the estimated asymptotic distribution of the model parameters, following Krinsky and Robb (1986). We use common random numbers, so that within an iteration all scenarios share both the parameter draw and the sequence of management draws, and differences across scenarios reflect only the changing fire and climate regime.

We run the simulation under four scenarios. In the baseline scenario, each stand's fire and VPD exposure are at observed levels. We then estimate three counterfactual scenarios, holding each stand's fire and weather fixed at a historical level.[24] These counterfactuals ask what management, standing volume, biomass, and fuels on non-federal western timberland would have looked like had fire activity and fire-related climate remained at turn-of-the-century levels. For each scenario, we record the cumulative number of partial cuts and clearcuts, the standing volume and biomass, and change in surface fuels at the end of the simulation. We present results for two simulation horizons, one ending in 2024, the last year of our estimating sample, and another ending in 2020, an exceptionally large fire year that we use in our emissions exercise in Section 7.2. These results are reported in Appendix F.

Figure 13 reports the results for 2024. Relative to each counterfactual, the observed fire and climate regime produces more clearcuts, more partial cuts, and less standing volume and biomass on the landscape. Clearcuts over 2001 to 2024 average 1,142 per iteration under the observed regime, compared with 862, 905, and 927 under the three counterfactuals. Partial cuts average 678 under the observed regime, compared with 485, 562, and 594. Standing volume in 2024 averages 4,165 cubic feet per acre under the observed regime, compared with 4,599, 4,521, and 4,474 cubic feet per acre under the counterfactuals, a reduction of 6.9 to 9.5 percent. Aboveground biomass falls similarly by 6.8 to 9.6 percent. Figure 14 reports the planting density responses to changing fire and climate. Post-clearcut planting density

[24] Our counterfactuals hold fire and fire-weather fixed at each plot's level in 2000, its 2000 to 2004 average, and its 2000 to 2009 average. Our fire variables measure cumulative fire activity over the fifteen years before the management year, and our VPD measure averages the three preceding fire seasons. Holding exposure fixed at an early-period level therefore holds it at a value built from earlier data. Fixing exposure at its 2000 to 2009 average, for example, uses fire activity observed from the mid-1980s onward.

averages 567 trees per acre under the observed regime, compared with 623 to 636 under the counterfactuals, a reduction of 56 to 69 trees per acre, or 9 to 11 percent.

These biomass differences imply a meaningful change in the carbon stored on the landscape. In 2024, standing aboveground biomass under the observed regime averages 69.8 tons per acre, 5.1 to 7.4 tons per acre less than under the counterfactuals (Table 22). Assuming carbon makes up half of dry biomass (Heath et al., 2009), this corresponds to 2.3 to 3.4 fewer metric tons of carbon per acre, or 8.5 to 12.4 fewer metric tons of CO_2.[25] At a social cost of carbon of $185 per ton (Rennert et al., 2022), the reduction in standing carbon is valued at roughly $1,560 to $2,290 per acre. This calculation overstates the net climate cost of adaptation, since much of the harvested carbon is stored in wood products and some would have otherwise been lost to future fire. However, this exercise illustrates that adaptation through additional cutting lowers the stock of carbon stored on the landscape, which may generate social costs.

We assume that no stand burns directly in the simulation. Therefore, the volume and biomass differences we estimate reflect purely a behavioral adaptation. Landowners respond to surrounding fire activity and hotter, drier fire seasons by holding less standing volume, which lowers their exposure to future loss and realizes the value of timber before fire can destroy it. This private response to elevated risk also changes the fuels available to future wildfires, with implications for smoke and carbon emissions, which we examine next.

7.2 Adaptation, smoke, and carbon

Fuel loads impact fire spread and severity (Agee and Skinner, 2005; Strabo et al., 2026), however the fuels available for fire include not only merchantable volume but also a number of surface fuels. A harvest removes merchantable volume, or crown fuels, but can leave branches and foliage, known as slash, on the forest floor. Before replanting, landowners may remove some of this slash through site preparation, or site prep. More cutting may therefore

[25] Biomass is reported in short tons, which we convert to metric tons (1 short ton = 0.907 metric tons). We convert carbon to CO_2 using the ratio of their molecular weights, 44/12.

lower crown fuels while raising surface fuels. We now ask how landowner adaptation changed emissions from recent wildfire seasons, holding the extent and severity of those fires fixed at observed levels.

We first estimate the effect of cutting on surface fuels using a difference-in-differences design. The sample consists of stands from the harvest choice sample that were clearcut, partial cut, or not cut between two FIA measurements, and we observe surface fuel loads at both measurements. We regress the change in each fuel load between measurements on indicators for a clearcut or partial cut, binned by years since the cut, controlling for site class and forest type, and we compare clearcuts and partial cuts against never-cut stands. The FIA also records whether a stand was site prepped after a clearcut, which allows us to estimate a separate effect of site preparation. We examine six fuel classes. The 1-, 10-, 100-, and 1000-hour classes are downed woody material on the forest floor of increasing size, named for the time each takes to respond to changes in moisture. Duff is a layer of decomposed organic material on the forest floor, and litter is a layer of recently fallen, undecomposed foliage.

Figures 18 and 19 in Appendix G report the estimates. Recent clearcuts raise all four woody fuel classes, consistent with the creation of slash, and the effects decay with time since the cut. Four or more years after a clearcut, 1-hour fuels fall below those of never-cut stands, possibly because a stand without mature trees no longer drops small branches. Clearcuts reduce duff, likely because harvest machinery disturbs the forest floor. Site preparation yields smaller point estimates for 1-hour and 1000-hour fuels, but apart from the 1000-hour class, these differences are not statistically significant. We treat this as suggestive evidence that site preparation reduces the surface fuels created by clearcutting, but these effects are not precisely estimated in our sample. Partial cuts have smaller but qualitatively similar effects on surface fuels. We do not estimate a site preparation effect for partial cuts, as site prep following a partial cut is uncommon.

These estimates enter the simulation at the surface fuel step for cut stands shown in

Figure 12 and described in Section 7.1. For each cut stand, we predict its change in surface fuels using the difference-in-differences estimates, and then compute the landscape-average change in each fuel class under the fire and climate regimes defined in Section 7.1. To examine the effect of site prep on fuels and emissions, we set site preparation at 30.7 percent of clearcuts, the rate observed in the FIA sample, and compare this to a scenario in which every clearcut is site prepared. Figures 20 and 21 show that, in both 2020 and 2024, the observed regime carries more woody surface fuel and less duff than the counterfactuals. Universal site preparation offsets part of this increase, lowering 1-hour and 1000-hour fuel loads, though it slightly raises 100-hour loads (Figures 22 and 23).

Taken together with Section 7.1, these results show that more cutting reduces standing volume and biomass but increases woody surface fuels. The net effect on emissions is therefore ambiguous. To quantify the impacts of our simulation results on wildfire emissions, we use the First Order Fire Effects Model (FOFEM) (Reinhardt et al., 1997; Lutes, 2020), a model developed by the U.S. Forest Service to predict the immediate effects of fire, including $PM_{2.5}$ and CO_2 emissions. We take every acre inside MTBS wildfire perimeters in a given fire year that burned on private and state timberland, identified using ownership data from Harris et al. (2025), and assign each acre its LANDFIRE Fuel Characteristic Classification System (FCCS) fuelbed and its MTBS burn severity.[26] We treat the observed fuelbeds from LANDFIRE as the adapted landscape, or in other words, the landscape with more harvesting. To construct each counterfactual landscape, we convert the simulated difference in each fuel class into a percent change by dividing it by the acre-weighted average load of that fuel type across the burned fuelbeds, and we scale crown fuels by the simulated change in aboveground biomass (Tables 22 and 23). We hold acres burned, burn severity, fire season, and fuel moisture fixed, so differences across scenarios reflect only changes in fuels. Therefore, this exercise does not capture any effect of adaptation on ignition, fire spread, or burn severity.

[26] We use the most recent FCCS release that predates each fire year: the 2016 release for 2020 and the 2023 release for 2024.

Table 24 reports the results. Adaptation lowers crown fuels and raises surface fuels, driven through increased harvesting. Because crown fuels burn mostly in the flaming phase while surface and ground fuels smolder, the adapted landscape emits more $PM_{2.5}$ and less CO_2 than the counterfactuals. Had fire activity and fire-related climate remained at turn-of-the-century levels, the 2020 fires on private and state timberland would have emitted 1,158 to 1,325 fewer metric tons of $PM_{2.5}$, or 0.27 to 0.30 percent less. Although these results are small in percentage terms, these changes are economically meaningful. Using the implied estimate of \$54,113.70 in damages per ton of $PM_{2.5}$ from western U.S. wildfires from Strabo et al. (2026), adaptation raised smoke damages from the 2020 fire season by roughly \$63 to \$72 million. At the same time, the counterfactual landscapes would have emitted 153,499 to 314,164 more metric tons of CO_2, or 0.43 to 0.89 percent more. At a social cost of carbon of \$185 per ton (Rennert et al., 2022), adaptation generated a climate benefit of roughly \$28 to \$58 million. The 2024 fire season, which burned far fewer acres of private and state timberland, shows the same pattern.

Table 25 examines the effect of site preparation. Under the observed fire and climate regime, landscape-wide site prep following every clearcut would have reduced 2020 emissions by 687 metric tons of $PM_{2.5}$, a benefit of roughly \$37 million, and by 62,173 metric tons of CO_2, a benefit of roughly \$11.5 million. Universal site prep does not fully offset the $PM_{2.5}$ increase from adaptation. Relative to the counterfactual with fire and climate held at 2000 levels and site preparation at its observed rate, $PM_{2.5}$ emissions remain 638 tons higher than the counterfactual world. Site prep does, however, add to the CO_2 reduction. Comparing the observed regime with the 2000-levels counterfactual, adaptation produced net damages of roughly \$13.6 million in 2020, as $PM_{2.5}$ damages exceed CO_2 benefits. Under universal site prep, these net damages become a net benefit of roughly \$35 million, given $PM_{2.5}$ damages implied by Strabo et al. (2026). This calculation omits the cost of site preparation to landowners, but it suggests that policies encouraging site preparation after clearcutting could generate sizable social benefits through lower wildfire emissions.

8 Discussion

Wildfire risk and intensity have increased, as have the climate conditions related to wildfire, and these trends are expected to continue under a warming climate (Westerling et al., 2006; Abatzoglou and Williams, 2016). Forest management can influence fire risk (Strabo et al., 2026), yet we know little about how management has responded to rising risk. This paper examines how non-federal forest landowners have adapted to changing wildfire risk through changes in forest management. Using forest inventory data for California, Oregon, and Washington, combined with local measures of fire exposure and climate, we find that landowners exposed to more nearby fire and hotter, drier weather conditions are more likely to partial cut and clearcut. Landowners also respond to rising fire activity over time by lowering their planting densities. Using a bio-economic Monte Carlo simulation, we find that observed changes in fire and climate this century have reduced standing volume on non-federal timberland by roughly 7 to 10 percent. Landowners are responding to rising wildfire risk by holding less volume on the landscape.

These changes in forest management are consistent with canonical economic theory of forest management under wildfire risk (Reed, 1984). Landowners respond to higher fire arrival rates by cutting more and cutting sooner. Reed's framework assumes clearcut harvests, but our partial cut results are qualitatively similar, where landowners respond to higher arrival rates by removing volume and selling it on the market. In other words, landowners act as if rising fire risk raises their discount rate. To show this consistency, we parameterize a Reed model for Douglas-fir using data from Hashida and Lewis (2019) and Wang and Lewis (2024), and find that an increase in the fire arrival rate implies reductions in standing volume of similar magnitude to our simulation results.[27] Our results are also consistent with Amacher et al. (2005), who find that higher risk should lower planting densities, and Patto

[27] We solve for the rotation that maximizes the Reed (1984) land expectation value, with zero salvage value and replanting costs, at annual fire arrival rates of 0.001 and 0.003 following Wang and Lewis (2024)], using the yield functions and stumpage prices of Hashida and Lewis (2019). Comparing average standing volume across the landscape under optimal rotations, the higher arrival rate reduces volume by roughly 4 to 12 percent across price regions, site classes, and discount rates of 1 to 5 percent.

and Rosa (2022), who show that commercial thinning happens sooner as arrival rates rise. Each of these behaviors is predicted to reduce a landowner's own exposure to fire. Lower planting densities, for example, mean less fuel, a common recommendation for reducing fire risk (Agee and Skinner, 2005; North et al., 2019).

Prior work shows that perceived changes in wildfire risk are capitalized into timberland markets (Wang and Lewis, 2024). Our work shows that these changes are also accompanied by changes in landowner behavior. Landowners adapt to local changes in fire activity and climate as those changes are observed. We interpret this as reactive adaptation (Massetti and Mendelsohn, 2018), consistent with prior studies of forest adaptation to climate risk (Hashida et al., 2025; Johnson and Lewis, 2024). To our knowledge, our findings provide the first large-scale empirical evidence that private landowners are adapting to changing fire risk through harvest actions predicted by natural resource economic theory.

Although landowners adapt according to their private net benefits, their adaptation generates both social benefits and social costs. Section 7 quantifies several of these. Additional cutting lowers the stock of carbon stored on the landscape, a reduction we value at roughly \$1,560 to \$2,290 per acre in 2024. Additional cutting also changes the fuels available to wildfire. It removes crown fuels but leaves slash on the forest floor, raising surface fuels. Holding the extent and severity of recent fires fixed, the adapted landscape emitted more $PM_{2.5}$ and less CO_2 during the 2020 fire season than it would have had fire and climate remained at turn-of-the-century levels. The additional smoke carries damages of roughly \$63 to \$72 million, while the reduction in fire emissions of CO_2 carries benefits of roughly \$28 to \$58 million. These effects depend in part on how landowners manage slash. Under universal site preparation following clearcuts, the net emissions damages of adaptation in 2020 become a net benefit.

Adaptation likely generates other externalities that we do not quantify. More cutting leads to more open canopies, which can benefit species such as elk and deer but harm closed-canopy species such as the federally threatened Northern Spotted Owl and Marbled Murrelet.

More clearcutting also plausibly leads to a larger share of young stands on the landscape. Young plantations are associated with more herbicide and fertilizer application, which may affect water quality through runoff, and may burn more severely in future fires (Zald and Dunn, 2018). Additional harvests raise short-run timber supply, while lower planting densities may reduce future timber supply. Finally, lower volumes on private land may spill over to neighboring private and public land by reducing fire spread and severity (Lauer et al., 2017). Because we document only a subset of these benefits and costs, we cannot sign the net welfare effect of private landowner adaptation to wildfire risk. Better understanding the full costs and benefits of adaptation is an important area for future research.

A number of caveats apply to our work. First, our emissions exercise holds the extent and severity of observed fires fixed, so forest management changes the fuels availabe for fire, but not how fires spread or how severely they burn. Recent evidence from federal land shows that fuel treatments reduce both fire spread and severity (Strabo et al., 2026). Whether private forest management has similar effects is an open question, particularly for practices such as clearcutting that are uncommon on public lands. Future work could estimate how private forest management influences fire spread and severity. Second, we do not trace the carbon stored in harvested wood products, so our estimate of the change in standing carbon overstates the net CO_2 cost of additional cutting. Third, our analysis does not consider the role of timber prices, although we control for macroeconomic shocks through time fixed effects. Fourth, we do not observe prescribed burning. Prescribed burning on western private land is relatively rare due to escape risk and liability laws (Miller et al., 2020). How it responds to changing wildfire risk in the West remains an open question, though Hashida et al. (2025) examine this for the southeastern United States. Fifth, we do not observe fuels management on neighboring private and public land, which may crowd out or encourage investment on a landowner's own land. Sixth, our estimates capture an average response across landowner types. Industrial owners and family forest owners may have different objectives, planning horizons, and information, and may adapt differently.

We do not observe these ownership groups in our data. Finally, our sample covers private and state timberland in Oregon, California, and Washington. It is unclear whether similar adaptation is occurring in other forestry-active regions, such as the southeastern United States, which holds considerably more private timberland but has lower wildfire risk.

We find that landowners adapt to risk through management changes that generate revenue. This creates an important tension with carbon pricing, which seeks to incentivize landowners to hold more volume. Fuel treatment cost-share programs, by contrast, seek to incentivize landowners to remove volume. These policies may interact in important ways. We find no evidence that landowners engage in more pre-commercial thinning, an activity that generates no immediate revenue but incurs fixed costs. This suggests cost-share programs may be most useful when targeted at costly activities whose benefits landowners do not fully internalize, rather than at revenue-generating harvests such as partial cuts and clearcuts that landowners already undertake. Our emissions results point to site preparation as one such activity. Site preparation is costly to landowners, but more widespread site preparation after clearcutting would reduce the surface fuels and smoke generated by adaptation.

Forest landowners are asset managers, and they are shifting their management to reduce their exposure to climate risk. Acting in their own self-interest, landowners are engaging in unsubsidized, landscape-scale changes in forest management that generate notable social benefits and social costs. Designing thoughtful policy requires understanding what landowners are already doing, and fully weighing the tradeoffs between these benefits and costs.

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Figures and Tables

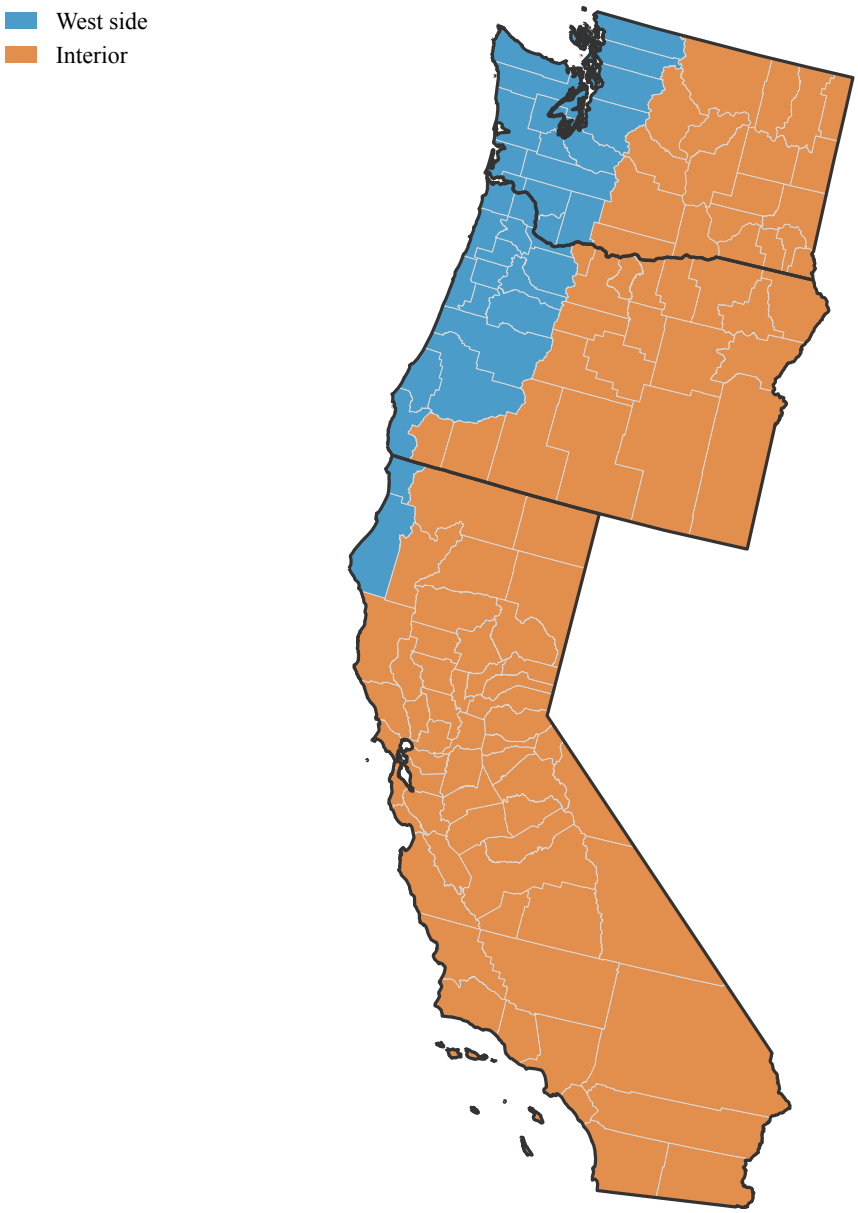


Figure 1: Study region

Notes: Coastal and western counties in northwest California, Oregon, and Washington are classified as west side. All other counties are classified as interior or "east-side".

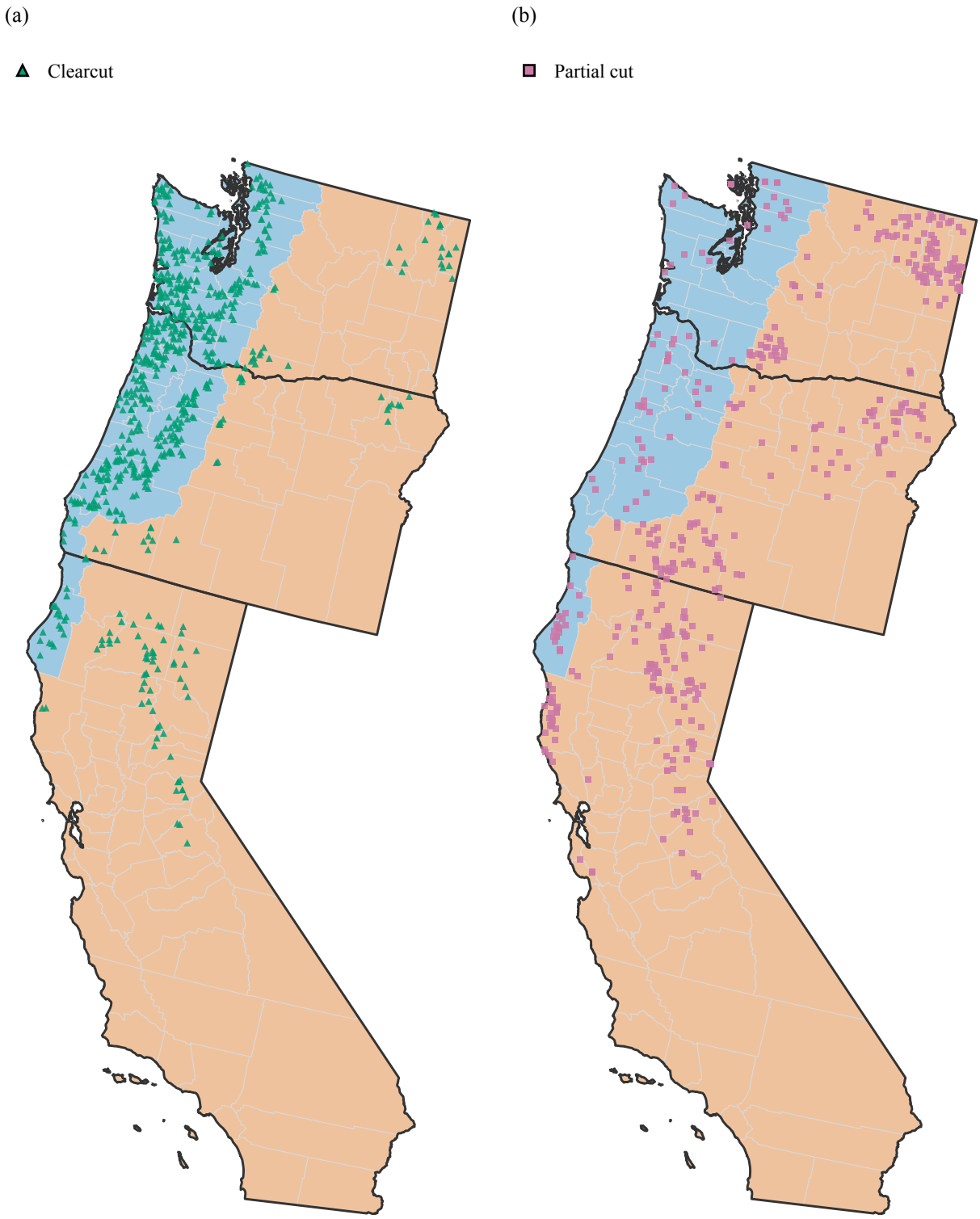


Figure 2: Observed clearcut and partial cut harvests by region

Notes: Locations of observed harvests on non-federal timberland in the sample. Panel (a): clearcut harvests. Panel (b): partial cut harvests. Blue denotes the west side and orange denotes the interior, as defined in Figure 1. Plot coordinates are approximate. The sample is described in Section 5.

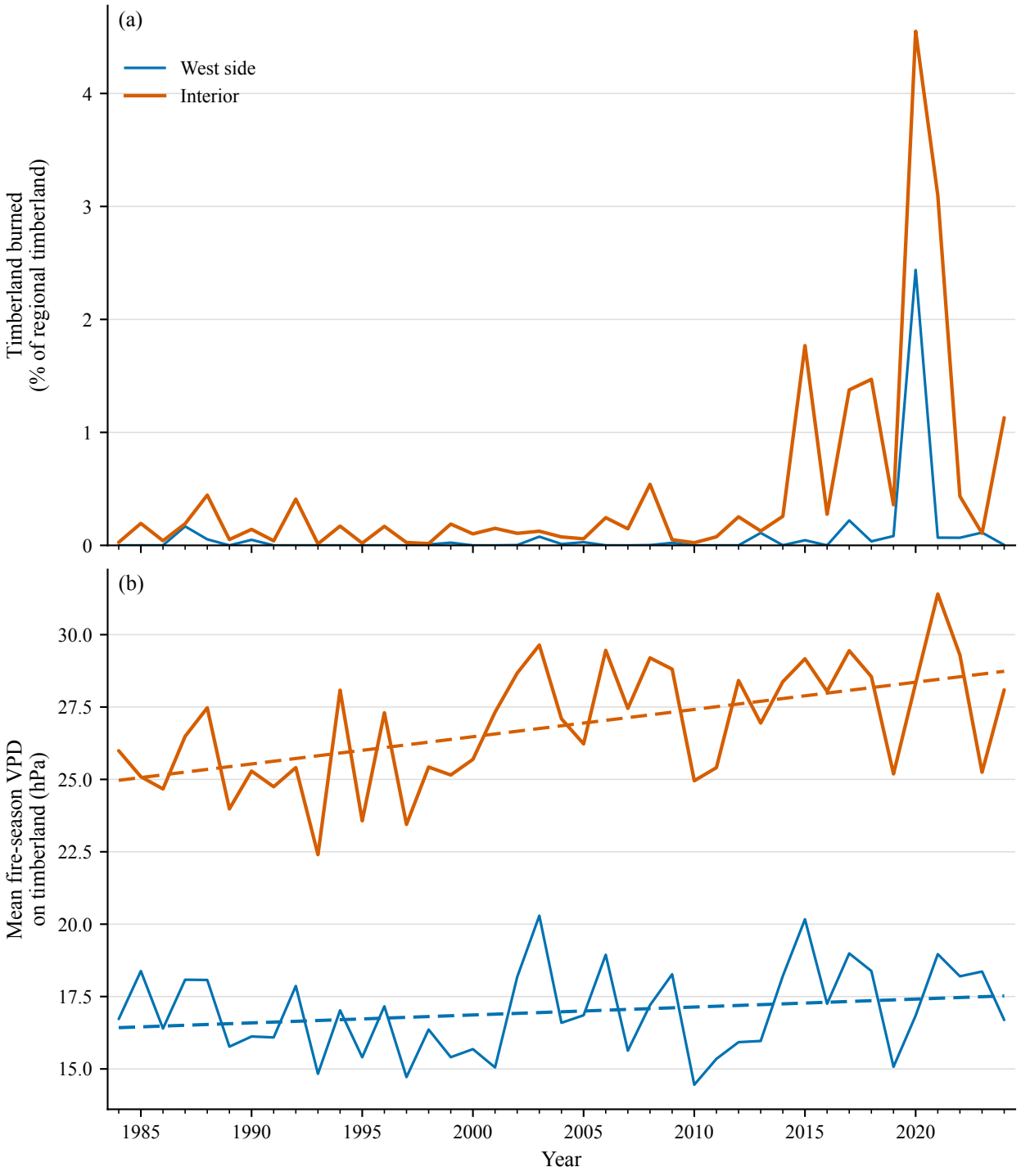


Figure 3: Fire and fire-season VPD by region, 1984–2024

Notes: Panel (a): non-federal timberland burned each year as a share of regional non-federal timberland. Burned area is measured from MTBS wildfire perimeters overlaid on the non-federal forest ownership map from Harris et al. (2025). Panel (b): mean June to September vapor pressure deficit on non-federal timberland, computed as a timberland-area-weighted average across counties, with fitted OLS trends. Non-federal timberland is measured as of roughly 2022 and held fixed throughout the period.

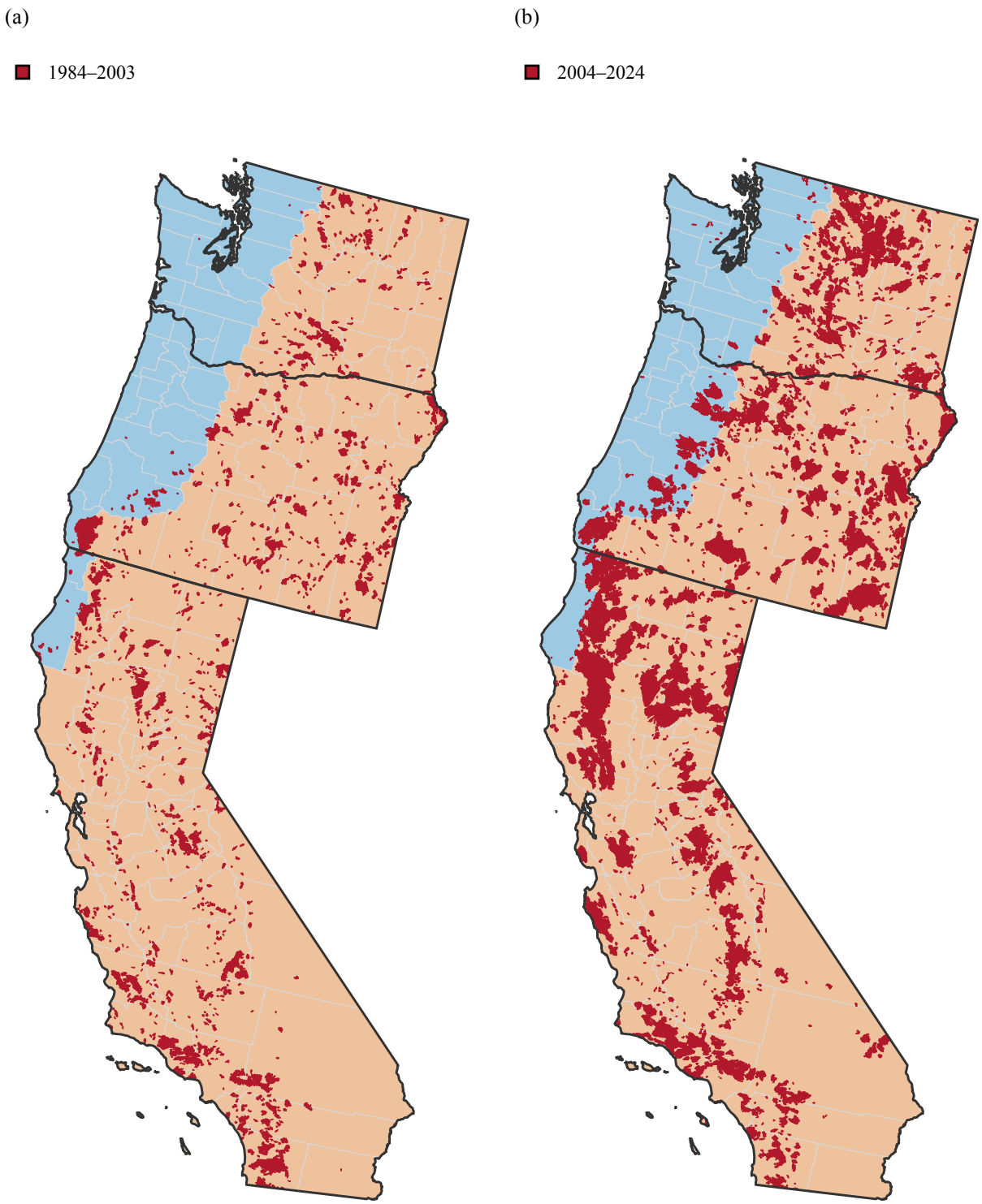


Figure 4: Burned area in the western United States, 1984–2003 vs. 2004–2024

Notes: MTBS wildfire perimeters greater than 1,000 acres in California, Oregon, and Washington. Panel (a): 1,414 fires, 11.6 million acres. Panel (b): 1,926 fires, 35.3 million acres.

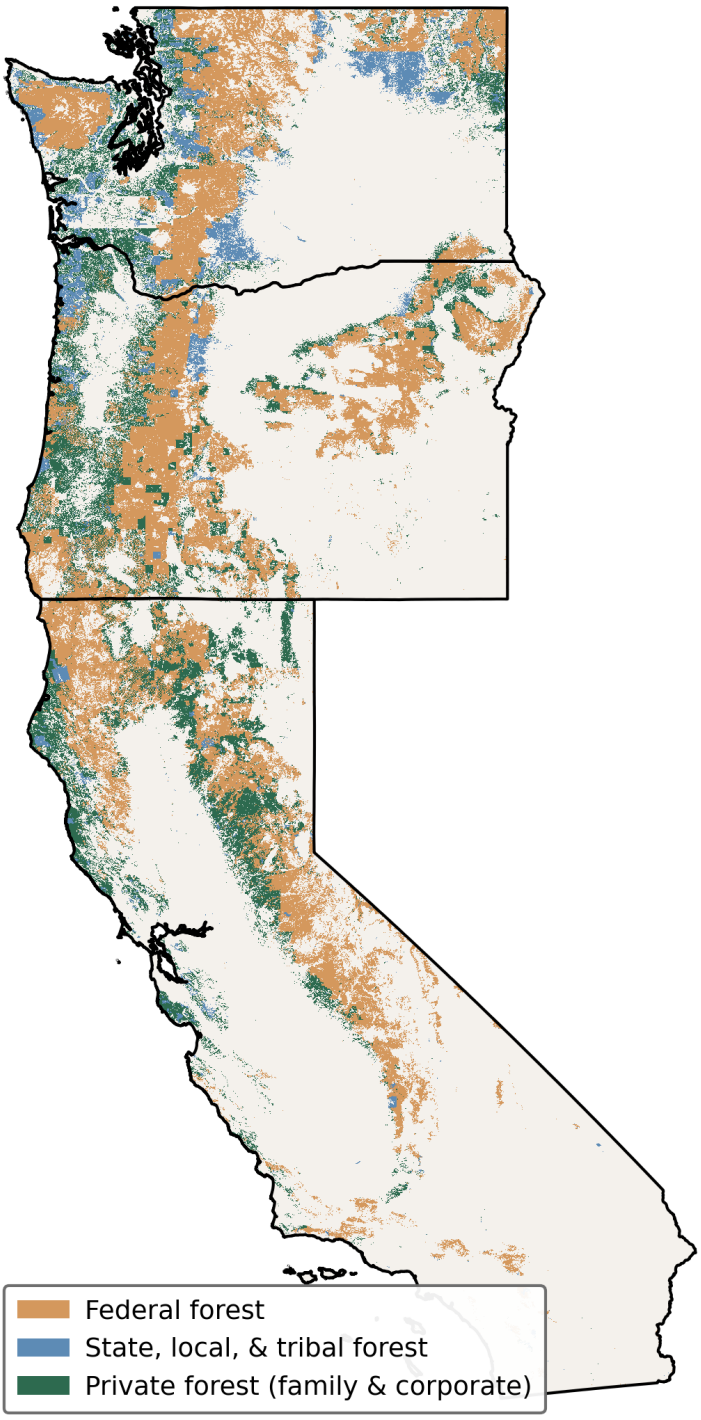


Figure 5: Forest land ownership in California, Oregon, and Washington

Notes: Forest land in California, Oregon, and Washington classified by ownership type. Western forests are a mosaic of federal and non-federal ownership. Our estimating samples (Figure 6 and Figure 7) are restricted to private and state timberland. Ownership data from Harris et al. (2025); state boundaries shown in black.

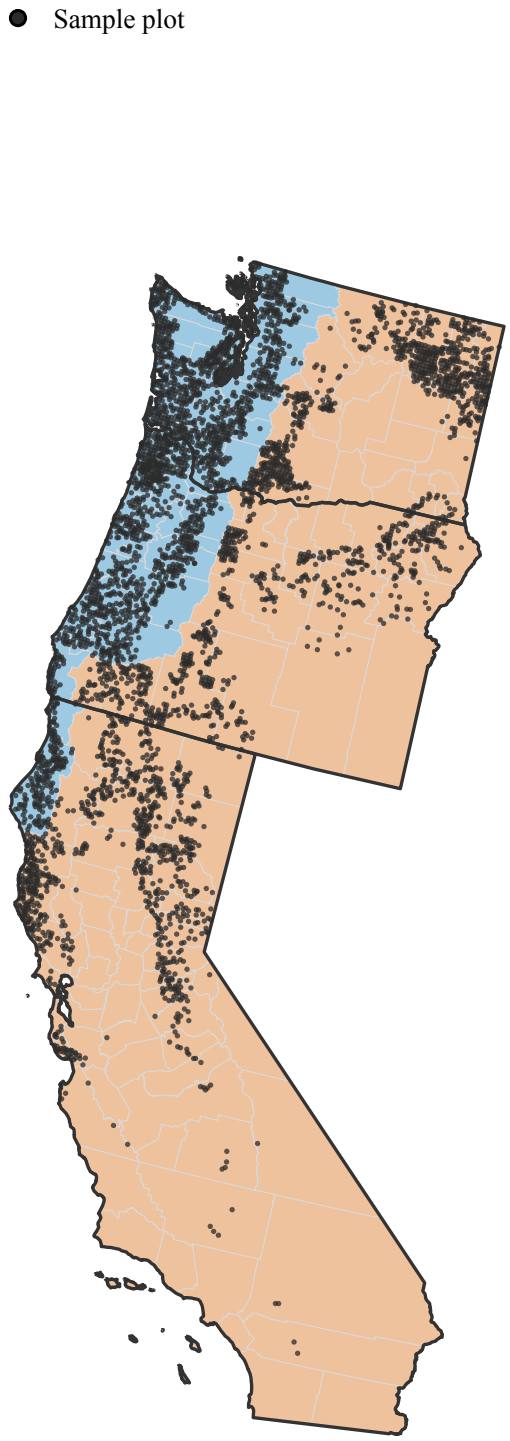


Figure 6: Non-federal FIA plots in the harvest-choice estimating sample

Notes: Each point represents one of the 4,367 unique FIA plots (5,277 stands) included in the harvest-choice estimating sample; 2,078 are on the west side and 2,289 in the interior. The sample restricts to non-federal stands aged 30 years or older, dropping any stands that recorded a salvage harvest. Plot coordinates are approximate.

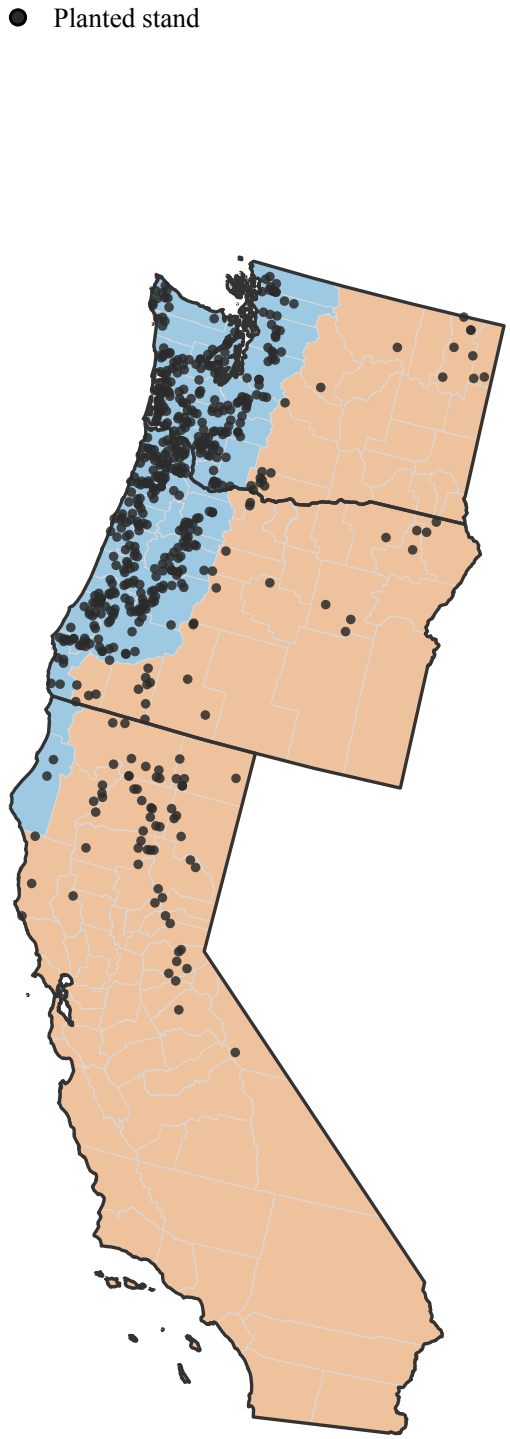


Figure 7: Planting density sample

Notes: Locations of recently planted stands in the planting density estimating sample. Plot coordinates are approximate.

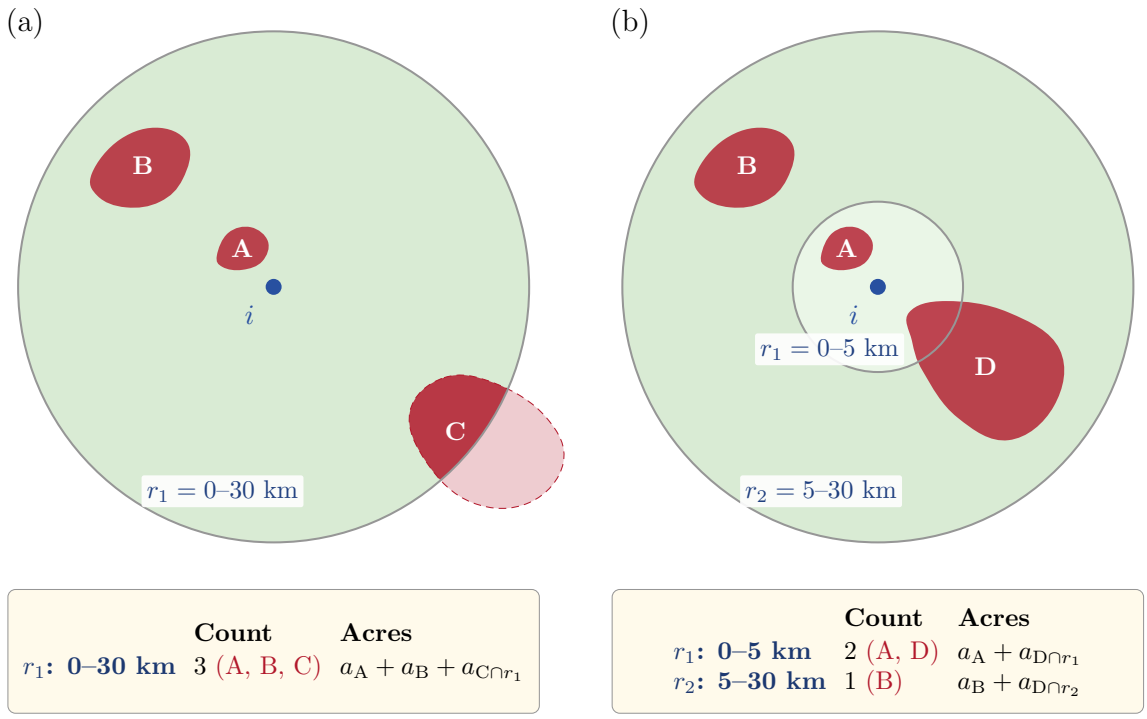


Figure 8: Construction of plot-level fire exposure variables

Notes: The figure illustrates how we measure fire exposure for a management decision observed on FIA plot i, over a look-back window ending a year before the management decision. Panel (a) shows the measure used in our main specifications: fires within 30 km of a plot. The fire count is the number of distinct fires whose perimeter falls at least partly within 30 km, and acreage burned sums only the burned area inside that distance. Panel (b) shows the decomposition used in robustness checks, which splits the area around the plot into an inner ring r_1 (e.g., 0–5 km) and an outer ring r_2 (e.g., 5–30 km). A fire that spans both rings, such as Fire D, is counted once, in the inner ring, while its acreage is split between the rings. Fire metrics are constructed from Monitoring Trends in Burn Severity (MTBS) fire perimeters, which include all fires of at least 1,000 acres. We construct each measure over 5, 10, and 15-year look-back windows.

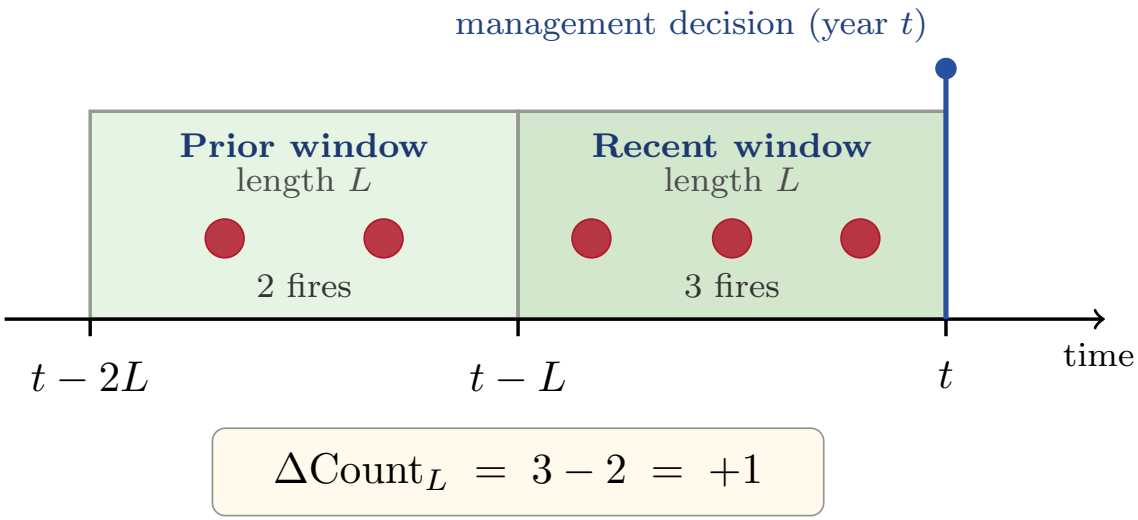


Figure 9: Construction of the change-in-fire-activity (Δ) variables

Notes: The figure illustrates how we measure changes in local fire activity over time, which define our Δ variables. For a management decision observed in year t and a look-back length L, we compare fire activity in the recent window $[t-L, t)$ against the immediately preceding window $[t-2L, t-L)$. Each red marker denotes one fire ($\geq$ 1,000 acres). In the example, three fires fall in the recent window and two in the prior window, so the change in fire count is $\Delta\text{Count}_L = 3-2 = +1$, indicating intensifying local fire activity. We construct these Δ measures separately within different spatial rings (e.g., r_1 and r_2 in Figure 8) and over 5, 10, and 15-year window lengths. Positive values indicate that recent fire activity exceeds that of the preceding period.

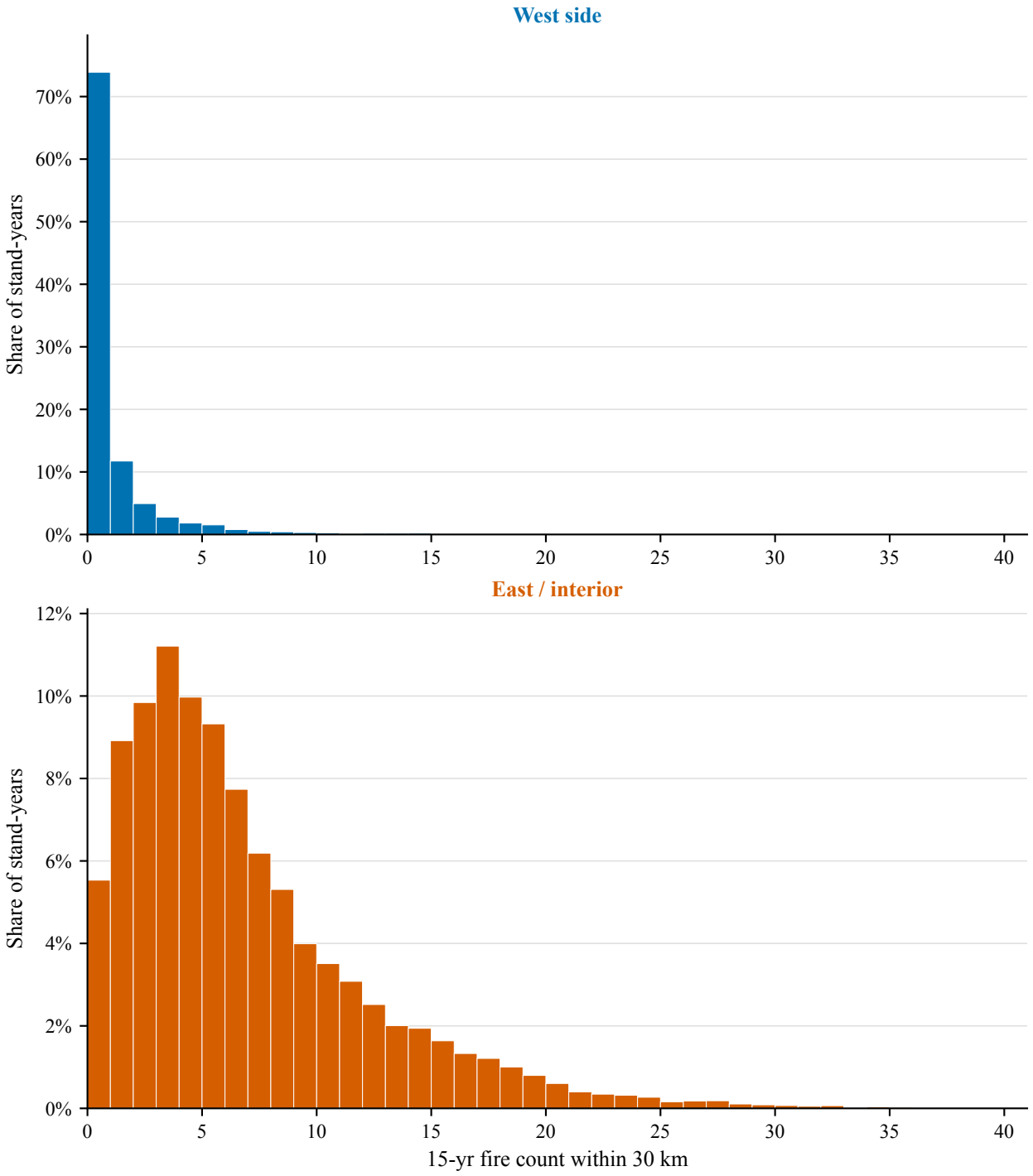


Figure 10: Distribution of fire count within 30 km, by region

Notes: Distribution of the 15-year fire count within 30 km, across stand-years in the harvest-choice estimating sample.

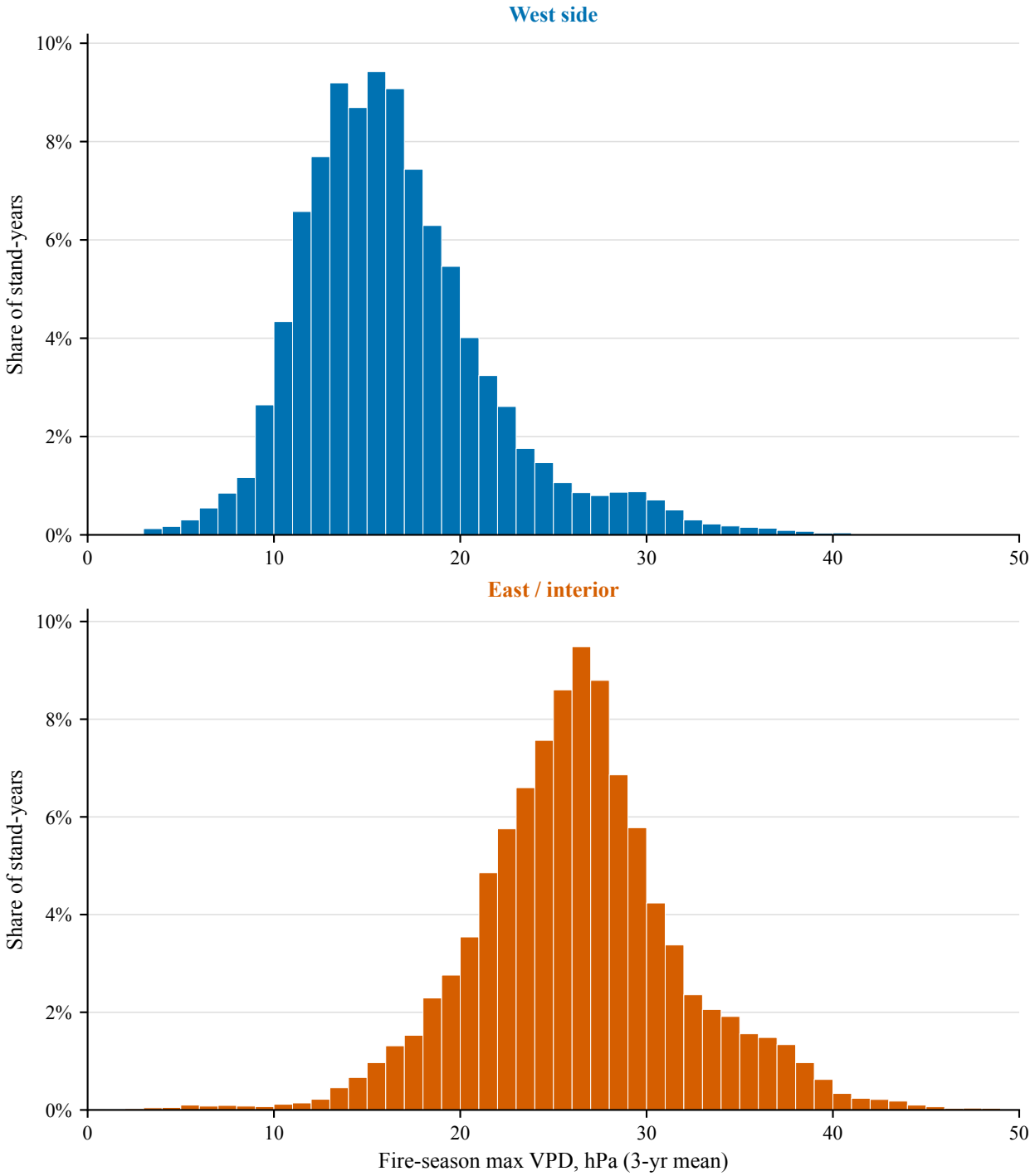


Figure 11: Distribution of fire-season maximum VPD, by region

Notes: Distribution of fire-season maximum vapor pressure deficit, measured as the three-year mean leading up to the management decision year, across stand-years in the harvest-choice estimating sample.

Table 1: Summary Statistics: Harvest Choice Sample, by Region

	Full sample	West side	East / interior
Sample size			
Stand-years	51,451	22,138	29,313
Stands	5,277	2,506	2,771
Plots	4,367	2,078	2,289
Stand-years in Washington	19,731	10,043	9,688
Stand-years in Oregon	18,360	9,859	8,501
Stand-years in California	13,360	2,236	11,124
Management actions (stand-years)			
No action	50,364	21,542	28,822
Partial cut	447	80	367
Clearcut	640	516	124
Wildfire exposure, mean (SD)			
Fire count, 0–30 km, 15-yr	3.97	0.76	6.39
	(5.10)	(2.00)	(5.39)
Acreage burned, 0–30 km, 15-yr (10,000 ac)	2.99	0.85	4.60
	(5.44)	(2.85)	(6.31)
Δ Fire count, 0–30 km, 10-yr	0.89	0.19	1.43
	(2.96)	(1.27)	(3.68)
Controls, mean (SD)			
Stand age (yrs)	74.7	60.5	85.4
	(39.2)	(34.0)	(39.6)
Fire-season max VPD, 3-yr (hPa)	22.12	16.56	26.32
	(7.27)	(5.20)	(5.61)

Notes: Summary statistics over the estimating sample of the multinomial logit of annual management action (Table 3): stand-years on private and state timberland in California, Oregon, and Washington, 2001–2024, for stands aged 30 years or older, dropping any stands that recorded a salvage harvest. West side and interior are as defined in Figure 1. Management actions are counts of stand-years with each action. Fire count and acreage burned are measured within 30 km of the plot over the 15 years before the management year, with acreage in units of 10,000 acres. The change in fire count is the number of fires within 30 km in the ten years before the management year minus the number in the ten years before that. Fire-season max VPD is the mean June–September maximum vapor pressure deficit over the three years leading up to the management decision year. The MTBS record begins in 1984, so the 10-year change in fire count, which requires twenty years of fire history, is undefined for management years 2001–2003 (1,974 stand-years); its statistics are computed over the remaining 49,477 stand-years.

Table 2: Summary Statistics: Planting Density Sample

Unit of observation: recently planted stands		
Sample size		
Stands	556	
Plots	530	
Counties	67	
States (OR / WA / CA)	280 / 214 / 62	
Planting years	2004–2023	
Measurement years	2004–2024	
	Mean	Std. Dev.
Outcome: planting TPA, dominant species		
All stands	524.6	865.0
Douglas-fir (n=401)	401.7	346.6
Other (n=155)	842.5	1497.8
Wildfire exposure		
Δ Fire count, 0–30 km, 10-yr	0.50	1.79
Fire count, 0–30 km, 15-yr	1.61	3.40
Acreage burned, 0–30 km, 15-yr (10,000 ac)	1.54	4.24
Post-burn planting (share)	7.2%	—
Controls		
Time since planting (yrs)	3.9	2.5
Elevation (km)	1.81	1.41
Slope (%)	32.0	23.2
Fire-season max VPD, 3-yr (hPa)	18.77	6.17
Site class (stands)		
Classes 1–2	234	
Classes 3–4	273	
Classes 5–7	49	

Notes: Summary statistics over the estimating sample of the planting density regression (column 1 of Table 4). Planting TPA is trees per acre of the dominant species, as measured by FIA field crews. Wildfire measures are defined as in Table 1, with the planting year in place of the management year. Post-burn planting equals one when the stand itself burned before planting. Fire-season max VPD is the mean June–September maximum vapor pressure deficit over the three years leading up to the planting year. Site class is the FIA site productivity class, from 1 (most productive) to 7, pooled into three bins as in Table 4. The MTBS record begins in 1984, so the 10-year change in fire count, which requires twenty years of fire history, is undefined for stands planted before 2004. This table therefore covers stands planted from 2004 onward and measured 2004–2024, the estimating sample of column 1 of Table 4; columns 2 and 3, which do not use the change, add 58 stands planted between 2001 and 2003.

Table 3: Multinomial Logit of Forest Management Action

	(1) Fire count (15-yr)	(2) Acreage burned (15-yr)	(3) Δ fire count (10-yr)
Panel A: Partial Cut vs. Do Nothing			
Wildfire exposure, 0–30 km	0.0862***	0.0638***	-0.0116
	(0.0272)	(0.0191)	(0.0193)
Fire-season Max VPD (hPa)	0.1379***	0.1364***	0.1342***
	(0.0446)	(0.0445)	(0.0475)
West side	-1.4284***	-1.3916***	-1.3072***
	(0.1664)	(0.1645)	(0.1642)
Stand age 50–69	0.2151	0.2194	0.2369
	(0.1580)	(0.1576)	(0.1602)
Stand age 70–89	0.4116***	0.4060***	0.3860**
	(0.1563)	(0.1560)	(0.1591)
Stand age 90+	0.1514	0.1651	0.1357
	(0.1625)	(0.1618)	(0.1618)
Panel B: Clearcut vs. Do Nothing			
Wildfire exposure, 0–30 km	0.2588***	0.0979***	0.0771***
	(0.0503)	(0.0155)	(0.0269)
Fire-season Max VPD (hPa)	0.1250***	0.1381***	0.0959**
	(0.0470)	(0.0464)	(0.0480)
West side	1.3398***	1.4338***	1.5282***
	(0.1345)	(0.1249)	(0.1222)
Stand age 50–69	0.2523***	0.2459**	0.2506**
	(0.0971)	(0.0971)	(0.0980)
Stand age 70–89	0.1358	0.1222	0.1181
	(0.1179)	(0.1179)	(0.1184)
Stand age 90+	-0.5409***	-0.5312***	-0.5568***
	(0.1587)	(0.1587)	(0.1595)
Baseline Pr(Partial Cut), pp	0.968	0.968	0.982
Baseline Pr(Clearcut), pp	1.334	1.334	1.372
Average marginal effects (pp per unit, SE in parens.)			
on Pr(Partial Cut):			
Wildfire exposure	0.080*** (0.026)	0.060*** (0.018)	-0.012 (0.019)
Fire-season Max VPD	0.131*** (0.043)	0.129*** (0.043)	0.129*** (0.046)
on Pr(Clearcut):			
Wildfire exposure	0.336*** (0.067)	0.127*** (0.021)	0.103*** (0.036)
Fire-season Max VPD	0.161*** (0.061)	0.178*** (0.061)	0.127** (0.064)
Observations	51,451	51,451	49,477
Plots (clusters)	4,367	4,367	4,362
Log pseudolikelihood	-4761.0	-4759.1	-4679.9
Pseudo R^2	0.060	0.060	0.054

Notes: Multinomial logit of annual forest management action on stand-year observations for private and state timberland in OR, CA, and WA, 2001–2024; do-nothing is the base outcome. Sample restricted to stand age ≥ 30 with salvage harvests excluded. Columns (1) and (2) are the count of distinct fires and the acres burned, in units of 10,000 acres, within 30 km over the 15 years before the management year. Column (3) is the 10-year change in the count of distinct fires within 30 km. Figures 8 and 9 illustrate how these measures are constructed. Because MTBS data begin in 1984, column (3) is estimated on management years 2004–2024. Fire-season Max VPD is the average June–September maximum vapor pressure deficit over the three preceding years, in hPa. West side is as defined in Figure 1. Stand age enters as 20-year bins, 30–49 omitted. All columns include period fixed effects and correlated random effects (Mundlak) group means for wildfire exposure and VPD, computed at the stand level over each stand's own observed window; the full coefficient set is in Appendix D. The baseline rows report the predicted probability of each outcome averaged over the estimating sample, in percentage points. Marginal effects are averaged over that same sample and are reported in percentage points per one-unit increase in the regressor: one additional fire within 30 km in column (1), 10,000 additional acres burned within 30 km in column (2), a one-fire increase in the 10-year change in column (3), and one hPa of fire-season max VPD in all three. Weighted by the forest stand size; standard errors clustered by plot. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Planting Density and Wildfire Exposure

	(1) Δ Fire count (10-yr)	(2) Fire count (15-yr)	(3) Acreage burned (15-yr)
Wildfire exposure, 0–30 km	-0.0808***	-0.0299	0.0056
	(0.0238)	(0.0345)	(0.0158)
Fire-season Max VPD (hPa)	0.0243	0.0272	0.0253
	(0.0206)	(0.0194)	(0.0202)
Post-burn planting	-0.1194	-0.2618	-0.3695
	(0.2123)	(0.2309)	(0.2345)
Time since planting (yrs)	0.0397*	0.0458**	0.0456**
	(0.0217)	(0.0204)	(0.0206)
Douglas-fir	-0.4761**	-0.4233**	-0.4098**
	(0.1888)	(0.1712)	(0.1719)
Site class (base: classes 1–2)			
Classes 3–4	-0.2735**	-0.2725**	-0.2744**
	(0.1252)	(0.1218)	(0.1241)
Classes 5–7	-0.5193**	-0.4663**	-0.4645**
	(0.2298)	(0.2261)	(0.2355)
Topographic controls			
Elevation (km)	0.0772	0.1047	0.0889
	(0.0783)	(0.0766)	(0.0747)
Slope (%)	-0.0008	-0.0004	-0.0004
	(0.0021)	(0.0021)	(0.0022)
Aspect (degrees)	0.0001	-0.0002	-0.0002
	(0.0004)	(0.0004)	(0.0004)
County FE	Yes	Yes	Yes
Planting period FE	Yes	Yes	Yes
Observations	556	614	614
Counties (clusters)	67	68	68
R^2	0.366	0.313	0.311

Notes: Weighted least squares estimates of log planting TPA (dominant species) on wildfire exposure for recently planted stands in OR, CA, and WA. Column (1) is the 10-year change in the count of distinct fires within 30 km. Columns (2) and (3) are the count of distinct fires and the acres burned, in units of 10,000 acres, within 30 km over the 15 years before planting. Figures 8 and 9 illustrate how these measures are constructed. Because MTBS data begin in 1984, column (1) is estimated on stands planted from 2004 onward. Post-burn planting is a dummy equal to one when the stand itself burned at some point before the planting year. Site class is the FIA seven-level site productivity classification pooled into three bins, with classes 1–2, the most productive, omitted. Douglas-fir is a dummy for the Douglas-fir forest type, the dominant planted type in the sample. Aspect is degrees of azimuth. Fire-season Max VPD is the average June–September maximum vapor pressure deficit over the three years before planting, in hPa. Planting-period fixed effects are bins for 2004 and earlier, 2005–08, 2009–12, 2013–16, and 2017 and later. Weighted by the forest stand size; standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

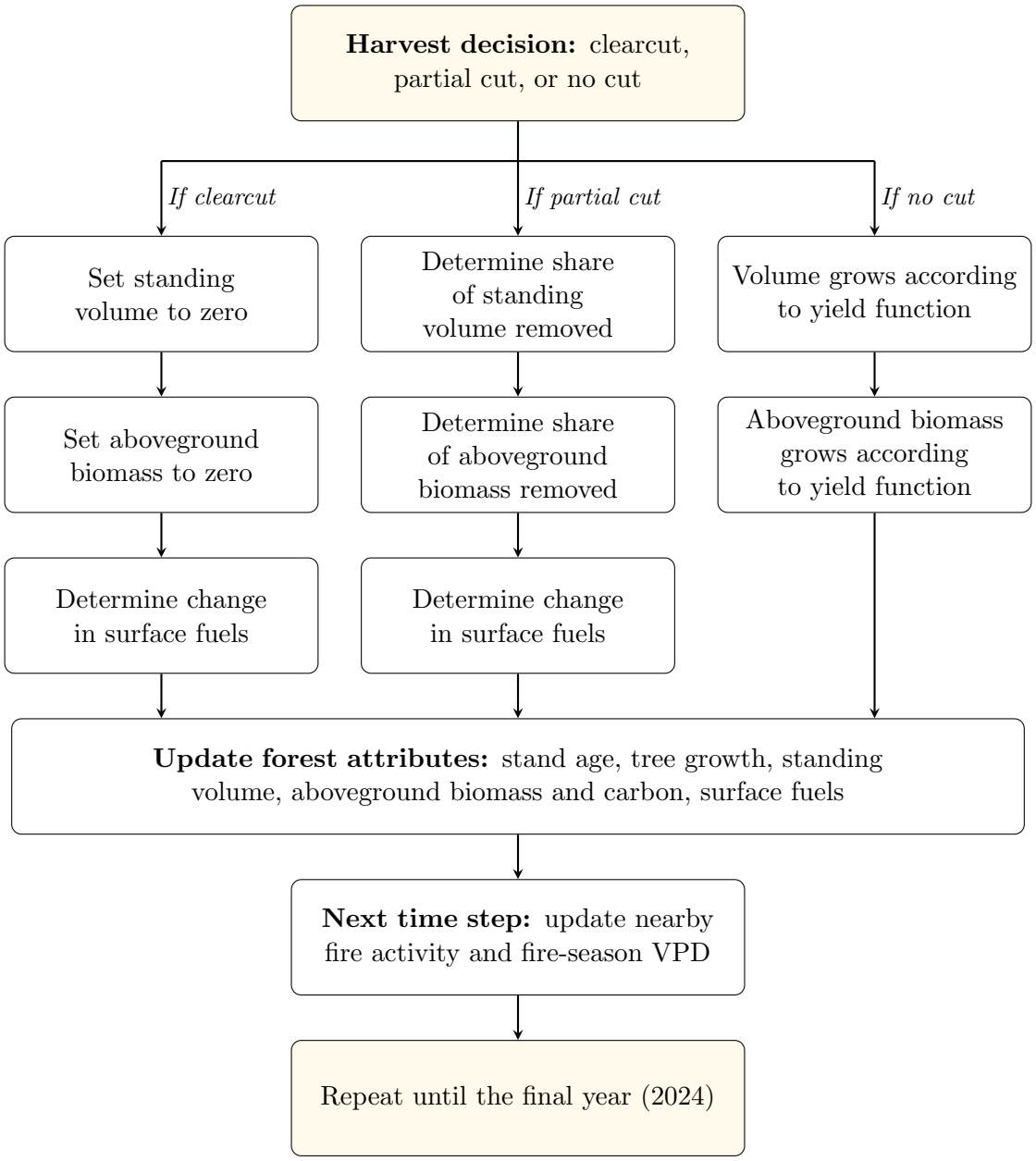


Figure 12: Structure of the bio-economic Monte Carlo simulation

Notes: The diagram shows one annual time step for a single stand. The harvest decision is drawn from the multinomial logit in column (1) of Table 3. After a clearcut, the stand is replanted at the density predicted by column (1) of Table 4. The share of standing volume and aboveground biomass removed by a partial cut is predicted by the intensive margin models in Tables 19 and 20. Uncut stands grow along the volume yield functions of Mihiar and Lewis (2021) and the biomass yield functions in Table 21. Changes in surface fuels after a cut follow the difference-in-differences estimates in Figures 18 and 19. Each of the 5,000 iterations runs every stand from 2001 to 2024 under the observed and counterfactual fire and climate regimes.

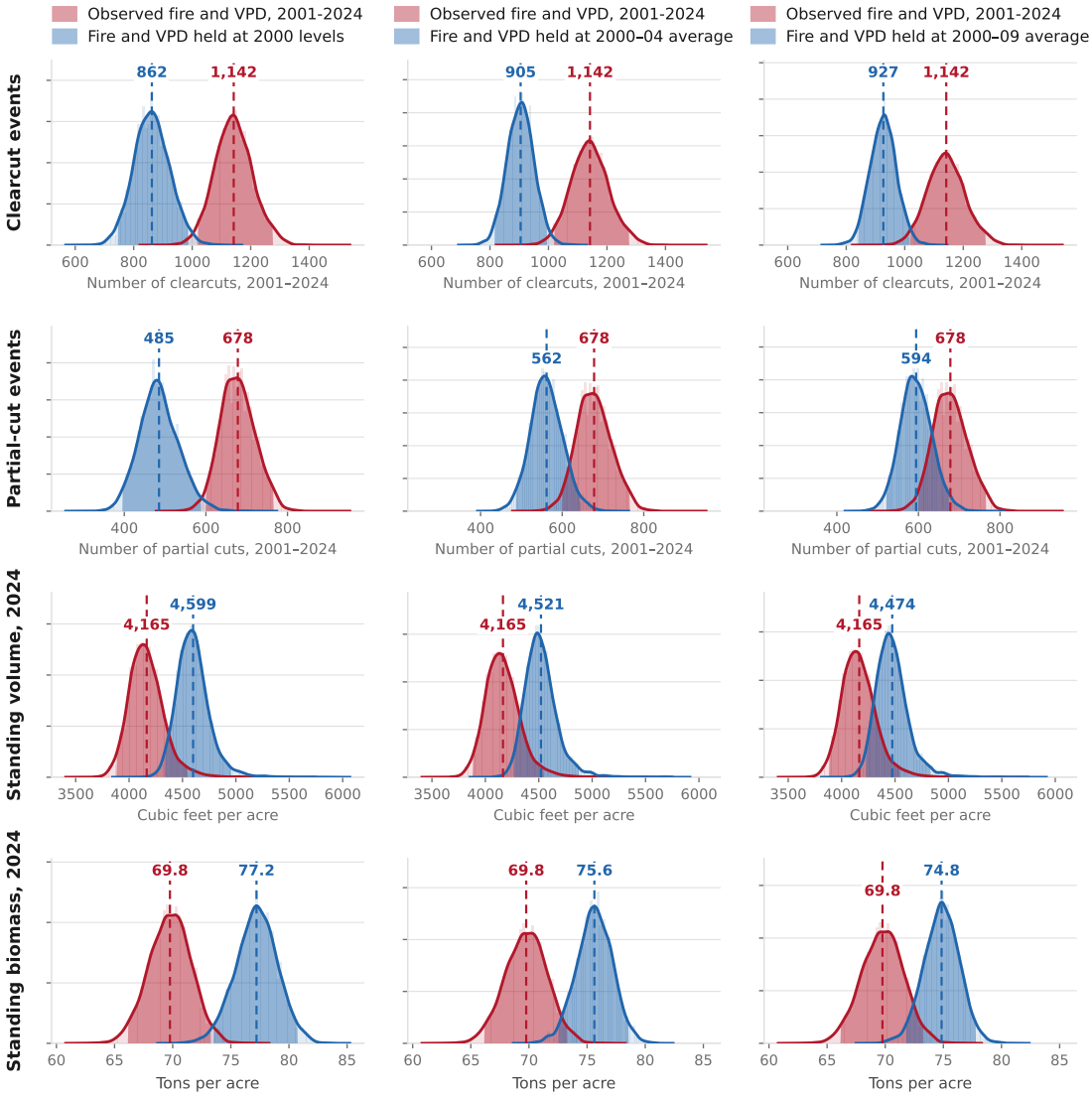


Figure 13: Simulated harvest activity, standing volume, and standing biomass under observed and counterfactual fire and climate regimes

Notes: Each panel plots the distribution of a simulated outcome across 5,000 Monte Carlo iterations. Every iteration draws a coefficient vector from the estimated asymptotic distribution of the parameters, following Krinsky and Robb (1986), and re-runs the full 2001–2024 stand-level simulation. Iterations share both the parameter draw and the sequence of random draws across arms. The three columns differ only in the level at which fire and fire-season VPD are held in the counterfactual arm. Dashed lines mark distribution means and shaded regions give the empirical 95% interval. Across the three counterfactual regimes, simulated standing volume in 2024 under the observed fire and climate regime is between 6.9% and 9.5% lower than it would have been under the counterfactual, with 95% intervals spanning 4.1% to 13.0%. The fourth row adds standing above-ground biomass, converted from pounds to tons per acre. It enters no harvest, planting or treatment decision and tracks volume closely, running 6.8% to 9.6% below the counterfactual in 2024.

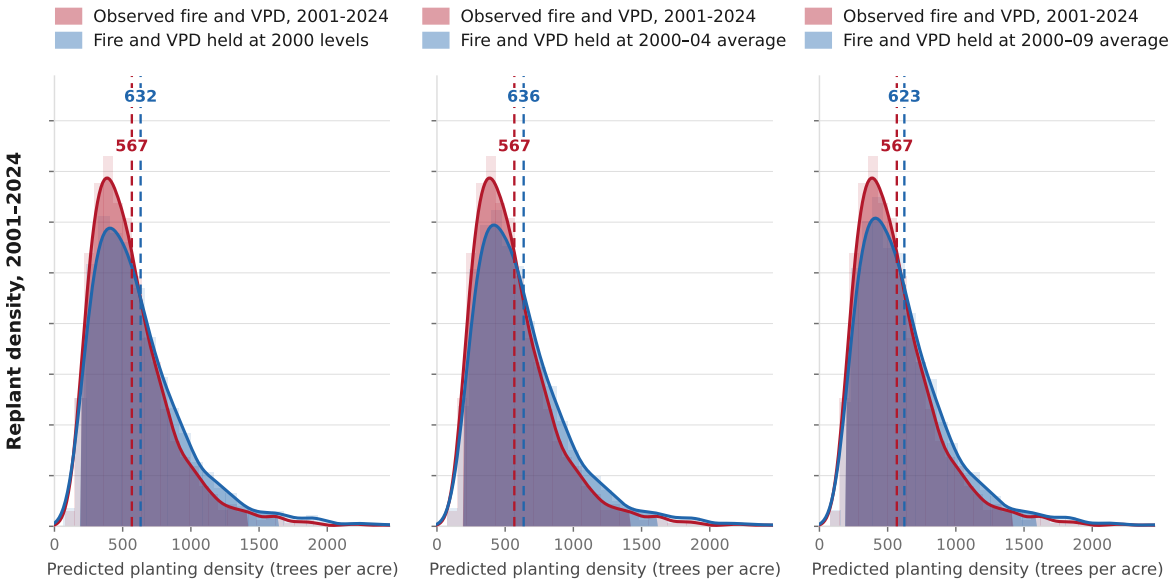


Figure 14: Simulated replanting density under observed and counterfactual fire and climate regimes

Notes: Each panel gives the distribution across the 5,000 simulation iterations of the mean predicted replanting density over the clearcuts occurring in that iteration, 2001–2024, in trees per acre. Replanting density is set by the stand-density model of Table 4, in which wildfire exposure reduces planted seedling density, so the counterfactual regimes replant more densely than the observed one: 632, 636 and 623 trees per acre against 567. Dashed lines mark distribution means. The three columns differ only in the level at which fire and fire-season VPD are held fixed, as in Figure 13. The horizontal axis is trimmed at 2,460 trees per acre for legibility, which omits 66 of 20,000 draw-values; all reported statistics use the full sample.

A Data and Sample Construction

A.1 Fire Variable Construction

Our wildfire data come from the Monitoring Trends in Burn Severity (MTBS) database (Eidenshink et al., 2007), which records the perimeter, ignition date, and burned area of every U.S. wildfire of at least 1,000 acres. We link MTBS fires to each FIA plot using the plot's confidential unfuzzed coordinates and construct plot-level measures of nearby fire exposure that vary across the spatial ring buffer, the look-back window, and the fire-activity measure (Table 5). Distances are circular buffers around the plot's true location with radii of 5 to 30 km in 5-km increments. Look-back windows are 5, 10, or 15 years and end the year before the management decision. We calculate the count of distinct fires, the acres burned inside the buffer, and the change in each between the most recent window and the window before it. Figures 8 and 9 illustrate how these measures are constructed. Because the MTBS data begins in 1984, the 10-year change measure is available from 2004 onward, and the 15-year change measure is available from 2014 onward. For this reason, we pair the 15-year count and acreage with the 10-year change measures in our main specifications.

Table 5: Fire Variables Constructed for Each Plot and Year

Dimension	Options constructed
Buffer radius (km)	5, 10, 15, 20, 25, 30
Look-back window (years)	5, 10, 15
Fire-activity measure	Fire count; acres burned; change in fire count; change in acres burned
Total	72 variables ($6 \times 3 \times 4$)

Notes: Every variable is constructed for each plot and management year over a window that ends the year before the decision.

A.2 FIA Sample Composition: Forest Type and Stand Origin

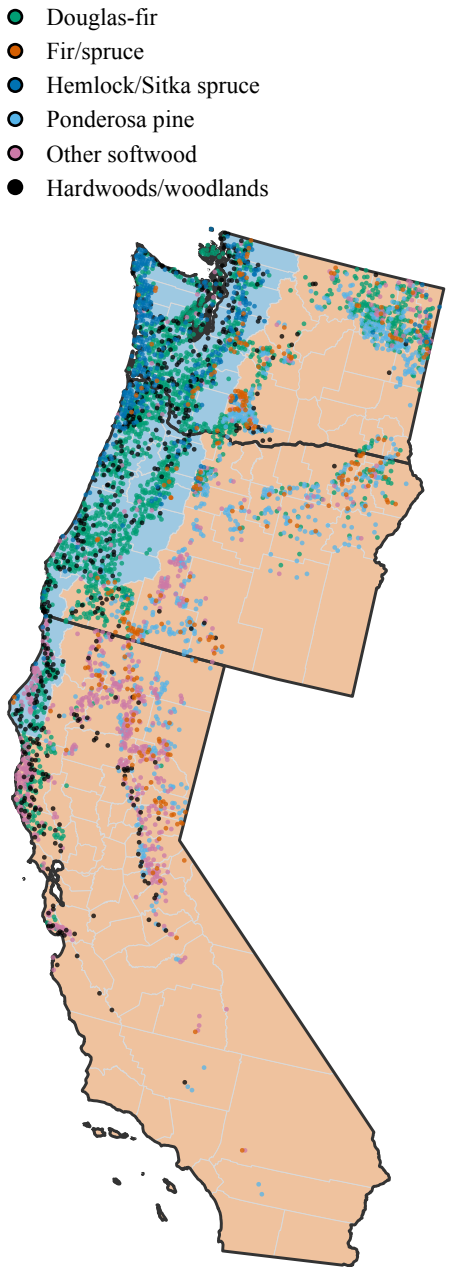


Figure 15: FIA plots by forest type group

Notes: The map shows the 4,367 unique FIA plots from the harvest-choice estimating sample as in Figure 6, colored by forest type group. Douglas-fir is the most common group, with 1,817 plots, followed by ponderosa pine (714), other softwoods (644), hardwoods and woodlands (501), hemlock and Sitka spruce (413), and fir and spruce (278). Forest type codes are recorded at FIA measurement. Plot coordinates are approximate.

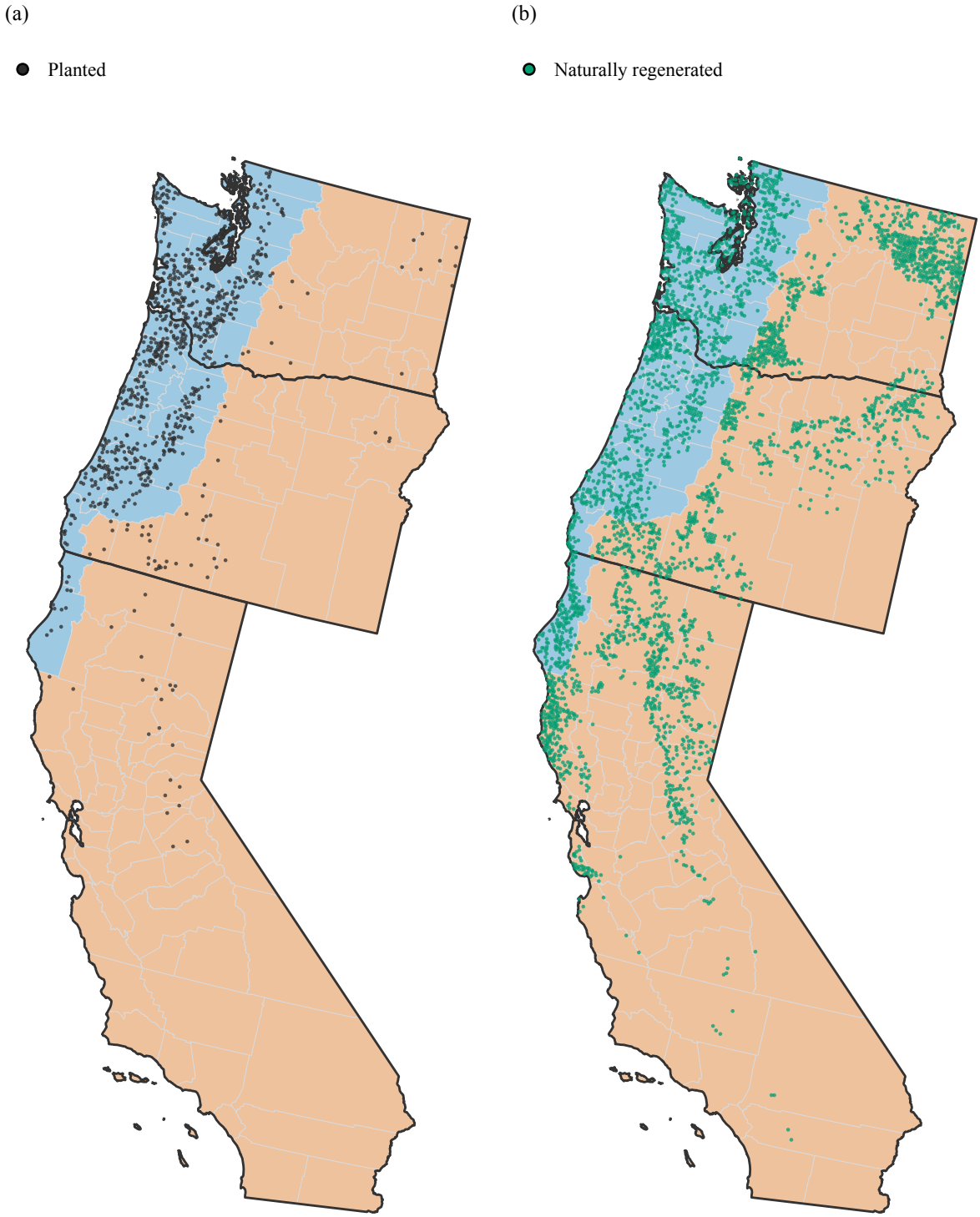


Figure 16: FIA plots by stand origin

Notes: The map shows the 4,367 unique FIA plots from the harvest-choice estimating sample as in Figure 6, split by stand origin. Most plots are naturally regenerated, with 3,573 plots in panel (b), compared with 794 planted plots in panel (a). Stand origin is determined by FIA field crews. Plot coordinates are approximate.

B Robustness Checks

B.1 Alternative Measures of Fire

Table 6 re-estimates the harvest choice model with wildfire exposure measured over 5-, 10-, and 15-year windows, as discussed in Section 6.3.

Table 6: Multinomial Logit of Forest Management Action: Alternative Fire Exposure Windows

	(1)	(2)	(3)	(4)	(5)	(6)
	Fire count			Acreage burned		
	5-yr	10-yr	15-yr	5-yr	10-yr	15-yr
Panel A: Partial Cut vs. Do Nothing						
Wildfire exposure, 0–30 km	0.1173***	0.0581*	0.0862***	0.0569***	0.0518**	0.0638***
	(0.0345)	(0.0320)	(0.0272)	(0.0214)	(0.0210)	(0.0191)
Fire-season Max VPD (hPa)	0.1361***	0.1382***	0.1379***	0.1337***	0.1358***	0.1364***
	(0.0448)	(0.0451)	(0.0446)	(0.0447)	(0.0447)	(0.0445)
Panel B: Clearcut vs. Do Nothing						
Wildfire exposure, 0–30 km	0.0608	0.2481***	0.2588***	0.0823***	0.1017***	0.0979***
	(0.0622)	(0.0478)	(0.0503)	(0.0265)	(0.0181)	(0.0155)
Fire-season Max VPD (hPa)	0.1397***	0.1202**	0.1250***	0.1337***	0.1351***	0.1381***
	(0.0468)	(0.0470)	(0.0470)	(0.0465)	(0.0463)	(0.0464)
Average marginal effects (pp per unit)						
on Pr(Partial Cut):						
Wildfire exposure	0.112***	0.053*	0.080***	0.054***	0.049**	0.060***
	(0.033)	(0.031)	(0.026)	(0.021)	(0.020)	(0.018)
Fire-season Max VPD	0.129***	0.131***	0.131***	0.127***	0.129***	0.129***
	(0.043)	(0.043)	(0.043)	(0.043)	(0.043)	(0.043)
on Pr(Clearcut):						
Wildfire exposure	0.078	0.322***	0.336***	0.107***	0.132***	0.127***
	(0.081)	(0.063)	(0.067)	(0.035)	(0.024)	(0.021)
Fire-season Max VPD	0.181***	0.155**	0.161***	0.173***	0.175***	0.178***
	(0.061)	(0.061)	(0.061)	(0.061)	(0.060)	(0.061)
Observations	51,451	51,451	51,451	51,451	51,451	51,451
Log pseudolikelihood	-4775.6	-4766.5	-4761.0	-4766.8	-4761.0	-4759.1

Notes: Each column re-estimates Table 3, column (1) or (2), with the fire count or acres burned within 30 km measured over the 5, 10, or 15 years before the management year. Columns (3) and (6) reproduce Table 3. The sample, controls, group means, fixed effects, and weights are as described in Table 3; each fire measure enters with its own group mean. Acres burned are in units of 10,000 acres. Average marginal effects are in percentage points per one-unit increase. Standard errors clustered by plot in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7 re-estimates the harvest choice model with wildfire exposure measured within 0–15 km and within 0–15 km and 16–30 km separately, as discussed in Section 6.3.

Table 7: Multinomial Logit of Forest Management Action: Distance Bands

	(1)	(2)	(3)	(4)	(5)	(6)
		Fire count			Acreage burned	
	0–30 km	0–15 km	0–15, 16–30 km	0–30 km	0–15 km	0–15, 16–30 km
Panel A: Partial Cut vs. Do Nothing						
Wildfire exposure, 0–30 km	0.0862***			0.0638***		
	(0.0272)			(0.0191)		
Wildfire exposure, 0–15 km		0.0631	0.0007		0.1648***	0.0946
		(0.0587)	(0.0643)		(0.0534)	(0.0701)
Wildfire exposure, 16–30 km			0.1227***			0.0511
			(0.0413)			(0.0329)
Fire-season Max VPD (hPa)	0.1379***	0.1452***	0.1386***	0.1364***	0.1405***	0.1365***
	(0.0446)	(0.0445)	(0.0446)	(0.0445)	(0.0446)	(0.0446)
Panel B: Clearcut vs. Do Nothing						
Wildfire exposure, 0–30 km	0.2588***			0.0979***		
	(0.0503)			(0.0155)		
Wildfire exposure, 0–15 km		0.4504***	0.3450***		0.1933***	0.0055
		(0.1035)	(0.1138)		(0.0607)	(0.1076)
Wildfire exposure, 16–30 km			0.2211***			0.1308***
			(0.0743)			(0.0403)
Fire-season Max VPD (hPa)	0.1250***	0.1351***	0.1253***	0.1381***	0.1416***	0.1381***
	(0.0470)	(0.0468)	(0.0470)	(0.0464)	(0.0467)	(0.0463)
Average marginal effects (pp per unit)						
on Pr(Partial Cut):						
Wildfire, 0–30 km	0.080***			0.060***		
	(0.026)			(0.018)		
Wildfire, 0–15 km		0.057	-0.002		0.156***	0.090
		(0.056)	(0.061)		(0.051)	(0.067)
Wildfire, 16–30 km			0.115***			0.048
			(0.040)			(0.031)
Fire-season Max VPD	0.131***	0.138***	0.131***	0.129***	0.133***	0.129***
	(0.043)	(0.043)	(0.043)	(0.043)	(0.043)	(0.043)
on Pr(Clearcut):						
Wildfire, 0–30 km	0.336***			0.127***		
	(0.067)			(0.021)		
Wildfire, 0–15 km		0.585***	0.449***		0.250***	0.006
		(0.136)	(0.149)		(0.080)	(0.140)
Wildfire, 16–30 km			0.286***			0.170***
			(0.097)			(0.053)
Fire-season Max VPD	0.161***	0.174***	0.162***	0.178***	0.183***	0.178***
	(0.061)	(0.061)	(0.061)	(0.061)	(0.061)	(0.061)
Test: 0–15 km = 16–30 km (p-value)						
Partial cut equation			0.17			0.65
Clearcut equation			0.43			0.38
Observations	51,451	51,451	51,451	51,451	51,451	51,451
Log pseudolikelihood	-4761.0	-4772.4	-4759.5	-4759.1	-4773.8	-4756.5

Notes: Each column re-estimates Table 3, column (1) or (2), with wildfire exposure measured within 0–30 km (as in Table 3), within 0–15 km, or within 0–15 km and 16–30 km entered together, all over the 15 years before the management year. The sample, controls, fixed effects, and weights are as described in Table 3; each fire measure enters with its own group mean. Acres burned are in units of 10,000 acres. Average marginal effects are in percentage points per one-unit increase. The test row reports the Wald test that the 0–15 km and 16–30 km coefficients are equal. Standard errors clustered by plot in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.2 Alternative Measures of Weather Shocks

Table 8 re-estimates the harvest choice model with maximum VPD measured over alternative months and averaging periods, as discussed in Section 6.3.

Table 8: Multinomial Logit of Forest Management Action: Alternative Measures of VPD

	(1) Baseline	(2)	(3) Alternative months	(4)	(5) Alternative averaging	(6)
Months	Jun–Sep	Apr–Jul	Jul–Sep	Jan–Dec	Jun–Sep	Jun–Sep
Years averaged	3	3	3	3	1	5
Panel A: Partial Cut vs. Do Nothing						
Wildfire count, 0–30 km	0.0862***	0.0835***	0.0890***	0.0844***	0.0878***	0.0858***
	(0.0272)	(0.0273)	(0.0272)	(0.0272)	(0.0270)	(0.0271)
Max VPD (hPa)	0.1379***	0.1009***	0.1352***	0.2253***	0.0578**	0.1909***
	(0.0446)	(0.0348)	(0.0483)	(0.0743)	(0.0243)	(0.0695)
Panel B: Clearcut vs. Do Nothing						
Wildfire count, 0–30 km	0.2588***	0.2597***	0.2635***	0.2589***	0.2590***	0.2574***
	(0.0503)	(0.0509)	(0.0499)	(0.0507)	(0.0499)	(0.0506)
Max VPD (hPa)	0.1250***	0.0703*	0.1417***	0.1987**	0.0849***	0.2339***
	(0.0470)	(0.0363)	(0.0547)	(0.0782)	(0.0211)	(0.0763)
Average marginal effects (pp per unit)						
on Pr(Partial Cut):						
Wildfire count	0.080***	0.078***	0.083***	0.078***	0.082***	0.080***
	(0.026)	(0.026)	(0.026)	(0.026)	(0.026)	(0.026)
Max VPD	0.131***	0.096***	0.128***	0.214***	0.055**	0.180***
	(0.043)	(0.033)	(0.046)	(0.071)	(0.023)	(0.067)
on Pr(Clearcut):						
Wildfire count	0.336***	0.337***	0.342***	0.336***	0.336***	0.334***
	(0.067)	(0.067)	(0.066)	(0.067)	(0.066)	(0.067)
Max VPD	0.161***	0.091*	0.183**	0.256**	0.110***	0.303***
	(0.061)	(0.047)	(0.071)	(0.102)	(0.028)	(0.100)
Mean of Max VPD (hPa)	22.31	16.40	24.00	12.25	22.34	22.28
SD of Max VPD (hPa)	7.21	5.03	7.72	3.75	7.39	7.15
Observations	51,451	51,451	51,451	51,451	51,451	51,451
Log pseudolikelihood	-4761.0	-4762.8	-4761.6	-4761.0	-4760.4	-4759.7

Notes: Each column re-estimates Table 3, column (1), with maximum vapor pressure deficit (VPD) averaged over the months and number of years before the management year shown in the column heading. Column (1) reproduces Table 3. The sample, controls, fixed effects, and weights are as described in Table 3; each VPD measure enters with its own group mean. The mean and SD of each VPD measure are weighted and computed over the estimating sample. Average marginal effects are in percentage points per one additional fire within 30 km or one hPa of VPD. Standard errors clustered by plot in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.3 Additional Harvest Choice Robustness Checks

Table 9 reports additional robustness checks for the harvest choice model, as discussed in Section 6.3.

Table 9: Multinomial Logit of Forest Management Action: Additional Robustness Checks

	(1) Year FE	(2) State × period FE	(3) Forest type × period FE	(4) Fire × VPD	(5) Direct burn	(6) All ages
Panel A: Partial Cut vs. Do Nothing						
Wildfire count, 0–30 km	0.0978***	0.0794***	0.0790***	0.0725**	0.0722***	0.0864***
	(0.0272)	(0.0282)	(0.0274)	(0.0335)	(0.0268)	(0.0266)
Fire-season Max VPD (hPa)	0.1192*	0.1388***	0.1352***	0.1234**	0.1320***	0.1252***
	(0.0622)	(0.0466)	(0.0450)	(0.0487)	(0.0448)	(0.0426)
Wildfire × VPD				0.0029		
				(0.0036)		
Direct burn					1.5194***	
					(0.2966)	
Panel B: Clearcut vs. Do Nothing						
Wildfire count, 0–30 km	0.2762***	0.2829***	0.2591***	0.3370***	0.2574***	0.2515***
	(0.0523)	(0.0512)	(0.0494)	(0.0501)	(0.0506)	(0.0479)
Fire-season Max VPD (hPa)	0.0848	0.1435***	0.1256***	0.1610***	0.1250***	0.1122**
	(0.0812)	(0.0479)	(0.0466)	(0.0485)	(0.0470)	(0.0452)
Wildfire × VPD				-0.0160***		
				(0.0043)		
Direct burn					0.2652	
					(0.7395)	
Average marginal effects (pp per unit)						
on Pr(Partial Cut):						
Wildfire count	0.091***	0.073***	0.073***	0.073***	0.067***	0.068***
	(0.026)	(0.027)	(0.026)	(0.028)	(0.026)	(0.022)
Fire-season Max VPD	0.114*	0.131***	0.128***	0.130***	0.125***	0.100***
	(0.060)	(0.045)	(0.043)	(0.043)	(0.043)	(0.034)
on Pr(Clearcut):						
Wildfire count	0.360***	0.367***	0.336***	0.516***	0.334***	0.265***
	(0.069)	(0.068)	(0.065)	(0.078)	(0.067)	(0.051)
Fire-season Max VPD	0.110	0.185***	0.162***	0.171***	0.161***	0.118**
	(0.106)	(0.063)	(0.061)	(0.061)	(0.061)	(0.048)
Observations	51,138	51,451	51,451	51,451	51,451	71,088
Plots (clusters)	4,365	4,367	4,367	4,367	4,367	5,040
Log pseudolikelihood	-4721.6	-4743.4	-4739.1	-4752.5	-4752.5	-5410.0
Pseudo R^2	0.066	0.063	0.064	0.061	0.061	0.068

Notes: Each column changes controls used in Table 3, column (1). Column (1) replaces the period bin fixed effects with management-year fixed effects and drops 2001 and 2024, the years with no partial cuts or no clearcuts. Column (2) replaces the period fixed effects with state × period fixed effects. Column (3) replaces them with forest type group × period fixed effects, using three groups: Douglas-fir, ponderosa pine, and all other forest types. These are coarser than the six groups in Figure 15 because finer forest type groups leave some group-periods with no harvests. Column (4) adds the interaction of the wildfire count with fire-season VPD, with VPD centered at its sample mean, so the wildfire coefficient is the effect of fire at the average VPD. Column (5) adds a dummy equal to one when the stand itself burned in either of the two years before the management year, measured by FIA field crews. Column (6) drops the restriction to stands aged 30 or older and adds an age bin for stands younger than 30. Marginal effects are averaged over the estimating sample and reported in percentage points per one additional fire within 30 km or one hPa of VPD. Standard errors clustered by plot in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The multinomial logit of Section 4 assumes independence of irrelevant alternatives across the three management actions. We test this assumption by re-estimating the harvest-choice model as a nested logit on the same sample, with the two harvest actions (partial cut and clearcut) grouped in a "cut" nest above the no-action alternative. Table 10 uses the specification of Table 3, column 1, and Table 11 drops the west-side dummy.

Table 10: Nested Logit of Forest Management Action

	(1) Multinomial logit	(2) Nested logit
Panel A: Partial Cut vs. Do Nothing		
Wildfire count	0.0862***	0.0489
	(0.0272)	(0.0394)
Max VPD (hPa)	0.1379***	0.2033***
	(0.0446)	(0.0645)
Panel B: Clearcut vs. Do Nothing		
Wildfire count	0.2588***	0.3907***
	(0.0503)	(0.0778)
Max VPD (hPa)	0.1250***	0.0757
	(0.0470)	(0.0632)
Panel C: Dissimilarity parameter		
λ_{cut}	—	12.162
		(13.627)
95% confidence interval	—	[−14.547, 38.871]
Wald test H_0: $\lambda_{\text{cut}} = 1$, $\chi^2(1)$	—	0.67
p-value	—	0.413
Observations	51,451	51,451
Plots (clusters)	4,367	4,367
Log pseudolikelihood	−4761.0	−4747.8

Notes: Column (1) reproduces Table 3, column 1. Column (2) estimates the same specification as a nested logit, with partial cut and clearcut grouped in a nest above the do-nothing alternative. The dissimilarity parameter λ_{cut} captures correlation in the unobserved utility of the two cut alternatives; $\lambda_{\text{cut}} = 1$ corresponds to the multinomial logit. Both models include the wildfire and VPD group means, stand-age bins, the west-side dummy, and period fixed effects, as in Table 3. Weighted by stand size; standard errors clustered by plot. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Nested Logit of Forest Management Action without the West-Side Dummy

	(1) Multinomial logit	(2) Nested logit
Panel A: Partial Cut vs. Do Nothing		
Wildfire count	0.0860***	0.0767***
	(0.0268)	(0.0286)
Max VPD (hPa)	0.1325***	0.1336***
	(0.0424)	(0.0483)
Panel B: Clearcut vs. Do Nothing		
Wildfire count	0.2601***	0.2922***
	(0.0504)	(0.0565)
Max VPD (hPa)	0.1206**	0.1204**
	(0.0492)	(0.0552)
Panel C: Dissimilarity parameter		
λ_{cut}	—	1.349
		(0.380)
95% confidence interval	—	[0.604, 2.093]
Wald test H_0: $\lambda_{\text{cut}} = 1$, $\chi^2(1)$	—	0.84
p-value	—	0.358
Observations	51,451	51,451
Plots (clusters)	4,367	4,367
Log pseudolikelihood	−4837.1	−4836.4

Notes: Column (1) re-estimates Table 3, column 1, without the west-side dummy. Column (2) estimates the same specification as a nested logit, with partial cut and clearcut grouped in a nest above the do-nothing alternative. The dissimilarity parameter λ_{cut} captures correlation in the unobserved utility of the two cut alternatives; $\lambda_{\text{cut}} = 1$ corresponds to the multinomial logit. Both models include the wildfire and VPD group means, stand-age bins, and period fixed effects. Weighted by stand size; standard errors clustered by plot. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12 varies how the stand-level group means are constructed, as discussed in Section 6.3.

Table 12: Multinomial Logit of Forest Management Action: Alternative Group Means

	(1) Baseline	(2) Period-bin group means	(3) 2001–2024 group means
Panel A: Partial Cut vs. Do Nothing			
Wildfire count	0.0862***	0.0868***	0.0908***
	(0.0272)	(0.0283)	(0.0259)
Max VPD (hPa)	0.1379***	0.1371***	0.1266***
	(0.0446)	(0.0451)	(0.0414)
Panel B: Clearcut vs. Do Nothing			
Wildfire count	0.2588***	0.0432	0.0464
	(0.0503)	(0.0505)	(0.0380)
Max VPD (hPa)	0.1250***	0.1182**	0.0609
	(0.0470)	(0.0512)	(0.0397)
Average marginal effects (pp per unit, SE in parens.)			
on Pr(Partial Cut):			
Wildfire count	0.080***	0.083***	0.086***
	(0.026)	(0.027)	(0.025)
Max VPD	0.131***	0.130***	0.120***
	(0.043)	(0.043)	(0.040)
on Pr(Clearcut):			
Wildfire count	0.336***	0.053	0.060
	(0.067)	(0.063)	(0.050)
Max VPD	0.161***	0.147**	0.078
	(0.061)	(0.064)	(0.052)
Wildfire and VPD group-mean window	Sample years	Sample years	2001–2024
Period-bin group means	No	Yes	No
Observations	51,451	51,451	51,451
Plots (clusters)	4,367	4,367	4,367
Log pseudolikelihood	-4761.0	-4432.3	-4776.8
Pseudo R^2	0.060	0.125	0.057

Notes: Robustness checks on the stand-level group means in the multinomial logit of forest management action; do-nothing is the base outcome. Column (1) reproduces Table 3, column 1, where the group means of the wildfire count and VPD are averaged over the years each stand is in the estimating sample. Column (2) adds group means of the period fixed effects, the share of each stand's sample years that falls in each period. Column (3) instead averages the wildfire count and VPD over a fixed 2001–2024 window, regardless of the years a stand is observed. Marginal effects are averaged over the estimating sample and reported in percentage points per one additional fire within 30 km or one hPa of VPD. Weighted by stand size; standard errors clustered by plot. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 Planting Density Robustness Checks

Table 13 reports robustness checks for the planting density model, as discussed in Section 6.3.

Table 13: Planting Density Model: Robustness Checks

	(1) Baseline	(2) State FE	(3) Planting year FE	(4) Forest-type FE	(5) 15-yr Δ count	(6) Annual 15-yr VPD	(7) Measured $\leq$ 3 yrs
Wildfire exposure, 0–30 km	-0.0808***	-0.0479*	-0.0824***	-0.0824***	-0.1082***	-0.0821***	-0.0840***
	(0.0238)	(0.0268)	(0.0250)	(0.0231)	(0.0355)	(0.0237)	(0.0314)
County FE	Yes	No	Yes	Yes	Yes	Yes	Yes
State FE	No	Yes	No	No	No	No	No
Planting-period FE	Yes	Yes	No	Yes	Yes	Yes	Yes
Planting-year FE	No	No	Yes	No	No	No	No
Douglas-fir dummy	Yes	Yes	Yes	No	Yes	Yes	Yes
Forest-type FE	No	No	No	Yes	No	No	No
Observations	556	556	556	556	232	556	274
Counties (clusters)	67	67	67	67	51	67	59
R^2	0.366	0.134	0.384	0.374	0.525	0.365	0.529

Notes: Weighted least squares estimates of log planting TPA for recently planted stands in OR, CA, and WA. Column (1) reproduces column (1) of Table 4. Column (2) replaces county fixed effects with state fixed effects. Column (3) replaces the planting-period bins with planting-year fixed effects. Column (4) replaces the Douglas-fir dummy with forest-type fixed effects, indicators for four forest-type groups assigned from the FIA field forest-type code recorded at measurement: Douglas-fir, ponderosa pine, hemlock and Sitka spruce, and all other types. Column (5) measures wildfire exposure as the 15-year rather than the 10-year change in the count of distinct fires within 30 km; because MTBS data begin in 1984, it is available only for stands planted from 2014 onward. Column (6) replaces fire-season Max VPD with the calendar-year maximum VPD averaged over 15 years; its coefficient remains statistically insignificant ($p = 0.37$). Column (7) restricts the sample to stands measured within three years of the planting year. Every column includes post-burn planting, time since planting, site class bins, and topographic controls, and every column other than (6) includes fire-season Max VPD, as in Table 4. Weighted by the forest stand size; standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Pre-Commercial Thinning Results

Table 14 reports the pre-commercial thinning model.

Table 14: Logit of Pre-Commercial Thinning

	(1) Fire count (15-yr)	(2) Acreage burned (15-yr)
Wildfire exposure, 0–30 km	0.0669	0.0390
	(0.0833)	(0.0347)
Fire-season Max VPD (hPa)	0.1858**	0.1854**
	(0.0783)	(0.0786)
West side	-0.3761	-0.3718
	(0.2404)	(0.2321)
Stand age (base: 0–9)		
10–19	1.5108***	1.5022***
	(0.2955)	(0.2964)
20–29	0.6591**	0.6506**
	(0.3147)	(0.3148)
30+	-1.7132***	-1.7195***
	(0.3398)	(0.3396)
Baseline Pr(Thin), pp	0.292	0.292
Average marginal effects (pp per unit)		
Wildfire exposure	0.019	0.011
	(0.024)	(0.010)
Fire-season Max VPD	0.054**	0.053**
	(0.023)	(0.023)
Observations	71,088	71,088
Pre-commercial thins	205	205
Plots (clusters)	5,040	5,040
Log pseudolikelihood	-932.0	-931.7
Pseudo R^2	0.140	0.140

Notes: Logit of annual pre-commercial thinning on stand-year observations. The wildfire and VPD measures, their group means, the west-side dummy, period fixed effects, weights, and clustering follow Table 3, columns (1) and (2). Unlike Table 3, the sample has no stand age restriction, since most thinning occurs before age 30, and stand age enters as bins for ages 0–9 (omitted), 10–19, 20–29, and 30 and older. As in Table 3, stands with any harvest or thinning coded as salvage are dropped. Marginal effects are averaged over the estimating sample and reported in percentage points per one-unit increase. Standard errors clustered by plot in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Harvest Choice Model Parameter Estimates

Tables 15 and 16 report the parameter estimates for the multinomial logit in Table 3.

Table 15: Multinomial Logit, Full Coefficient Set: Partial Cut Equation

	(1) Fire count (15-yr)	(2) Acreage burned (15-yr)	(3) Δ fire count (10-yr)
Wildfire exposure, 0–30 km	0.0862***	0.0638***	-0.0116
	(0.0272)	(0.0191)	(0.0193)
Wildfire, CRE bar	-0.1295***	-0.1257***	-0.0069
	(0.0302)	(0.0258)	(0.0258)
Fire-season Max VPD (hPa)	0.1379***	0.1364***	0.1342***
	(0.0446)	(0.0445)	(0.0475)
Max VPD, CRE bar	-0.1413***	-0.1396***	-0.1453***
	(0.0461)	(0.0460)	(0.0489)
Stand age 50–69	0.2151	0.2194	0.2369
	(0.1580)	(0.1576)	(0.1602)
Stand age 70–89	0.4116***	0.4060***	0.3860**
	(0.1563)	(0.1560)	(0.1591)
Stand age 90+	0.1514	0.1651	0.1357
	(0.1625)	(0.1618)	(0.1618)
West side	-1.4284***	-1.3916***	-1.3072***
	(0.1664)	(0.1645)	(0.1642)
Period 2005–08	0.3179	0.3298	0.2353
	(0.2392)	(0.2394)	(0.3158)
Period 2009–12	0.1451	0.2205	0.1271
	(0.2273)	(0.2281)	(0.3135)
Period 2013–16	0.3286	0.3924*	0.3028
	(0.2274)	(0.2277)	(0.3176)
Period 2017–24	0.1360	0.1773	0.1156
	(0.2457)	(0.2473)	(0.3178)
Constant	-4.3922***	-4.4693***	-4.3692***
	(0.3217)	(0.3248)	(0.3884)
Observations	51,451	51,451	49,477
Plots (clusters)	4,367	4,367	4,362
Log pseudolikelihood	-4761.0	-4759.1	-4679.9
Pseudo R^2	0.060	0.060	0.054

Notes: Full coefficient set for the partial cut equation of the three specifications reported in Table 3. Wildfire exposure and VPD each enter with their within-stand group mean, computed over the stand's own observed window, so the reported level coefficient is the effect of a deviation from that stand's own average. Stand age enters as 20-year bins with no group means and 30–49 the omitted category. The omitted period category is 2001–04. Weighted by the forest stand size; standard errors clustered by plot. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: Multinomial Logit, Full Coefficient Set: Clearcut Equation

	(1) Fire count (15-yr)	(2) Acreage burned (15-yr)	(3) Δ fire count (10-yr)
Wildfire exposure, 0–30 km	0.2588***	0.0979***	0.0771***
	(0.0503)	(0.0155)	(0.0269)
Wildfire, CRE bar	-0.3331***	-0.1691***	-0.1278**
	(0.0671)	(0.0307)	(0.0517)
Fire-season Max VPD (hPa)	0.1250***	0.1381***	0.0959**
	(0.0470)	(0.0464)	(0.0480)
Max VPD, CRE bar	-0.1291***	-0.1416***	-0.1058**
	(0.0483)	(0.0475)	(0.0488)
Stand age 50–69	0.2523***	0.2459**	0.2506**
	(0.0971)	(0.0971)	(0.0980)
Stand age 70–89	0.1358	0.1222	0.1181
	(0.1179)	(0.1179)	(0.1184)
Stand age 90+	-0.5409***	-0.5312***	-0.5568***
	(0.1587)	(0.1587)	(0.1595)
West side	1.3398***	1.4338***	1.5282***
	(0.1345)	(0.1249)	(0.1222)
Period 2005–08	-0.0459	-0.0771	-0.4516
	(0.2546)	(0.2502)	(0.2753)
Period 2009–12	0.4476**	0.4796**	0.0338
	(0.2239)	(0.2215)	(0.2591)
Period 2013–16	0.7690***	0.8059***	0.3685
	(0.2177)	(0.2169)	(0.2655)
Period 2017–24	0.3292	0.3543	0.0029
	(0.2453)	(0.2441)	(0.2683)
Constant	-5.3346***	-5.4535***	-5.0603***
	(0.3120)	(0.3111)	(0.3272)
Observations	51,451	51,451	49,477
Plots (clusters)	4,367	4,367	4,362
Log pseudolikelihood	-4761.0	-4759.1	-4679.9
Pseudo R^2	0.060	0.060	0.054

Notes: Full coefficient set for the clearcut equation of the three specifications reported in Table 3. Wildfire exposure and VPD each enter with their within-stand group mean, computed over the stand's own observed window, so the reported level coefficient is the effect of a deviation from that stand's own average. Stand age enters as 20-year bins with no group means and 30–49 the omitted category. The omitted period category is 2001–04. Weighted by the forest stand size; standard errors clustered by plot. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Harvest Timing Results

E.1 Partial Cut Timing

Table 17 reports the partial cut timing analysis, which estimates how stand age at the time of partial cut varies with prior nearby wildfire exposure.

Table 17: Stand Age at Partial Cut and Wildfire Exposure

	0–15 km		0–30 km	
	(1)	(2)	(3)	(4)
	Fire count	Acres burned	Fire count	Acres burned
Wildfire exposure (15-yr)	-2.140***	-1.600*	-0.779	-0.487
	(0.822)	(0.818)	(0.484)	(0.374)
Fire-season Max VPD (hPa)	0.377	0.300	0.380	0.306
	(0.461)	(0.461)	(0.470)	(0.462)
County FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
Topographic controls	Yes	Yes	Yes	Yes
Mean stand age at partial cut (years)	71.0	71.0	71.0	71.0
Observations	537	537	537	537
Plots (clusters)	491	491	491	491
Counties	77	77	77	77
R^2	0.404	0.402	0.401	0.400

Notes: OLS estimates of stand age in the year of a partial cut, one observation per partial cut. Wildfire exposure is the count of distinct fires or the acres burned, in units of 10,000 acres, within 15 km (columns 1–2) or 30 km (columns 3–4, the measures in Table 3) over the 15 years before the partial cut. Fire-season Max VPD is measured as in Table 3. As in Table 3, stands with any harvest coded as salvage are dropped, but there is no stand age restriction because stand age is the outcome. The west-side dummy of Table 3 is absorbed by the county fixed effects. Topographic controls are slope, aspect, and elevation. Weighted by stand size; standard errors clustered by plot in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E.2 Clearcut Timing

Table 18 reports the clearcut timing analysis, which estimates how stand age at the time of clearcut varies with prior nearby wildfire exposure.

Table 18: Stand Age at Clearcut and Wildfire Exposure

	0–15 km		0–30 km	
	(1)	(2)	(3)	(4)
	Fire count	Acres burned	Fire count	Acres burned
Wildfire exposure (15-yr)	-0.177	0.457	-0.686	-0.432
	(1.419)	(2.085)	(0.581)	(0.437)
Fire-season Max VPD (hPa)	0.203	0.184	0.258	0.257
	(0.347)	(0.357)	(0.356)	(0.362)
County FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
Topographic controls	Yes	Yes	Yes	Yes
Mean stand age at clearcut (years)	57.1	57.1	57.1	57.1
Observations	730	730	730	730
Plots (clusters)	630	630	630	630
Counties	65	65	65	65
R^2	0.396	0.396	0.398	0.397

Notes: OLS estimates of stand age in the year of a clearcut, one observation per clearcut. Wildfire exposure is the count of distinct fires or the acres burned, in units of 10,000 acres, within 15 km (columns 1–2) or 30 km (columns 3–4, the measures in Table 3) over the 15 years before the clearcut. Fire-season Max VPD is measured as in Table 3. As in Table 3, stands with any harvest coded as salvage are dropped, but there is no stand age restriction because stand age is the outcome. The west-side dummy of Table 3 is absorbed by the county fixed effects. Topographic controls are slope, aspect, and elevation. Weighted by stand size; standard errors clustered by plot in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Monte Carlo Simulation

The simulation combines three estimated models. The extensive margin, reported in Table 3, determines whether a stand is left alone, partially cut, or clearcut in a given year. When a clearcut occurs, replanting density is set by the stand-density model of Table 4. When a partial cut occurs, the share of standing volume removed is set by the intensive-margin model reported in Table 19.

Table 19: Intensive Margin: Share of Standing Volume Removed in a Partial Cut

	Share of volume removed
Wildfire exposure, 0–30 km (15-yr count)	-0.0029
	(0.0072)
Fire-season Max VPD (hPa)	0.0091*
	(0.0050)
Regrowth gap (yrs)	-0.0110
	(0.0092)
Stand age at cut (yrs)	0.0010*
	(0.0005)
Slope (%)	-0.0007
	(0.0010)
Aspect (degrees)	0.0002
	(0.0001)
Elevation (km)	0.0284
	(0.0323)
Douglas-fir forest type	-0.0549
	(0.0583)
Stand origin (base: natural)	
Planted	0.0486
	(0.1121)
Site class (base: classes 1–2)	
Classes 3–4	0.0713
	(0.0808)
Classes 5–7	-0.0167
	(0.0886)
County FE	Yes
Period FE	Yes
Observations	312
Counties (clusters)	65
R^2	0.343

Notes: Ordinary least squares estimates of the share of standing volume removed in a partial-cut event, removed = max(0, $1 - V_{\text{post}}/V_{\text{pre}}$), on partial cuts observed in OR, CA, and WA. This is the intensive-margin model used by the Monte Carlo simulation: when the multinomial logit assigns a partial cut, the predicted share here determines how much volume comes off the stand, applied to trees per acre in every species slot and bounded to [0, 0.999]. County and period fixed effects are included as explicit dummies rather than absorbed, because the simulation applies the county effect out of sample. We estimate the intensive margin on the same partial cuts the harvest-choice model uses. Of the 444 stands the multinomial logit records as receiving a partial cut, we are left with 312 usable observations, dropping any stands that did not have measured pre-cut volume, dropping stands that were burned between measurements, and dropping stands where plots had multiple stands at the first measurement and one stand at the second measurement, as we are unable to determine the correct pre- and post-cut volume. Site class is the FIA seven-level classification pooled into three bins, and forest type enters as a Douglas-fir dummy, matching the planting density specification in Table 4. Regrowth gap is the number of years between the harvest and the post-harvest remeasurement. Wildfire exposure is not statistically significant here, consistent with the harvest response to fire operating through the extensive margin rather than through how much is removed once a partial cut occurs. Fire-season VPD is significant at the ten percent level. Standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The simulation also tracks aboveground biomass. Table 20 repeats the specification of Table 19 on the same stands, replacing net merchantable cubic-foot volume with oven-dry aboveground biomass as the outcome.

Table 20: Intensive Margin: Share of Aboveground Biomass Removed in a Partial Cut

	Share of biomass removed
Wildfire exposure, 0–30 km (15-yr count)	-0.0014
	(0.0069)
Fire-season Max VPD (hPa)	0.0088**
	(0.0044)
Regrowth gap (yrs)	-0.0105
	(0.0085)
Stand age at cut (yrs)	0.0010*
	(0.0005)
Slope (%)	-0.0006
	(0.0009)
Aspect (degrees)	0.0003*
	(0.0002)
Elevation (km)	0.0360
	(0.0326)
Douglas-fir forest type	-0.0532
	(0.0549)
Stand origin (base: natural)	
Planted	0.0362
	(0.1110)
Site class (base: classes 1–2)	
Classes 3–4	0.0913
	(0.0752)
Classes 5–7	0.0134
	(0.0839)
County FE	Yes
Period FE	Yes
Observations	312
Counties (clusters)	65
R^2	0.326

Notes: Ordinary least squares estimates of the share of aboveground biomass removed in a partial-cut event, removed = $\max(0, 1 - B_{\text{post}}/B_{\text{pre}})$, on partial cuts observed in OR, CA, and WA. Biomass is oven-dry aboveground biomass summed over live trees and expressed per acre, in place of the net merchantable cubic-foot volume used in Table 19. This is the intensive-margin model the Monte Carlo simulation uses for biomass. When a stand partial cuts, the predicted share here determines how much aboveground biomass comes off the stand, bounded to [0, 0.999]. County and period fixed effects are included as explicit dummies rather than absorbed, because the simulation applies the county effect out of sample. We estimate the intensive margin on the same partial cuts the harvest-choice model uses. Of the 444 stands the multinomial logit records as receiving a partial cut, we are left with 312 usable observations, dropping any stands that did not have measured pre-cut volume, dropping stands that were burned between measurements, and dropping stands where plots had multiple stands at the first measurement and one stand at the second measurement, as we are unable to determine the correct pre- and post-cut volume. Site class is the FIA seven-level classification pooled into three bins, and forest type enters as a Douglas-fir dummy, matching the planting density specification in Table 4. Regrowth gap is the number of years between the harvest and the post-harvest remeasurement. Wildfire exposure is not statistically significant here, consistent with the harvest response to fire operating through the extensive margin rather than through how much is removed once a partial cut occurs. Fire-season VPD carries a coefficient of similar size to the volume model but a smaller standard error, and is significant at the five percent level. Standard errors clustered by county. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The simulation also grows aboveground biomass forward from stand age. Unlike the volume curves, which are taken from Mihiar and Lewis (2021), no published biomass curves match this setting, so they are estimated directly from FIA and reported in Table 21.

Table 21: Biomass Growth Curve Estimates

Region	Site class	n	Curve	α (tons/acre)	β	R^2
Douglas-fir						
East	I–II	19	pooled sc	74.8 (3.4)	0.0510 (0.0053)	0.074
East	III–IV	528	own	108.0 (8.5)	0.0423 (0.0051)	0.164
East	V–VII	833	own	76.8 (6.7)	0.0272 (0.0029)	0.111
West	I–II	1,565	own	240.4 (13.4)	0.0401 (0.0019)	0.631
West	III–IV	2,169	own	189.7 (8.2)	0.0398 (0.0016)	0.549
West	V–VII	193	own	127.7 (12.0)	0.0414 (0.0050)	0.423
Fir/Spruce						
East	I–II	16	pooled sc	93.7 (8.1)	0.0268 (0.0027)	0.199
East	III–IV	300	own	103.1 (11.1)	0.0275 (0.0033)	0.250
East	V–VII	205	own	90.0 (14.2)	0.0204 (0.0031)	0.230
West	I–II	48	pooled sc	251.0 (30.0)	0.0267 (0.0024)	0.628
West	III–IV	74	own	283.3 (41.7)	0.0244 (0.0027)	0.686
West	V–VII	16	pooled sc	251.0 (30.0)	0.0267 (0.0024)	0.628
Hemlock/Sitka spruce						
East	I–II	3	pooled sc	132.3 (17.9)	0.0198 (0.0030)	0.346
East	III–IV	33	pooled sc	132.3 (17.9)	0.0198 (0.0030)	0.346
East	V–VII	33	pooled sc	132.3 (17.9)	0.0198 (0.0030)	0.346
West	I–II	649	own	191.8 (14.1)	0.0469 (0.0041)	0.420
West	III–IV	287	own	191.1 (15.2)	0.0356 (0.0032)	0.412
West	V–VII	48	pooled sc	180.7 (9.8)	0.0456 (0.0030)	0.382
Ponderosa pine						
East	I–II	32	pooled sc	45.5 (1.8)	0.0482 (0.0035)	0.114
East	III–IV	471	own	62.3 (4.2)	0.0477 (0.0047)	0.229
East	V–VII	1,215	own	43.9 (2.5)	0.0358 (0.0031)	0.133
West	I–II	3	pooled sc, reg	45.2 (1.7)	0.0492 (0.0035)	0.108
West	III–IV	7	pooled sc, reg	45.2 (1.7)	0.0492 (0.0035)	0.108
West	V–VII	3	pooled sc, reg	45.2 (1.7)	0.0492 (0.0035)	0.108
Other softwood						
East	I–II	228	own	241.6 (45.0)	0.0245 (0.0047)	0.294
East	III–IV	726	own	125.2 (9.6)	0.0299 (0.0030)	0.225
East	V–VII	393	own	81.6 (10.4)	0.0229 (0.0028)	0.224
West	I–II	129	own	247.8 (35.2)	0.0407 (0.0065)	0.591
West	III–IV	93	own	316.3 (99.9)	0.0249 (0.0064)	0.553
West	V–VII	24	pooled sc	248.7 (44.3)	0.0337 (0.0060)	0.516
Hardwoods/woodlands						
East	I–II	27	pooled sc	85.9 (8.1)	0.0607 (0.0109)	0.095
East	III–IV	189	own	95.3 (10.7)	0.0623 (0.0131)	0.148
East	V–VII	140	own	59.4 (5.6)	0.0718 (0.0152)	0.076
West	I–II	283	own	114.0 (7.7)	0.0710 (0.0069)	0.212
West	III–IV	586	own	112.3 (5.8)	0.0626 (0.0048)	0.212
West	V–VII	114	own	127.1 (17.1)	0.0433 (0.0076)	0.233

Notes: Estimates of biomass$(a) = \alpha\,(1 - e^{-\beta a})^3$, aboveground oven-dry biomass per acre at stand age a, fitted by nonlinear least squares. Standard errors, clustered by plot, are in parentheses; α and its standard error are rescaled from pounds to tons per acre. These yield functions are estimated across forest type, region and site class groups, grouping site class as in Table 4. Some stand types have too few observations to support their own curve: where a group has fewer than 50 observations we fall back to one estimated by pooling site classes within region ("pooled sc"), and where that is still too thin, by pooling regions as well ("pooled sc, reg").

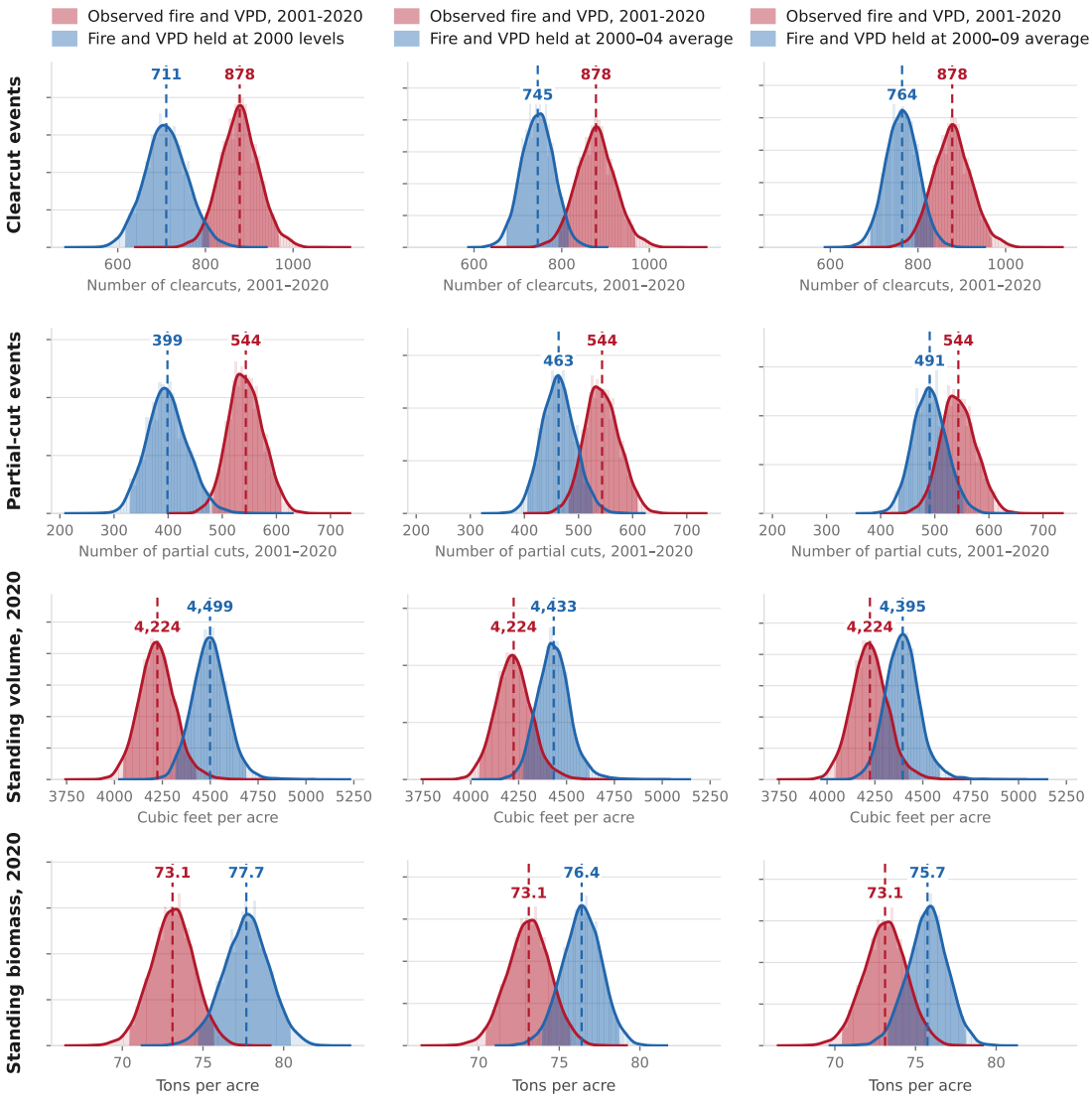


Figure 17: Simulated harvest activity, standing volume, and standing biomass in 2020 under observed and counterfactual fire and climate regimes

Notes: Figure 13 at a 2020 horizon to match the fire year used in the FOFEM exercise. The simulation, the parameter draws, and the random draws are identical to those in Figure 13. Cut counts are cumulative over 2001–2020 rather than 2001–2024, and standing volume is measured at the end of 2020. Dashed lines mark distribution means and shaded regions give the empirical 95% interval across the 5,000 iterations. Across the three counterfactual regimes, simulated standing volume in 2020 under the observed fire and climate regime is between 3.9% and 6.1% lower than it would have been under the counterfactual, with 95% intervals spanning 2.1% to 8.8%. The corresponding range in 2024 is 6.9% to 9.5%. The fourth row reports standing above-ground biomass in tons per acre, 3.5% to 5.9% below the counterfactual in 2020 against 3.9% to 6.1% for volume.

G FOFEM Estimates

Figures 18 and 19 report how surface fuel loads change after a clearcut and after a partial cut, relative to stands that were never cut.

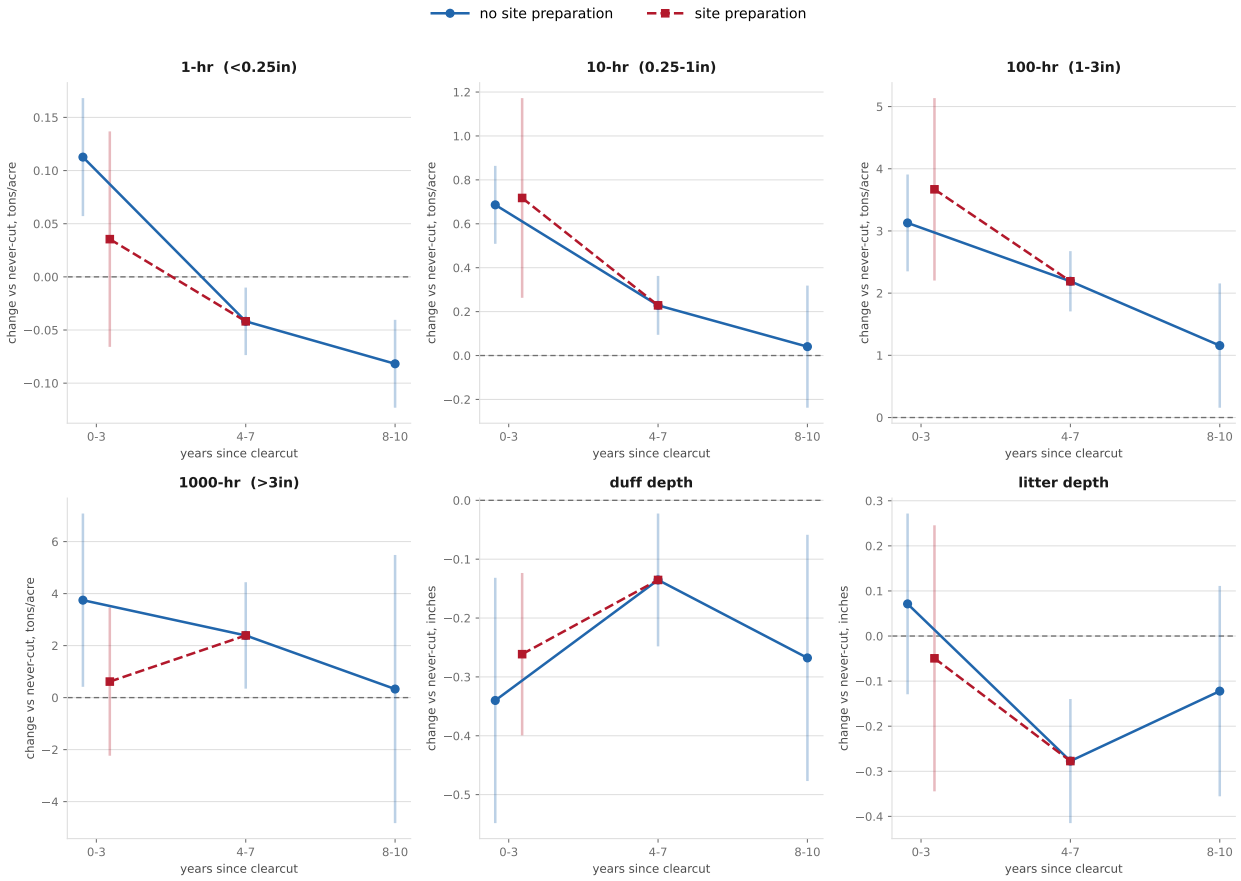


Figure 18: Surface fuel loads after a clearcut, relative to never-cut stands

Notes: Each point is a coefficient from a difference-in-differences regression of the change in fuel load between a stand's two FIA measurements, measured in tons per acre for the four downed-wood size classes and in inches for duff and litter depth. The sample has 420 clearcuts and 2,478 never-cut stands, dropping any stand recording a treatment code between the two visits. Stands recording a fire disturbance are excluded from both groups. The regressions condition on site class and forest type group, standard errors are clustered by plot, and the bars report 95% confidence intervals. Site preparation is identified only in the 0–3 bin, where 63 of the clearcuts were site prepped, so the two effects share the same estimate from 4–7 onward.

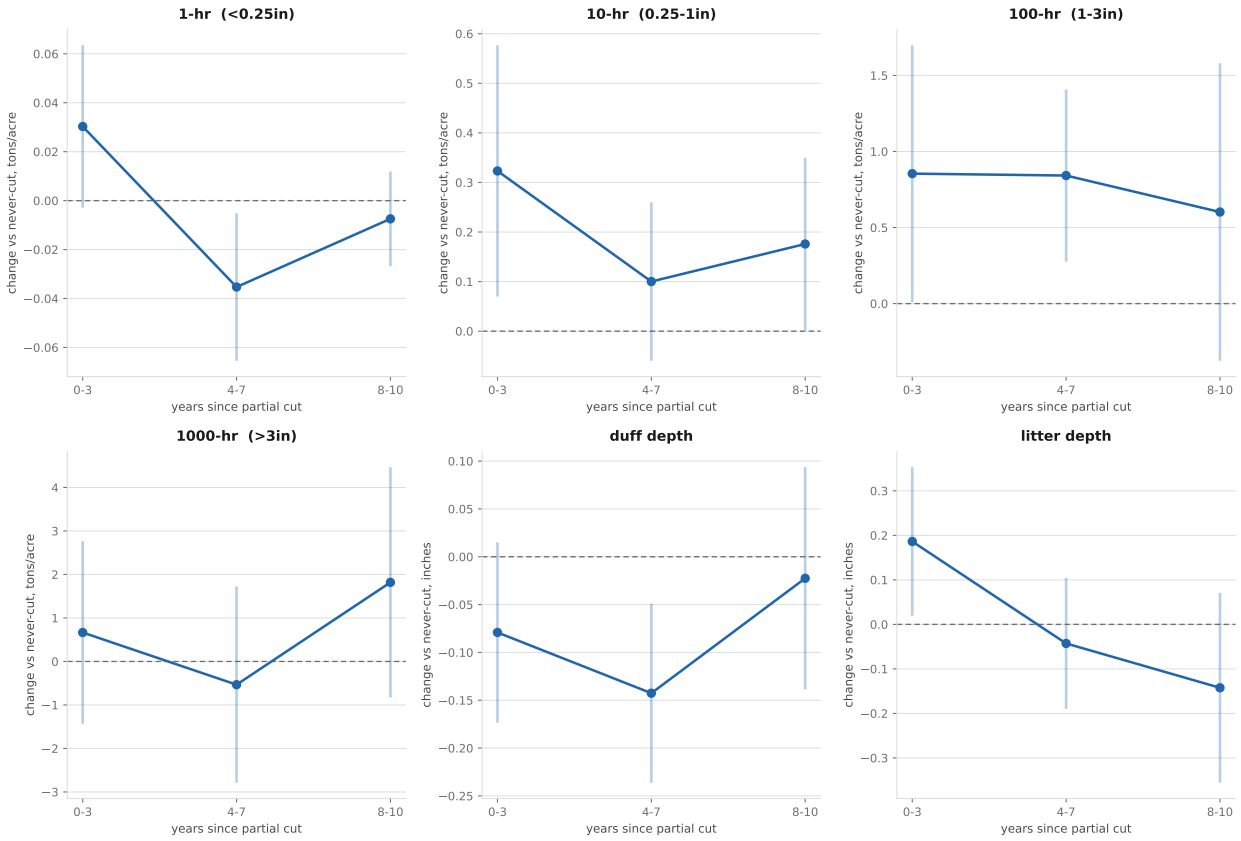


Figure 19: Surface fuel loads after a partial cut, relative to never-cut stands

Notes: As in Figure 18, with partial cuts as the treatment and the same 2,478 never-cut stands as the control group; no clearcuts enter this sample. The 311 partial cuts split 108 in the 0–3 bin, 135 in 4–7, and 68 in 8–10, and one stand measured 11 years after its cut is dropped for parsimony. There is no site-preparation effect because only 9 of these stands (2.9%) were site prepped.

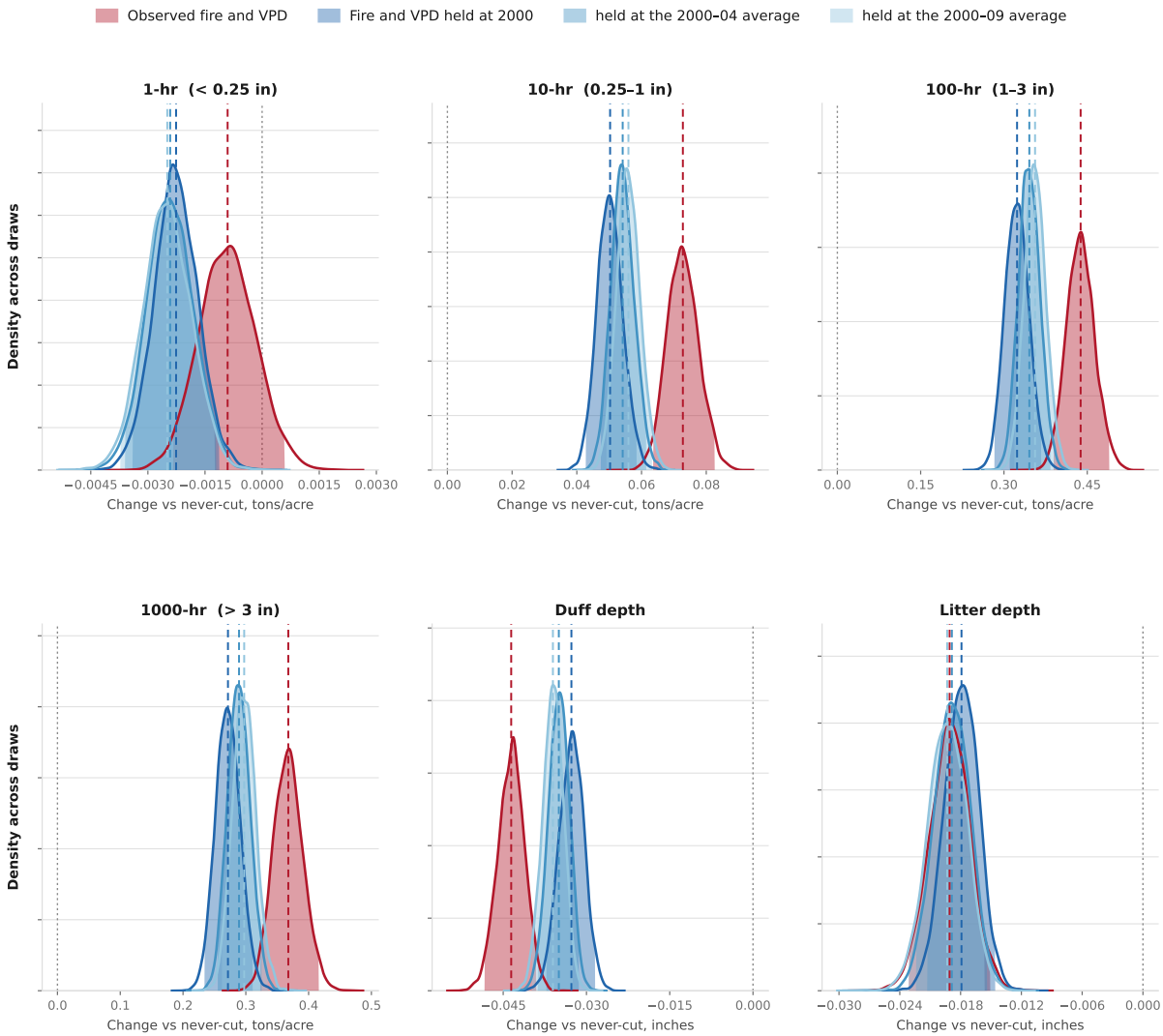


Figure 20: Simulated landscape surface fuel loads in 2020 under observed and counterfactual fire and climate regimes

Notes: Each panel gives the distribution across the 5,000 simulation iterations of the landscape-average change in one class of surface fuels, measured against a landscape that was never cut. Units are tons per acre for the four downed-wood size classes and inches for duff and litter depth. Cut stands take the coefficients in Figures 18 and 19 according to their years since cut, and site preparation is set at 30.7% of clearcuts in every iteration, matching the rate observed in the FIA sample. Dashed lines mark distribution means and shaded regions give the empirical 95% interval. Fuel loads are higher under the observed regime than under the counterfactual in all four woody classes, because more cutting leaves more slash, while duff and litter are lower, because cutting disturbs the forest floor. Relative to a never-cut landscape the three coarser woody classes are elevated, while 1-hour fuel sits slightly below never-cut under both regimes. Percent changes are formed by dividing the simulated change in load by the average loading of that fuel class across the LANDFIRE FCCS fuelbeds falling inside the observed fire perimeters, weighted by burned acres. For example, simulated 100-hour fuel in 2020 averages 0.4381 tons per acre above a never-cut landscape under the observed regime, against 0.3236 under the counterfactual. The difference of 0.1145 tons per acre, divided by the 1.8699 tons per acre average 100-hour loading of the burned fuelbeds, gives a 6.1% change.

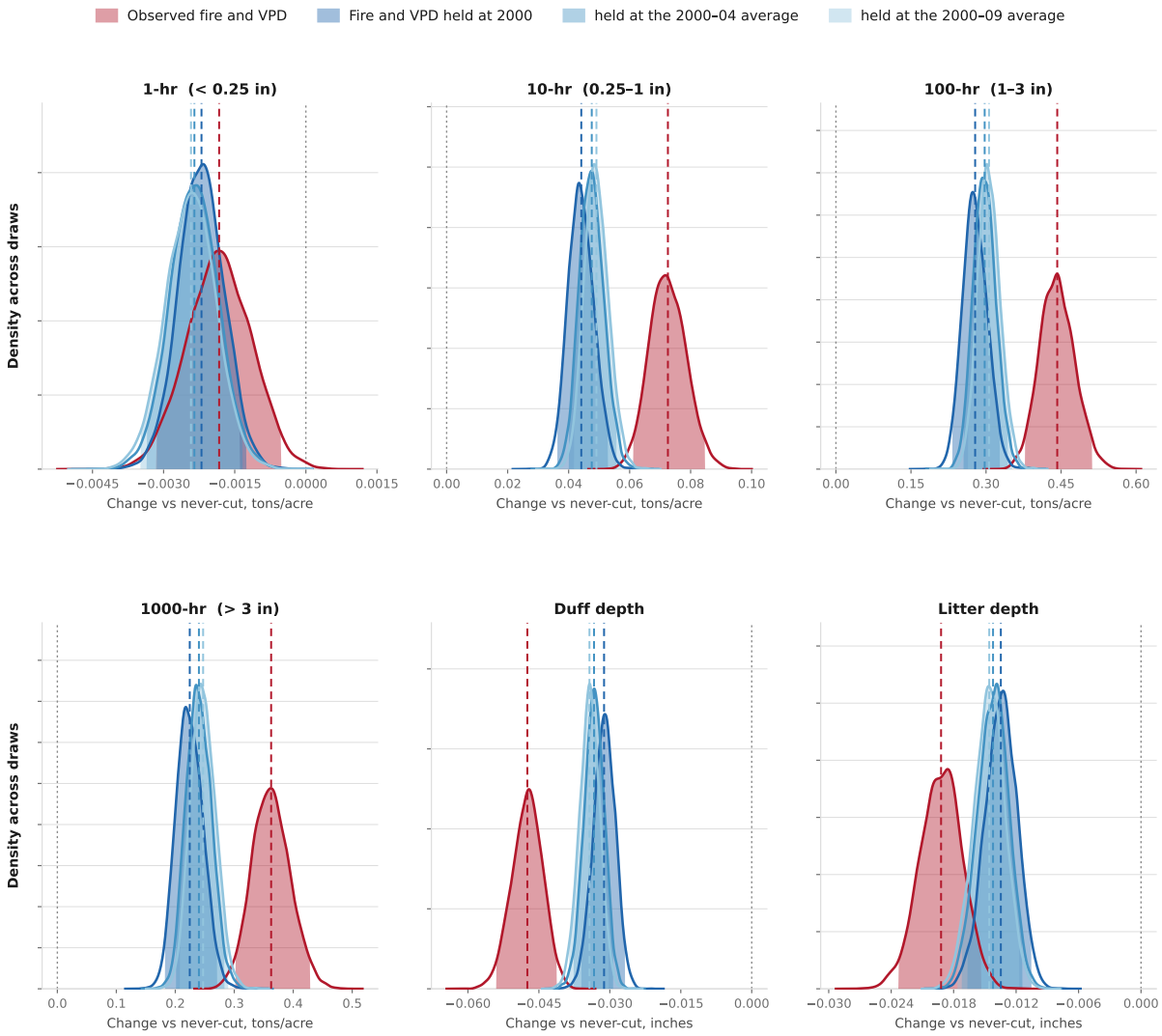


Figure 21: Simulated landscape surface fuel loads in 2024 under observed and counterfactual fire and climate regimes

Notes: Each panel gives the distribution across the 5,000 simulation iterations of the landscape-average change in one class of surface fuels at the end of 2024, measured against a landscape that was never cut. Units are tons per acre for the four downed-wood size classes and inches for duff and litter depth. Cut stands take the coefficients in Figures 18 and 19 according to their years since cut, and site preparation is set at 30.7% of clearcuts in every iteration, matching the rate observed in the FIA sample. Dashed lines mark distribution means and shaded regions give the empirical 95% interval. Fuel loads are higher under the observed regime than under the counterfactual in all four woody classes, because more cutting leaves more slash, while duff and litter are lower, because cutting disturbs the forest floor. Percent changes are formed by dividing the simulated change in load by the average loading of that fuel class across the LANDFIRE FCCS fuelbeds falling inside the observed fire perimeters, weighted by burned acres, shown in Table 22

Table 22 converts the changes in landscape-average fuels shown in Figures 20 and 21 into percent changes. We do this by taking the gap between the observed and counterfactual distributions and dividing it by the average fuel load of that class across the FCCS fuelbeds that burned in the corresponding fire year. The crown row instead uses the simulated change in standing above-ground biomass, over its baseline.

Table 22: Percent Changes in Surface and Crown Fuels Applied to the FOFEM Counterfactual Scenarios

	FCCS Avg.	CF1		CF2		CF3	
Fuel class	Fuel Loads	Change	%	Change	%	Change	%
Panel A. 2020 fire year							
1-hour (< 0.25 in)	0.433	0.0014	0.31	0.0015	0.35	0.0016	0.37
10-hour (0.25–1 in)	1.131	0.0225	1.99	0.0186	1.64	0.0168	1.49
100-hour (1–3 in)	1.870	0.1145	6.12	0.0924	4.94	0.0821	4.39
1000-hour (> 3 in)	20.847	0.0961	0.46	0.0785	0.38	0.0702	0.34
Duff depth	1.959	−0.0108	−0.55	−0.0086	−0.44	−0.0075	−0.38
Litter depth	0.836	−0.0012	−0.14	−0.0002	−0.03	0.0002	0.03
Crown fuels	73.111*	−4.5770	−6.26	−3.2795	−4.49	−2.6324	−3.60
Panel B. 2024 fire year							
1-hour (< 0.25 in)	0.339	0.0004	0.11	0.0005	0.15	0.0006	0.17
10-hour (0.25–1 in)	1.245	0.0284	2.28	0.0250	2.01	0.0234	1.88
100-hour (1–3 in)	1.152	0.1641	14.25	0.1451	12.60	0.1364	11.84
1000-hour (> 3 in)	9.490	0.1380	1.45	0.1225	1.29	0.1153	1.22
Duff depth	1.010	−0.0162	−1.61	−0.0141	−1.40	−0.0131	−1.30
Litter depth	0.799	−0.0057	−0.72	−0.0050	−0.63	−0.0046	−0.58
Crown fuels	69.763*	−7.4444	−10.67	−5.8623	−8.40	−5.0834	−7.29

Notes: Each percentage is the landscape-average change in fuel load under the observed fire and climate regime minus the change under the counterfactual, divided by the acre-weighted average fuel load across the LANDFIRE FCCS fuelbeds falling inside the observed fire perimeters for that year that burned on private and state timberland. The three counterfactuals differ only in the level at which fire and fire-season VPD are held fixed: CF1 at their 2000 levels, CF2 at the 2000–04 average, and CF3 at the 2000–09 average. Site preparation is set at 30.7% of clearcuts. A positive percent change means the observed regime carries more of that fuel than the counterfactual, so a FOFEM counterfactual scenario applies a percent reduction in that fuel load. FOFEM takes duff and litter weights as inputs, while our estimates are depths, so we apply the same proportional change to those weights. The crown row is formed differently from the rest, because standing biomass is not an FCCS field. It is the simulated change in standing above-ground biomass, and its denominator, marked *, is the simulated baseline biomass rather than an FCCS loading. We assume the change in above-ground biomass carries through proportionally to FOFEM's crown foliage and crown branch inputs. Above-ground biomass excludes foliage, so applying it to crown foliage assumes foliage scales with woody crown. All figures come from the same 5,000-iteration simulation as Figures 20 and 21.

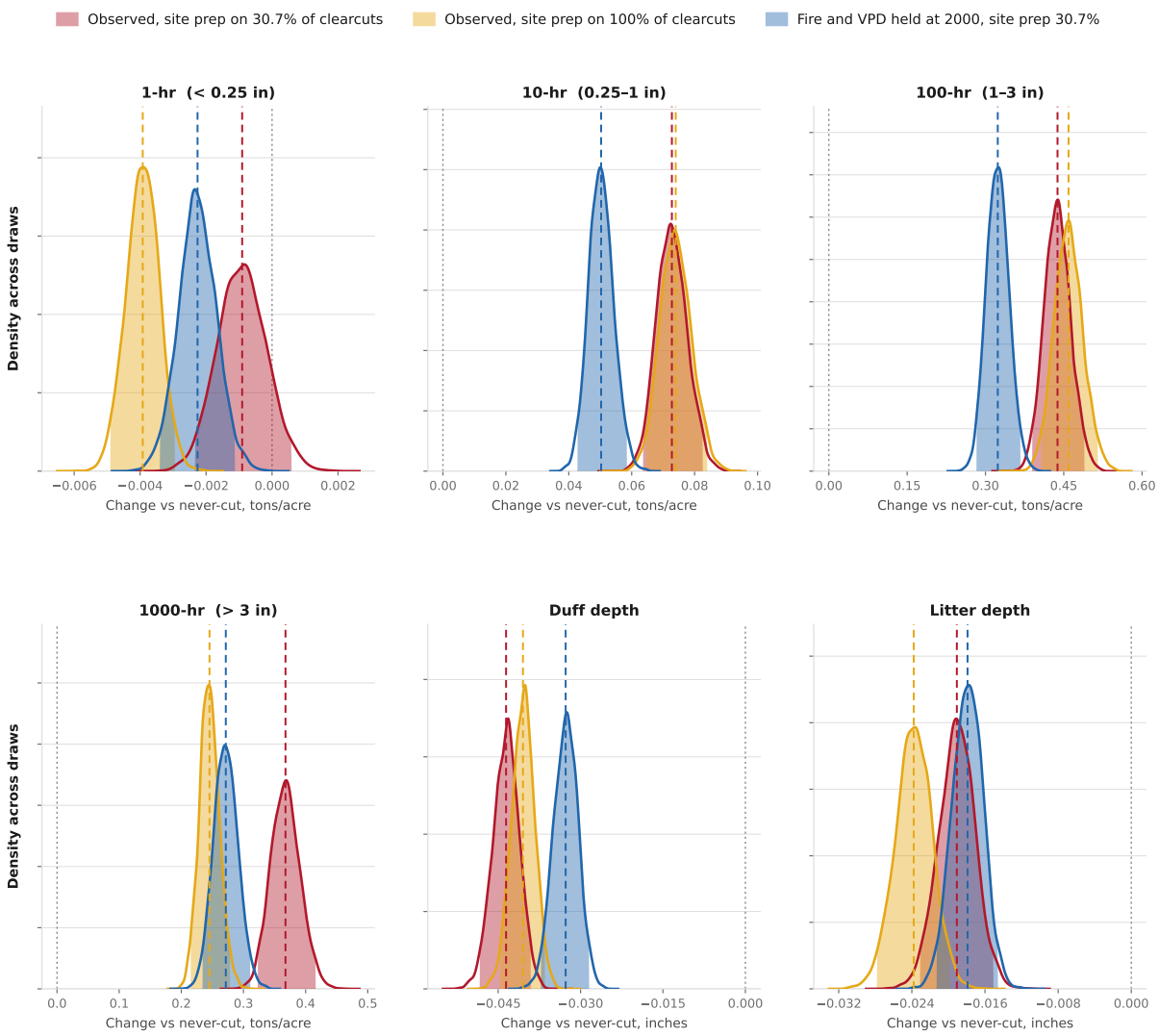


Figure 22: Simulated landscape surface fuel loads in 2020 under two rates of site preparation

Notes: Each panel gives the distribution across the 5,000 simulation iterations of the landscape-average change in one class of surface fuels at the end of 2020, measured against a landscape that was never cut, on the same basis as Figure 20. The three distributions separate two margins. The first two hold the observed fire and climate regime and differ only in how often a clearcut is site prepared, 30.7% as observed in the FIA sample against 100%. The gold-to-red distance is the effect of universal site prep. The third holds fire and fire-season VPD at their 2000 levels at the observed site prep rate, so the red-to-blue distance is the fire-regime effect. Dashed lines mark distribution means and shaded regions give the empirical 95% interval. Site preparation removes larger fuels, taking 1000-hour loads from 0.368 to 0.245 tons per acre, but it raises 100-hour loads from 0.438 to 0.459 against a counterfactual of 0.324.

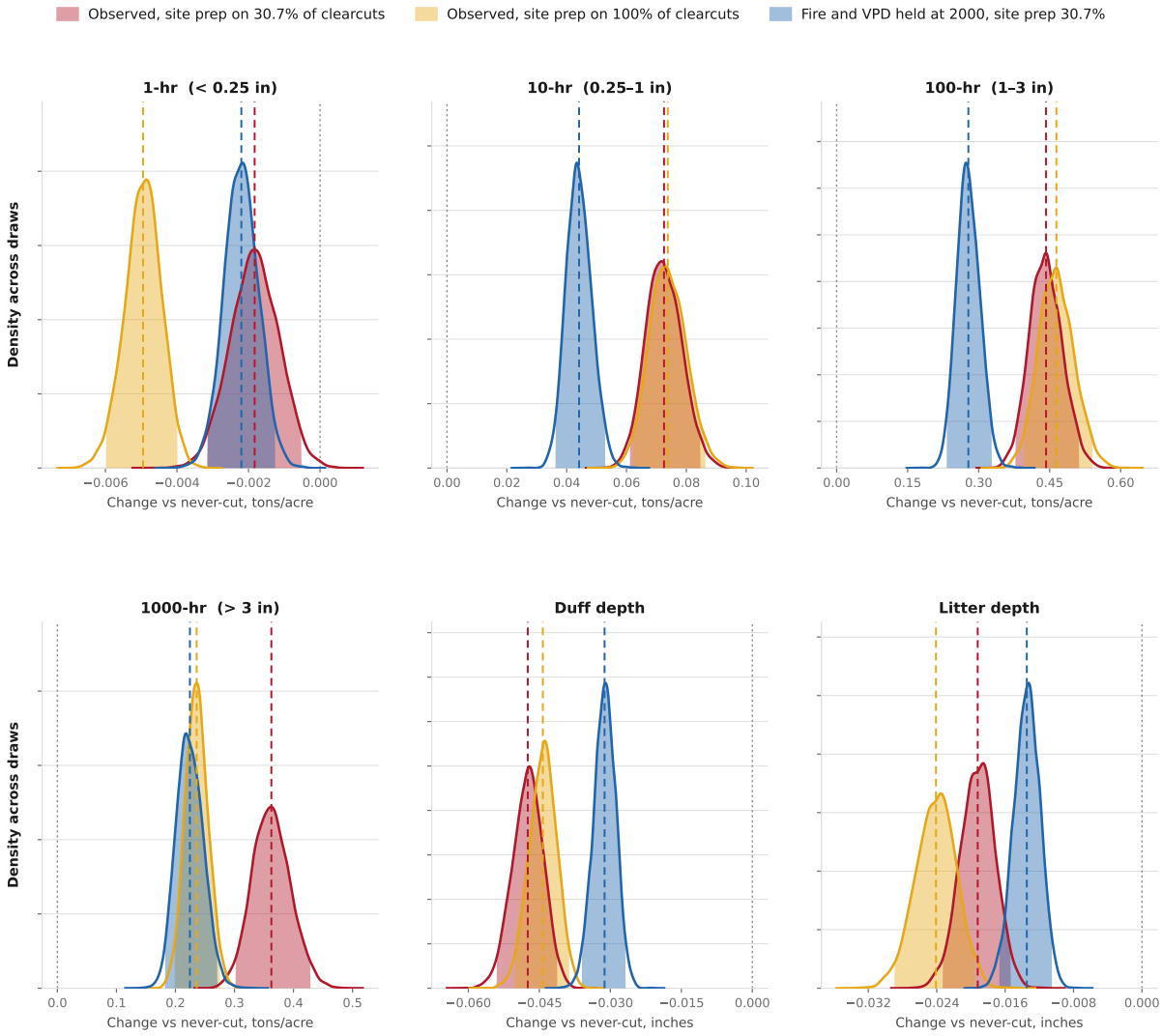


Figure 23: Simulated landscape surface fuel loads in 2024 under two rates of site preparation

Notes: Each panel gives the distribution across the 5,000 simulation iterations of the landscape-average change in one class of surface fuels at the end of 2024, measured against a landscape that was never cut, on the same basis as Figure 21. The three distributions separate two margins. The first two hold the observed fire and climate regime and differ only in how often a clearcut is site prepared, 30.7% as observed in the FIA sample against 100%. The gold-to-red distance is the effect of universal site-prep. The third figure holds fire and fire-season VPD at their 2000 levels at the observed site prep rate, so the red-to-blue distance is the fire-regime effect. Dashed lines mark distribution means and shaded regions give the empirical 95% interval. Site prep takes 1000-hour loads from 0.363 to 0.236 tons per acre while 100-hour loads rise from 0.442 to 0.464 against a counterfactual of 0.278.

Table 23: Site Preparation and Adaptation as Separate Margins on Surface Fuels

	FCCS Avg.	Site prep		Adaptation		Combined	
		100% vs 30.7%		Baseline vs CF1		100% prep vs CF1	
Fuel class	Fuel Loads	Change	%	Change	%	Change	%
Panel A. 2020 fire year							
1-hour (< 0.25 in)	0.433	−0.0030	−0.70	0.0014	0.31	−0.0017	−0.38
10-hour (0.25–1 in)	1.131	0.0012	0.11	0.0225	1.99	0.0237	2.10
100-hour (1–3 in)	1.870	0.0211	1.13	0.1145	6.12	0.1356	7.25
1000-hour (> 3 in)	20.847	−0.1223	−0.59	0.0961	0.46	−0.0262	−0.13
Duff depth	1.959	0.0031	0.16	−0.0108	−0.55	−0.0078	−0.40
Litter depth	0.836	−0.0047	−0.56	−0.0012	−0.14	−0.0059	−0.70
Panel B. 2024 fire year							
1-hour (< 0.25 in)	0.339	−0.0031	−0.92	0.0004	0.11	−0.0027	−0.81
10-hour (0.25–1 in)	1.245	0.0013	0.10	0.0284	2.28	0.0297	2.38
100-hour (1–3 in)	1.152	0.0218	1.90	0.1641	14.25	0.1860	16.15
1000-hour (> 3 in)	9.490	−0.1266	−1.33	0.1380	1.45	0.0115	0.12
Duff depth	1.010	0.0032	0.31	−0.0162	−1.61	−0.0130	−1.29
Litter depth	0.799	−0.0049	−0.61	−0.0057	−0.72	−0.0106	−1.33

Notes: Each percentage divides a landscape-average change in fuel load by the acre-weighted average fuel load across the LANDFIRE FCCS fuelbeds falling inside the observed fire perimeters for that year that burned on private and state timberland, as in Table 22. The site preparation columns compare the baseline scenario, in which 30.7% of clearcuts undertake site preparation, with a scenario in which 100% of clearcuts undertake site preparation. The adaptation columns compare the baseline, in which fire and fire-season VPD are as observed and landowners cut more in response, against CF1, which holds fire and VPD at their 2000 levels, with site preparation at 30.7% in each. The combined columns compare the baseline at 100% site preparation against CF1, which remains at the observed 30.7% rate, and show how surface fuels would compare under the adaptation scenario if every clearcut were site prepped; site preparation can offset the adaptation-driven increase in some fuel classes. A positive percent change means more of that fuel than the comparison world.

Table 24 reports the result of the FOFEM exercise. For each fire year we report the tons of $PM_{2.5}$ and CO_2 released by the acres that burned on private and state timberland, under the observed fuel loadings and then under each of the three counterfactual fuel regimes. We observe fuels from LANDFIRE FCCS fuelbeds, using the most recent version available that pre-dates our chosen fire years, corresponding to 2016 LANDFIRE FCCS for 2020 and 2023 LANDFIRE FCCS for 2024. A world in which landowners had cut less would have carried more standing crown biomass but less surface fuels related to harvesting. Because crown burns in the flaming phase while surface and ground fuels smoulder, that world would have released more carbon and less $PM_{2.5}$.

Table 24: Estimated Wildfire $PM_{2.5}$ and CO_2 Emissions on Burned Private and State Timberland, Observed and Counterfactual Fire Regimes

		$PM_{2.5}$ (metric tons)			CO_2 (metric tons)		
Scenario		Emitted	Difference	%	Emitted	Difference	%
Panel A. 2020 fire year	85 fires, 811,714 acres						
Observed regime (baseline)		436,359	—	—	35,434,539	—	—
CF1	2000 levels	435,034	−1,325	−0.30	35,748,703	+314,164	+0.89
CF2	2000–04 average	435,126	−1,233	−0.28	35,641,725	+207,186	+0.58
CF3	2000–09 average	435,202	−1,158	−0.27	35,588,038	+153,499	+0.43
Panel B. 2024 fire year	52 fires, 140,921 acres						
Observed regime (baseline)		29,409	—	—	3,505,985	—	—
CF1	2000 levels	29,267	−142	−0.48	3,612,960	+106,975	+3.05
CF2	2000–04 average	29,243	−167	−0.57	3,587,569	+81,584	+2.33
CF3	2000–09 average	29,212	−198	−0.67	3,575,162	+69,177	+1.97

Notes: Emissions are estimated with FOFEM 6.7 (First Order Fire Effects Model). We take every acre that burned on private and state timberland from the MTBS wildfires from California, Oregon, and Washington in the given fire year, assign it the LANDFIRE FCCS fuelbed and MTBS burn severity recorded at that location, and run the resulting fuels through FOFEM to obtain emissions per acre, which are then weighted up by acres burned. Ownership comes from Harris et al. (2025); we keep family, corporate and state forest and exclude federal, tribal, local and unknown-ownership land. The baseline row uses the observed FCCS fuels. Each counterfactual row rescales those fuels by the simulated difference in landscape-average fuels reported in Table 22, holding acres burned, burn severity, season and fuel moisture fixed, so that the difference between rows is attributable to the changes in fuels. The three counterfactuals differ only in the level at which fire and fire-season vapour pressure deficit are held fixed: CF1 at their 2000 levels, CF2 at the 2000–04 average and CF3 at the 2000–09 average. Site preparation is set at 30.7% of clearcuts throughout. We report differences as the counterfactual minus the baseline. We find that the counterfactual world would have emitted less smoke than was actually observed and more CO_2. Herb and shrub fuels are held at observed levels in every scenario because the simulation produces no estimate for them. FOFEM reports emissions in pounds per acre; we report metric tons throughout, converting at 1 short ton = 0.907185 metric tons. We assume a dry regime in FOFEM. Emissions levels are sensitive to this choice, but the sign and rough magnitude of the differences across scenarios are robust to different moisture regimes. The two years are not comparable in scale. 2020 was an extreme season in western Oregon while 2024 burned less than a fifth as many acres of private and state timberland.

Table 25 separates the two margins. The site preparation columns hold the observed fire and climate regime and ask what would happen if every clearcut were site prepped rather than the observed 30.7%. The adaptation columns repeat the baseline against CF1 comparison from Table 24, and the combined columns put both margins together. Site preparation lowers both pollutants, because it removes more coarse woody fuel than it adds in the finer size classes and leaves standing crown biomass untouched. Adaptation raises $PM_{2.5}$ and lowers CO_2.

Table 25: Site Preparation and Adaptation as Separate Margins on Wildfire $PM_{2.5}$ and CO_2 Emissions

	Site prep 100% vs 30.7%		Adaptation Baseline vs CF1		Combined 100% prep vs CF1	
Pollutant	Change	%	Change	%	Change	%
Panel A. 2020 fire year	85 fires, 811,714 acres					
$PM_{2.5}$	−687	−0.16	+1,325	+0.30	+638	+0.15
CO_2	−62,173	−0.18	−314,164	−0.89	−376,337	−1.06
Panel B. 2024 fire year	52 fires, 140,921 acres					
$PM_{2.5}$	−69	−0.23	+142	+0.48	+73	+0.25
CO_2	−7,394	−0.21	−106,981	−3.05	−114,375	−3.26

Notes: Metric tons of $PM_{2.5}$ and CO_2 emitted by the acres that burned on private and state timberland from the MTBS wildfires from California, Oregon, and Washington in the given fire year, estimated with FOFEM 6.7 as in Table 24, which reports the underlying levels. Each pair of columns differences two of those scenarios, on the same basis as Table 23, and in the same order: the first named world minus the second. The site preparation columns hold the observed fire and climate regime and raise site preparation from the observed 30.7% of clearcuts to 100%. The adaptation columns hold site preparation at 30.7% and compare the observed regime against CF1, which fixes fire and fire-season vapour pressure deficit at their 2000 levels. The combined columns compare the observed regime at 100% site preparation against CF1 at the observed rate, and so show whether site preparation could offset the emissions consequence of the additional harvesting. Percentages divide by observed baseline emissions in all three cases, so that the columns are additive and the combined change is the sum of the other two. A negative entry means the first named world emits less. Site preparation lowers both pollutants, while adaptation raises $PM_{2.5}$ and lowers CO_2, so the two margins reinforce each other for carbon and work against each other for smoke. We assume a dry regime in FOFEM; emissions levels are sensitive to this choice, but the sign and rough magnitude of these differences are robust to changes in moisture.